SECTION TP20
MATERIALS AND WORKMANSHIP

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TP20: MATERIALS AND WORKMANSHIP

TP20.01 GENERAL

A. General

Materials and workmanship shall be in accordance with the requirements of the Technical Provisions unless otherwise approved by the Engineer.

Inclusion of a material or method in this Section shall not necessarily indicate approval for application or use in a specific situation. When a material or method is specified in these Technical Provisions, it shall be applicable; however, specific requirements detailed in applicable Technical Provisions shall take precedence.

Material Safety Data Sheets (MSDS) shall be submitted for all materials, used in the fabrication of the cars. This requirement shall include lubricants but excludes non-hazardous metallic materials. Information shall be in a form compliant with ANSI Z400.1-1993 **[CDRL 20-001]**.

B. Quality

In accordance with TP21, the Contractor's Quality Assurance (QA) Program shall assure that all aspects of the Contract are in conformance with the design, materials, and workmanship requirements provided in the Technical Provisions.

C. Standards

The following domestic standards and specifications shall define the materials for this Contract: Federal or Military Specifications or Standards, the Specifications of the Aluminum Association of America, AAR, AISI, ANSI, ASME, ASTM, FRA, IEEE, APTA Standards and Recommended Practices for Rail Passenger Equipment, and others, as specified herein. Where other or foreign standards are proposed by the Contractor, the Contractor shall submit documentation providing a section by section comparison for RTD review and approval to demonstrate that the proposed standards are the equivalent of the foregoing standards and specifications. Proposed substitute specifications shall be submitted in both English and the language of the country of origin. The most recent standards and specifications applicable at the time of issuance of the Notice to Proceed (NTP) shall apply unless otherwise approved by the Engineer.

D. Marking

All materials intended for use on these vehicles shall be marked or stored so as to be readily identifiable and shall be adequately protected during handling and storage. Rejected material shall be clearly marked and stored in an area specifically designated for that purpose.

E. Cleaning Agents

A list of recommended cleaning agents shall be provided to the Engineer for all materials exposed to normal cleaning operations. The Contractor shall make every attempt to minimize the number of different cleaning agents proposed. Where possible, the cleaning agents recommended shall be among those currently used by RTD for similar applications **[CDRL 20-002]**. This information shall also be included in the maintenance documentation for the vehicle.

F. Prohibited Materials

The following materials shall be prohibited from use on the cars:

- PVC
- Asbestos
- Cadmium (except for battery)
- Lead, all applications including in paint and coatings, except for electronics solder and diesel engine starting applications
- PCBs
- Carcinogenic materials as listed by current Publication of the American Conference of Governmental Industrial Hygienists (ACGIH)
- All CFC and HCFC compounds
- Urethane foam
- Materials listed in 29 CFR Sec. 1910.9

G. Material Handling

The Contractor shall exercise care in the handling of all material used in the construction of the EMU and DMU railcars. Examples of mishandling material include but are not limited to the following:

- Exposure to corrosives or other elements that may deteriorate the subject material
- Inappropriate storage (stacking beyond packaged limits, careless placement of material, temperature extremes beyond the manufacturer's recommendations)
- Dragging materials across rough surfaces that may cause damage (e.g. dragging cables)
- Stretching materials during handling
- Exposure of certain materials to excessive UV light

RTD reserves the right to direct the Contractor to change material handling methods if it is deemed necessary.

TP20.02 JOINING AND FASTENING

A. Joining

1. General

Certain combinations of materials require particular care in joining to avoid the possibility of corrosion. Isolating and moisture-proofing materials, appropriate to the materials being joined, shall be employed at all times where these combinations exist. The contractor shall design the vehicles to minimize the number of such combinations, and to minimize the accumulation of water, cleaning chemicals, and chemicals present in the environment, at or near combination joints. Isolating and moisture-proofing materials, appropriate to the materials being joined, shall be used at all times where these combinations exist. Detailed requirements for protection of materials against effects of joining dissimilar materials are contained in sections TP20.02.A.3, TP20.02.A.4, TP20.02.A.5, TP20.03.C, and TP20.06.D. The contractor shall provide a detailed plan for installation of dissimilar materials in accordance with these sections and applicable industry practice. This plan shall be submitted for approval by the Engineer prior to beginning of assembly of the first car **[CDRL 20-003].**

2. Joint Fitting

Joints shall be properly fitted, whether exposed or concealed. When not otherwise specified in Contractor drawings or specifications, gaps between joints shall be held to a dimension less than 10% of the thinner material being joined or 0.002 inch, whichever is greater. Gaps shall be uniform in width. Edges shall be free of burrs and sharp edges and shall have a smooth, finished appearance.

Where excessive gaps (greater than those permitted by approved drawings or standards) are found to exist at the faying surfaces of structural bolted or riveted connections, metal shims of the same material as that of the deficient part may be used, but only with the written approval of the Engineer. Shims, if used, shall be permanently fastened to one of the base parts being joined. The use of epoxy or other plastic filler at such locations is expressly prohibited.

3. Metal-to-Metal Connections

Where metals contact each other, the contact surfaces shall be free of dirt, grease, rust, and scale. Unless specified otherwise, the contact surfaces shall be coated with a metal based primer which conforms to Society for Protective Coatings Specification SSPC-Paint 25. Metal primer may be omitted for austenitic stainless steel to austenitic stainless steel joints.

For proper treatment of a connection involving aluminum, refer to TP20.06.D.

4. Wood-to-Metal Connections

Where wood and ferrous metal surfaces are placed together, the wood shall be coated with aluminum paint conforming to Federal Specification TT-P-38, and the metal shall be coated with a primer which conforms to Society for Protective Coatings Specification SSPC-Paint 25.

All bolts or rods passing through wood shall be coated with aluminum paint conforming to Federal Specification TT-P-38.

5. Wood-to-Wood Connections

Where wood and wood are placed together, both abutting surfaces shall be coated with aluminum paint conforming to Federal Specification TT-P-38.

B. Fasteners

1. General

The Contractor and suppliers shall be responsible for selecting fastener types, sizes, styles, lengths, materials, grades, and finishes that will meet the requirements of the Technical Provisions. The Contractor shall minimize the number of different sizes and styles of fasteners used. Use of self-tapping fasteners is prohibited, unless expressly approved by the Engineer.

Fasteners used throughout the car shall be inch-standard fasteners, except that ISO metric fasteners may be used in conformance with TP20.02.B.2.b.

All fasteners used on this car shall be specified under one of three categories:

- Category 1: Electrical and electronic,
- Category 2: Structural and safety-related,
- Category 3: Decorative.

Design submittals and drawings shall indicate, where appropriate, the type of fastener (Category 1, 2 or 3) that is proposed for the relevant application. Fasteners internal to electrical or electronic components are specified in appropriate materials and workmanship subsections for electrical devices and wiring. All structural and safety-related fasteners are specified under TP20.02.B.3. Fasteners used to attach interior lining or trim and exposed to passenger view are specified under TP20.02.B.4. All fasteners in any category, which attach to car structure, shall be in accordance with the requirements of the Technical Provisions.

Safety-related fasteners include, but are not limited to, those applied to trucks, bolsters, brake equipment, couplers, and power collection devices. A fastener is safety related if failures cannot be tolerated, that is, if even a single fastener fails there is a possibility of brake failure, derailment, or an accident. In the event of a dispute, the Engineer will be the final arbitrator in determining which fasteners are to be considered safety related.

All fasteners used to secure access covers, doors, or panels to equipment boxes or interior panels shall be made captive to the panel in which they are used. Where access for service is expected more often than every five years, access panels shall be equipped with quarter-turn fasteners. Quarter-turn fasteners shall have a minimum shank diameter of 1/4 inch, be of adequate strength, and as manufactured by Southco, or approved equal.

When making connections to heat producing apparatus, thermal expansion of the components shall be taken into consideration for selection of fastener materials. If the joined

components are high expansion alloys such as copper or austenitic stainless steel, austenitic stainless steel fasteners shall be used. If the joined components are low expansion materials such as carbon steel or ferritic stainless steel, zinc plated carbon steel fasteners of minimum Grade 5 shall be used.

When bolts are used to secure apparatus, where the bolt head is not accessible, a reusable mechanical locking device shall be used to prevent the bolt head from turning when the nut is being turned.

At least 1 1/2 full screw threads shall be visible beyond all nuts. When used without elastic stop nuts, bolts shall not project more than 1 1/2 full threads plus 1/4 inch for bolts, 1/4-inch diameter or less, and shall not project more than eight threads for larger diameter bolts, unless otherwise approved by the Engineer. With elastic stop nuts, bolt threads shall not project more than 1/4 inch, regardless of bolt size.

Undercar equipment shall not be supported by bolts in tension.

All-metal prevailing torque-type locknuts shall only be used where there is insufficient clearance to install ESNA type locknuts, or where the locknut is exposed to temperatures above 200 degrees Fahrenheit.

2. Threaded Fasteners

a. Inch-Standard Fasteners

All inch-standard threaded fasteners shall conform to ANSI B1.1 Standard, Unified Inch Screw Threads (UN and UNR Thread Form), or Industrial Fasteners Institute Fastener Standards.

Prevailing torque-type locknuts shall be nylon insert type, ESNA, or approved equal, conforming to IFI Fastener Standards or Military Standard MS-21044.

b. Metric Fasteners

Upon approval by the Engineer, specific Line Replacement Units (LRUs) that are supplied by a supplier or sub-supplier to the Contractor may be supplied with metric fasteners to ANSI B1.13M (ISO-metric) standards. All internal fasteners and threaded components of the approved unit shall have ISO-metric threads. Internal to components, there shall be no mixing of metric and inch-threaded fasteners. External mounting fasteners and threaded connecting components shall have ISO-inch threads to ANSI B1.1 Standards. Each unit, component, or group containing ISO-metric threads shall be indelibly identified in an approved manner and in a conspicuous approved location, to signify that the unit contains metric threaded fasteners. All repair and maintenance manuals shall be conspicuously marked on each page to indicate where metric threaded fasteners are used within the unit. Replacement, repair, and maintenance parts supplied under the Technical Provisions shall include all necessary replacement fasteners of the correct size and grade.

Metric fasteners shall be marked as required in “Metric Fastener Standards,” Industrial Fasteners Institute’s latest edition.

3. Structural Fasteners

All structural fasteners shall have documentation identifying manufacturer and purchase specifications available for examination by the Engineer at the Contractor's QA department. This documentation shall include the fastener material or grade and finish including plating material and specifications, when applicable. Whether the buyer is a sub-contractor, supplier, or the Contractor, the Contractor shall obtain and hold this documentation for a period ending no sooner than the termination date of the warranty period for the last car.

Where applicable, safety related fasteners shall comply with all of the requirements of 49CFR231 – Railroad Safety Appliance Standards.

All safety-related fasteners shall conform to the requirements of SAE J429, and a representative sample of each production lot of fasteners shall be tested for conformance to purchase specifications by an independent laboratory accredited by the American Association of Laboratory Accreditation (AALA) or approved equal. A production lot is defined as one size of fastener, from one manufacturer, produced during one continuous production run. Fasteners not meeting this definition of production lot shall be treated as separate lots. Testing shall be performed using sample quantities as proposed by the Contractor and approved by the Engineer. Tests conducted shall confirm that fastener material meets specified chemistry and strength requirements. The Contractor shall obtain certified test results from the testing laboratory and hold the documents for a period ending no sooner than the termination date of the warranty period for the last car.

All safety-related fasteners that are plated or chemically cleaned shall have certifications showing freedom from hydrogen embrittlement, based on a representative sample of the actual production fasteners which have been tested for hydrogen embrittlement by the OEM Contractor or a supplier following ASTM F519 procedures. An ASTM F606 wedge-test sample may be used in place of the F519 standard samples. Test loads shall be a minimum of 80% of yield strength or proof load and held for a minimum of 168 hours. Any failures shall reject the entire lot.

All structural bolts shall be a minimum Grade 5, and for undercar and overhead mounted equipment, the bolt diameter shall be no less than 3/8 inch, regardless of design load. Stronger fasteners shall be used if required by the application.

The mounting and attachment bolts for undercar- and overhead-mounted equipment and equipment support structures and brackets shall be sized to the design strengths for Grade 2 bolts and Class A nuts.

4. Decorative and Appearance Fasteners

All interior fasteners exposed to passengers shall be either bright or finished to match the surfaces being joined and installed such that the fastener head is flush with the mating surface. Bright finished fasteners used for stanchions shall be austenitic grade stainless steel. Bright finished interior fasteners may be either austenitic or plated martensitic stainless steel.

All exterior fasteners visible to passengers shall be austenitic stainless steel for the stainless steel carbody. Exterior aluminum shall be joined by austenitic stainless steel or aluminum alloy fasteners, as appropriate to the design and appearance requirements. Fasteners used on

the side sill to attach heavy equipment brackets are structural fasteners specified under the structural fasteners section.

The Contractor shall prepare a fastener plan identifying all interior fasteners and detailing which of those fasteners are recommended to be tamper-proof fasteners. Tamper-proof fasteners shall have pin-in-head torx style heads. Tamper-proof fasteners should be considered for safety sensitive equipment access panels. Fasteners on access panels, plates, covers, or other components accessible by passengers shall be of a single style of fastener. The fastener plan shall be submitted for review and approval of the Engineer **[CDRL 20-004].**

All decorative and appearance fasteners shall have documentation that identifies the manufacturer, base material, plating, or finish, if applied, and the fastener type. The Contractor or supplier shall maintain this documentation on file for the Engineer to review for a period ending no sooner than the termination date of the warranty period for the last car.

5. Torquing

All safety-related fasteners, including truck and brake equipment bolts and all fasteners exposed to fatigue loads, shall be torqued to a minimum preload equal to 75% of their proof load and "torqued striped" after torquing by paint or other approved means. All other fasteners shall be torqued to a value appropriate to the application, so that they do not loosen in service.

Fastener installation torque for standard oiled or waxed bolts with standard or heavy hex nuts shall be calculated from Industrial Fasteners Institute, Fastener Standards equations using values for "K" of 0.18 for unplated and 0.15 for plated threads. Locknuts shall be torqued in accordance with their manufacturer's recommendations, or the Contractor may conduct tests to determine the proper installation torque. For those nuts or bolts requiring "torque striping," the Engineer may require bolt torque-tension tests to verify that installed preload is equivalent to 75% of proof loads.

6. Washers and Lock Washers

Washers shall be used under the heads of all bolts, screws and under all nuts. Washers shall ANSI B18.22.2 or ANSI B18.22M, latest revisions, as is appropriate for the application. Where high strength fasteners are applied, washers shall be hardened and comply with IFI Fastener Standards.

Lock washers, when applied, shall conform to IFI Fastener Standards. Lock washers shall not be used for fatigue applications where the fastener must be torqued and marked. If applicable, prevailing torque nuts shall be used for these applications.

Other types of washers, including Belleville washers, may be used for special applications with approval by the Engineer.

7. Rivets and Lock Pins

Rivets and lock pins exposed to passengers shall be austenitic stainless steel or aluminum, as appropriate to the materials being joined. Structural steel rivets shall conform to ASTM A

502 or American National Standard B 18.1.2 standards. Rivets may be hand driven when hot and shall completely fill the rivet holes. Rivets driven cold shall be mechanically driven. Exposed heads shall be concentric with the shank and free from rings, fins, pits, and burrs.

Swage-locking (Huckbolt type) fasteners shall conform to Military Specification MIL-P-23469. All rough surfaces of the collar end of these fasteners shall be machined or ground smooth where accessible to passengers, crew, or maintenance personnel performing routine maintenance functions. The Engineer shall be the final arbiter in determining whether an application is hazardous to maintenance personnel.

8. Plating of Fasteners

All carbon, alloy, and martensitic stainless steel fasteners shall be plated with zinc, unless specifically waived by the Engineer. Zinc plating shall conform to ASTM-B-633, Type II, and SC2, SC3, or SC4 or ASTM B695, Class 8, Type II. Cadmium plated fasteners will not be permitted.

Grade 8, Metric 10.9, or stronger fasteners shall not be plated if the OEM finish is other than plating.

Alternate fastener coatings are permissible if qualified by testing per ASTM B117 with no red rust or visible corrosion products after 96 hours of exposure. The Contractor shall submit qualification results for each process used at each subcontractor applying the proposed coating. In order to use an alternate coating, the vendor shall submit:

1) Coating manufacturer's product data including required thickness,

2) ASTM B117 test results from an accredited third party laboratory,

3) Documentation of torque/tension characteristics, and

4) A statement from the coating manufacturer regarding the propensity for the coating process to cause hydrogen embrittlement of the fastener during coating.

Regardless of the coating's propensity for hydrogen embrittlement, each lot of high strength fasteners, including OEM plated zinc or yellow bolts (Grade 5 or Metric Grade 8.8 or higher) shall be tested for hydrogen embrittlement in accordance with Section 15.4.2.3.f. Each lot of lower strength fasteners shall be tested for hydrogen embrittlement if the coating has the possibility of causing hydrogen embrittlement.

If the proposed coating results in a change in the K-value for the plated fastener to outside the range of 0.13-0.15, as defined by Industrial Fasteners Institute Standard IFI-543, the vendor shall use the alternate coating on all fasteners within the particular LRU. The LRU shall contain an indelible label identifying the coating type used within the LRU and the required torque values for each size fastener used therein. Fasteners internal to a subcomponent within an LRU may use the standard coating system if they are not subject to removal during Owner's maintenance activities.

9. Rivet and Bolt Holes

Rivet and bolt holes shall be accurately located and aligned, and, when necessary, during assembly, holes shall be reamed round to specified size in position. This allowance does not relieve the Contractor of the interchangeability requirements of the Technical Provisions. Bolt hole clearances shall not exceed the Industrial Fasteners Institute's requirements. All removed and replaced rivets shall have the holes reamed to the size required such that the next larger rivet may be driven securely.

All holes for mechanical fasteners shall be clean and free of burrs. The Contractor shall devised a method for removing the burrs on the far (blind) side of the fastener hole and a method for inspection.

TP20.03 STAINLESS STEEL

A. General

Permitted uses of structural stainless steels are specified throughout the Technical Provisions. Ferritic stainless steels shall be painted where exposed to passengers or the weather. Austenitic stainless steels may be unpainted. Unpainted stainless steels exposed to passengers shall be a single grade of austenitic stainless steel in which both the color and surface finish of abutting pieces shall match, except where the design specifically calls for contrasting appearance.

B. Austenitic Stainless Steel

Structural austenitic, stainless steel components assembled by fusion or resistance welding, shall be of AISI type 201L (UNS S20103), 301L (UNS S30103), 301LN (UNS S30153), or JIS SUS301L (with Nitrogen) and shall conform to the requirements of ASTM A 666, except that the carbon content shall not exceed 0.03% and the nitrogen content of type 301LN and SUS301L (with Nitrogen) shall not exceed 0.25%. Other stainless steels conforming to ASTM A 666 are acceptable for non-welded applications.

Stainless steel used in structural applications covered by the Technical Provisions shall also conform to APTA SS-C&S-004-98, *"Standard for Austenitic Stainless Steel for Railroad Passenger Equipment."*

General requirements for delivery of stainless steel shall be as required by the Certification Provisions of ASTM A 666, and stainless steel to be used in structural applications shall be tested for susceptibility to intergranular corrosion in accordance with ASTM A 262, latest revision. Practice A of ASTM A 262 can be used to accept material only; Practice E is required for final determination of acceptance or rejection of material that is not acceptable by Practice A.

C. Ferritic Stainless Steels

When specified, ferritic stainless steel conforming to ASTM A 176 may be used for sheeting up to 0.20-inch thickness. Ferritic stainless steel sheet shall have a ductile-to-brittle transition temperature (DBTT) or nil-ductility temperature (NDT) below 0 degrees Fahrenheit. Weld heat-affected zones shall also have a DBTT or NDT below 0 degrees Fahrenheit. Ferritic stainless steel sheet shall have a balanced composition (low carbon and/or suitable titanium content) that will, for all conditions of fabrication and assembly into the carbody, inhibit formation of martensite and

limit chromium depletion in weld heat-affected zones so that material shall meet ASTM A 763 requirements for resistance to intergranular corrosion.

General requirements for delivery of stainless steel shall be per ASTM A 480.

Where ferritic stainless steels are welded to other structural steels, the less-noble steel shall be painted with weld-through primer.

Ferritic stainless steel shall be used only with the specific written approval of the Engineer.

D. Testing

The Contractor shall prepare (or have prepared), submit, and receive approval of a test and inspection plan for acceptance of all stainless steel to be used in welded applications prior to purchasing any such material. The tests and inspections shall verify that the stainless steel conforms to specified requirements. For austenitic stainless steels, the test and inspection plan shall include frequency of submittal of certifications in accordance with Certification Provision of ASTM A 666 and frequency of submittal of checks for susceptibility to intergranular corrosion in accordance with ASTM A 262. For ferritic stainless steels, the test and inspection plan shall include frequency of submittal of checks for susceptibility to intergranular corrosion in accordance with ASTM A 763 **[CDRL 20-005]**.

TP20.04 LOW-ALLOY, HIGH-TENSILE STEEL

A. General

Low Alloy High Tensile (LAHT) steel structural shapes, plates, and bars shall, as a minimum, conform to the requirements of ASTM A 588, where available. Plate steel may alternatively conform to ASTM A 710, Grade A, Class 1, 2, or 3. Where not available in A 588, hot rolled or formed structural shapes conforming to ASTM A 36 may be used for limited applications, including equipment supports and jack pads. General requirements for delivery of LAHT shapes, plates, and bars shall be as required by ASTM A 6.

Cold and hot rolled LAHT sheet and strip shall, as a minimum, conform to the requirements of ASTM A 606, Type 4. General requirements for delivery of these products shall be as required by ASTM A 568.

Other low-alloy, high-tensile steels which meet or exceed the above minimum requirements may be used, provided their detailed specifications are submitted and approved as equivalent or better material for the proposed applications. All LAHT steels shall be applied according to their specification properties.

Welded LAHT steel shall develop 15 ft-lbs Charpy V Notch impact strength in the CGHAZ (Coarse grain heat affected zone) 0.039 inches from fusion area at -20 degrees Fahrenheit.

B. Testing

The Contractor shall prepare (or have prepared), submit, and receive approval of a test and inspection plan for acceptance of all structural steels in accordance with the requirements of this Section before purchasing any such material. The test and inspection plan shall include provisions

for submission of reports and certification to the Engineer for each shipment in accordance with the applicable material requirements and specified CGHAZ impact tests **[CDRL 20-006]**.

TP20.05 STRUCTURAL CASTINGS

A. General

The Contractor shall be responsible for selecting casting grade, composition, strength, and finishing. However, steel castings, used in the carbody structure and truck assemblies shall meet AAR Specification M-201 latest revision, Grade "B," plus 2% nickel, minimum. These castings shall be heat treated to develop a minimum tensile strength of 75,000 psi, a minimum yield strength of 48,000 psi, elongation of not less than 25% in 2 inches, and reduction of area of not less than 50%. Also, steel castings used for coupler, drawbars, and anchors shall meet AAR Specification M-201, latest revision, Grade "C" or "E," quenched and tempered.

Where cast steel of superior properties is required for a specific application, the Contractor may propose such castings for review and approval of the Engineer.

Stainless steel composition and processing must be selected such that the castings shall be able to meet or exceed the strength required by the specified application, as determined by the Engineer. Stainless steel castings shall be made in accordance with appropriate ASTM standard(s), depending on the type of stainless steel used. Other standards may be used upon Engineer approval.

B. Design Qualification of Structural Castings

One steel casting, selected by the Engineer from the first lot of production steel castings, shall be subjected to a qualification test of the casting design by the Contractor. A statistical sample of stainless steel castings, as agreed upon by the Engineer and the Contractor, from the first lot of production stainless steel castings, shall be subjected to a qualification test of the casting design by the Contractor. Qualification tests shall include radiographic examination for material soundness using reference radiographs to ASTM E 446 and any mechanical testing.

Acceptance levels for the design qualification radiographic examinations shall be selected by the Contractor as appropriate for the service intended, subject to the approval by the Engineer before any castings are produced. Radiographs shall meet the requirements of ANSI/ASTM E 94 and E 142 for steel castings, and ASTM E 1742 for stainless steel castings, and the quality level in the area of inspection shall be at least 2% (2-2T).

A qualification test report shall be prepared and submitted to the Engineer for approval. The production of any castings before receipt of the Engineer's approval of this report shall be at the Contractor's risk. All radiographs from the qualification test shall be made available to the Engineer for review. If the casting selected for qualification fails to qualify, a plan of action, including details of how failed material will be handled, shall be included in the qualification test report. Once a design is qualified and accepted by the Engineer, no changes shall be made in the casting pattern, technique, heat treatment, or material composition without re-qualification in accordance with the requirements of this Section **[CDRL 20-007]**.

C. Quality of Structural Castings

All structural castings supplied shall be equal to or better than the design qualification castings in all respects. The casting supplier or Contractor shall test, inspect, and accept castings in

accordance with procedures described in AAR Specification M-201 for steel castings. For stainless steel castings, the procedures shall be proposed by the Contractor and approved by the Engineer. In addition, the inspections below, as applicable, shall be performed and a written report of the results of the tests and inspections shall be furnished for each lot of castings produced **[CDRL 20-008]**.

1. Magnetic Particle Inspection

Magnetic particle inspections of all surfaces of each casting shall be conducted, according to ASTM E 709, by personnel certified to MIL-STD-410. With respect to structural castings, including coupler castings, the maximum permissible magnetic particle indications shall be not more than 1/4 inch in the direction transverse to the usual direction of loading, and no more than 3/4 inch in the direction parallel to the usual direction of loading.

For martensitic and ferritic stainless steel castings, acceptance criteria shall be in accordance with ASTM A 903.

2. Radiographic Inspection

Radiographic inspection of steel castings shall be conducted according to the requirements of ASTM Standards E 94 and E 142, using reference radiographs to ASTM E 446. Radiographic inspection of stainless steel shall be conducted according to the requirements of ASTM Standards E 94, E 1742 and E 1030. A sampling frequency shall be proposed by the Contractor and submitted for approval by the Engineer.

Structural castings shall not exceed severity level 3 of ASTM E 446 in all critical areas of such castings and shall not exceed level 5 in all other areas of the castings. During demonstration that the stated severity level requirements of ASTM E 446 have been met, successively produced castings shall be re-inspected by radiography in the defective areas shown in the prior radiographic inspection. After such severity levels have been proved, the sampling frequency for structural castings shall be one casting out of each 10 produced. If no castings are rejected by radiographic inspection, this frequency may be extended to one casting in 25.

3. Liquid Penetrant Inspection

For non-magnetic stainless steel castings, liquid penetrant inspections of all surfaces of each casting shall be conducted according to ASTM E 165, by personnel certified to MIL-STD-410. Acceptance criteria shall be established in accordance with ASTM A903.

D. Repair Welding and Cast-Weld Design

Repair welding of steel castings is permitted, provided the casting supplier performs all repair welds according to the structural welding requirements of TP20.23. Castings requiring repair or modification by welding after completion of heat treatment may be stress relieved locally by using electrically-controlled heating to a temperature not greater than 1,150 degrees Fahrenheit and slow cooling. Manual torch stress relief shall not be permitted except for cosmetic welds and only then after the procedures have been submitted for review and approval. For cast-weld designs, the entire length of all assembly welds on any welded assembly of several separate castings selected for design qualification shall be radiographically inspected to ANSI/ASTM E 94 and E 142, using reference radiographs from the International Institute of Welding's *"Collection of Reference*

Radiographs of Welds," quality level Green. Portions of assembly welds stressed in tension by service loads shall meet quality level Blue.

No repair welding of stainless steel castings is permitted without express written approval of the Engineer.

E. Disposal of Non-Conforming Castings

If castings are found to be non-conforming to requirements determined by the design qualification castings, the material shall be repaired, retested, and re-inspected or destroyed at the Contractor's expense.

TP20.06 ALUMINUM

A. General

Aluminum alloy mill products shall be identified by Unified Numbering System designations and shall conform to The Aluminum Association specifications contained in the Association's publication *"Aluminum Standards and Data."* Aluminum alloy castings used for door thresholds shall conform to ASTM B 26, B 85, or B 108 for, respectively, sand, die, or permanent mold castings, respectively. Aluminum alloy forgings shall conform to ASTM B 247. Copies of all test reports for sheet, extrusion, and forgings shall be submitted to the Engineer for approval **[CDRL 20-009]**.

B. Design Stresses

All aluminum structural members shall be designed so that calculated stresses under the specified AW3 passenger load do not exceed the allowable stresses per APTA SS-C&S-015-99, *"Standard for Aluminum Alloys for Passenger Equipment Car Body Construction"*. Proper allowance shall be made for the effects of fatigue, for column and plate stability effects, and for strength reduction at welded regions. Permissible fatigue stresses under the specified AW3 passenger load shall be established, with approval based on available relevant research data or on prototype testing under the variable load patterns expected to occur in service.

C. Fabrication and Fastening

The forming of aluminum parts, joining of parts by bolting, riveting, and welding, and the protection of contact surfaces shall, as a minimum, conform to the requirements of the Aluminum Company of America's Technical Report Number 524, *"Specification Covering Use of Aluminum in Passenger Carrying Railway Vehicles,"* except as otherwise specified herein.

Fabrication techniques shall be such that the strength and corrosion resistance of the aluminum shall not be impaired nor the surface finish permanently marred or discolored during construction.

D. Protection of Contact Surfaces

The specific measures to be taken by the Contractor to prevent the risk of direct metal-to-metal contact and resultant possible electrolytic corrosion shall be subject to approval and shall depend upon the determination of the most suitable method which can be adapted to the design involved. The following instructions shall be the minimum protection.

Aluminum alloy surfaces shall not be secured to or make direct contact with the surfaces of copper, copper-bearing aluminum alloy, brass, bronze, silver, nickel, nickel alloys, nickel plated parts, lead, tin, or wood.

The contact surfaces of aluminum alloy with aluminum alloy shall be painted with zinc chromate primer or approved equal before securement.

The surfaces of aluminum alloy parts secured to steel parts, where exposed to weathering or harsh environments, shall be protected with a one-part polysulphide sealant, zinc chromate paste, mica insulation joint material, or an approved equivalent material which completely covers the faying surfaces. The insulating material shall be non-hygroscopic, and, if fibrous, shall be impregnated with bitumen or an approved, non-corrosive, water- and moisture-repellant substance. After driving, fasteners shall be primed and painted with red oxide or aluminum paint.

Stainless steel and carbon steel fasteners plated with zinc shall be coated with zinc chromate paste or approved equal before installation. Where possible, only the head and the shank of the bolt shall be in contact with the aluminum part when secured in place. Suitable bushings may be used in place of the zinc-chromate paste.

E. Interior Trim

Where unpainted aluminum is exposed to contact by passengers, it shall have a clear (natural) anodic finish. The finish process shall be the Aluminum Company of America's "Alumilite 204" with a minimum coating thickness of 0.0004 inch and a minimum coating weight of 21 mg/square inch, or approved equal process.

TP20.07 ELASTOMERS

A. General

Elastomers shall be compounded and cured to perform satisfactorily in the environment specified in TP02. The elastomers shall have high resistance to ultraviolet radiation, weather, and all proposed car washing and other cleaning fluids. All elastomeric parts shall be resistant to ozone, oxidation, heat, oil, grease, and acid, and have the longest possible life consistent with the other characteristics specified.

The following elastomeric parts shall be of neoprene and shall have a minimum service life of 10 years, unless otherwise specified or approved:

- Glazing Rubber
- Door Seals
- Door Nosing
- Isolation Tapes/Pads
- Other parts exposed to the outdoor ambient environment, except where otherwise specified

All resilient mounts and elastomeric truck components shall be of natural rubber. Synthetic rubber compounds may be substituted for natural rubber only when approved for a specific application.

Elastomeric parts within pneumatic or hydraulic equipment shall be as necessary to meet the performance requirements of this Specification for the pneumatic or hydraulic device.

B. Tests

All tests shall be conducted according to the latest revisions of the specified ASTM test procedures, unless otherwise specified. All resilient, natural rubber mounts and elastomeric truck suspension components shall be tested in accordance with the procedures outlined for elastomers in TP20.25; the results of the testing shall be submitted to the Engineer for approval **[CDRL 20-010]**.

The test specimens shall be cut out from the proposed material, and at least one tensile strength and elongation test and one accelerated aging test shall be made on the material used for each order. If the compound or cure, or both, are changed during the production of material for one order, at least one test of each type shall be made for each different batch.

When testing the 6 inch by 1/2 inch ASTM "dumb bell" type test specimen (or smaller size if the size of the part necessitates) by the methods specified in ASTM D 3182, D 3183, D 3190, and D 412, the tensile strength shall not be less than 1,500 psi and elongation shall be a minimum 350%. The tensile strength of the elastomer shall not be reduced more than 25% when subjected to accelerated aging by the methods specified in ASTM D 573 for a period of 96 hours in an air oven at 158 degrees Fahrenheit.

The ozone resistance of the elastomer shall be tested in accordance with ASTM D 1149 using an ozone concentration of 100 pphm, an exposure time of 100 hours at 100 degrees Fahrenheit, and a specimen elongation of 20%. The elastomer shall not exhibit any cracks during the test period.

C. Life Expectancy

For all parts made by vulcanizing an elastomer to metal, any premature failure (less than six years) between metal and the elastomer or in the elastomer, occurring when the parts are used in normal service and according to the provisions of the Technical Provisions, shall be considered as having been caused by defect of materials or workmanship.

D. Metal Parts

Metal parts to which elastomeric material is vulcanized shall be made of SAE 1020 or 1045 hot-rolled steel, unless otherwise approved by the Engineer.

E. Bonding

The joining of elastomeric pieces shall be conducted by the hot vulcanization process. Bonding of elastomers by other processes shall not be allowed unless the Contractor submits the application, bonding procedure and bonding agent technical data for approval prior to the purchase of any materials.

F. Truck Parts

Truck bumpers, snubbers, and the exterior surfaces of air springs shall be made of natural rubber or approved equal. They shall be compounded to be resistant to abrasion, oil, grease, and acid.

G. Seals

Glazing strips shall be of neoprene conforming to ASTM C 542, or approved equal material.

All door mating edges, door and window seals, and glazing strips shall be of neoprene material. The durometer hardness measured with a Shore Type "A" durometer at a temperature between 70 degrees Fahrenheit to 90 degrees Fahrenheit shall be 70 ± 5, unless otherwise approved.

TP20.08 GLAZING MATERIALS

A. Safety Glass

Safety glass shall meet the requirements under Item 1, Table 1 of ANSI Z26.1, *"American National Standard for Safety Glazing Materials for Glazing Motor Vehicles and Motor Vehicle Equipment Operating on Land Highways – Safety Code,"* and 49 CFR 223 and 238 FRA Type I or II test as appropriate for the application.

1. Glass Type

All safety glass shall be of the laminated type.

2. Flatness

When an individual light of glass is laid on a truly flat surface such as a surface plate, the glass shall not indicate a bow of more than 0.030 inch per linear foot in any direction.

3. Dimension Tolerance

The overall dimensions of individual lights as supplied shall be held within 0.060 inch of the dimensions ordered. The thickness of individual lights shall be held within 0.020 inch.

4. Overlap Tolerance

The overlap of one laminate of the light with respect to the other at an edge shall not exceed 1/32 inch. Corners and burrs shall be ground smooth and all edges shall be treated in accordance with ANSI Z26.1, Section 6.

5. Color

The color of the glass shall be as required by TP04.07. When new, there shall be no more than +/- 4% variation in the color of individual lights of laminated sheet glass when examined over a white background.

6. Haze

All the laminates of the safety glass shall be so nearly free from haze that the glass shall have approximately the same clarity as a light of the same nominal thickness of plate glass when viewed against a north light.

7. Specks and Scratches

Occasional specks of foreign material and scratches will be permissible, provided such specks do not exceed 0.020 inch in greatest dimension and scratches do not exceed a total of 3 inches in length, and neither are within the central three-quarters area of the light.

8. Bond Separation

The bond between two sheets of glass and the membrane shall be of such quality that when the glass is broken by twisting or by direct impact, there will be no separation between the glass sheets. Lights that contain unbonded areas shall not be used.

9. Light Transmission

Average visible light transmission through clear safety glass shall be a minimum of 85%.

10. Quality Assurance

The Contractor shall be responsible for the performance of all inspection and test requirements. Except as otherwise specified, the Contractor may utilize the facilities of its supplier or any approved commercial laboratory.

11. Shipping

The material shall be carefully prepared for shipping and shall be properly protected to prevent damage. If a pressure sensitive masking is used, it shall be easily stripped from the material and not leave a gummy or sticky residue.

12. Identification

Each light shall be marked for identification in accordance with the requirements of ANSI Z26.1, Section 7 and 49 CFR 223 by the supplier in legible letters 0.1-inch high in the lower right hand corner as viewed from the inside of the vehicle. This identification shall be no closer than 3/4 inch to the edge. The identification shall give the product name, the manufacturer, the serial number, and FRA Type I or II designation. Marking shall be legible and permanent for this application and shall be applied in such a manner so as not to reduce the integrity of the coating. The light shall be installed so that the marking can be read from the inside.

13. Documentation

The Contractor shall certify that the shipped material complies with the requirements in TP20.08.A **[CDRL 20-011]**.

B. Plastic Glazing

1. General

Plastic glazing shall meet the requirements under Item 4, Table 1, of the latest revision of ANSI Z26.1, *"American National Standard for Safety Glazing Materials for Glazing Motor Vehicles and Motor Vehicle Equipment Operating on Land Highways – Safety Code."* The material shall be selected from General Electric Company "Margard," Atohaas "Tuffak CM-2," or approved equal, meeting 49 CFR 223 and 238.

2. Applicable Documents

The latest issue of the following documents, in addition to those specified in this Section, shall form a part of the Technical Provisions:

- ASTM D 673 – *Mar Resistance Test.*
- ANSI Z26.1 – *Item 4, Table 1, "American National Standard for Safety Glazing Materials for Glazing Motor Vehicles and Motor Vehicle Equipment Operating on Land Highways – Safety Code."*
- ASTM G 23 – *Recommended Practice for Operating Light – and Water – Exposure Apparatus (Carbon Arc Type) for Exposure of Nonmetallic Materials.*
- ASTM D 1499 – *Recommended Practice for Operating Light – and Water – Exposure Apparatus (Carbon Arc Type) for Exposure of Plastics.*
- ASTM D 1003 – *Recommended Practice for Determining Haze Factor.*
- ASTM E 162 – *Radiant Panel Test For Determining Flame Propagation Index (Is).*
- 49 CFR Part 223 – *Impact Requirements.*
- ASTM E 662 – *Test to Determine Smoke Emission Values (Ds).*

3. Material Physical Properties

Plastic glazing shall meet the following requirements.

a. Strength

Samples shall be prepared and tested according to 49 CFR 223 and 238 and Test Number 10, Article 5.10, of ANSI Standard Z26.1. Samples shall not shatter or break when subjected to the falling dart impact requirements of Test Number 10. The dart tip shall be no more than 0.50-inch radius. Denting or marring of the surface of the tested piece in this test is permissible.

b. Light Transmission

Visible light transmission through clear plastic glazing shall not be less than 85% in 0.125-inch thickness, 82% in 0.250-inch thickness, 80% in 0.375-inch thickness, and 78% in 0.5-inch thickness.

c. Weathering Test

The materials shall pass the long arc Xenon lamp-weathering test Number 16 of ANSI Standard Z26.1.

d. Color

The color of the plastic glazing shall be as specified. The materials shall have UV stabilizer additives to inhibit fading and loss of properties due to extended exposure to direct sunlight. When new, there shall be no more than 4% variation in the color between lights of plastic material of a specified color and thickness, when examined

over a white background, and measured by the appropriate light transmission or colormeter inspection and test instruments.

e. Abrasion Resistance

Plastic glazing materials shall be silicone-coated on both sides to increase resistance to abrasion. The coated plastic shall meet the following specifications:

Property	Test Method	Requirements
Abrasion Resistance	ANSI Z26.1 Test 17	The change in percent haze after 100 cycles shall be less than 6

The glazing material supplier is required to perform the following test prior to qualification **[CDRL 20-012]**: The plastic glazing material after 300 hours of weatherometer testing shall pass abrasion resistance ANSI Z26.1 Section 5.17. The foregoing test is to be performed for initial product certification.

f. Chemical Resistance

Samples shall be prepared and tested according to Test 19, Article 5.19, of ANSI Z26.1. The exposed fabricated edges of the test samples may be coated with the same material as the face surfaces by the manufacturer. In addition to those chemicals specified in this test, the test shall also include, but not be limited to, such cleaning solutions as dilute oxalic acid solution (3% by weight), 1/2 strength Neleco Products Subway Soil Solvent – Parts 1 and 2, and Electrosol. The exposure time of the test shall be increased to 1-hour intimate contact with the test chemicals on the faces of the test sample. The contaminants shall be either wiped or sprayed onto the coated faces of the test sample. Any tackiness, crazing or apparent loss of transparency shall be cause for rejection. After immersion, a change of greater than 5% in haze, as measured by Test 17 on ANSI Z26.1, shall be cause for rejection.

To produce the specified 1/2 strength solution of Subway Soil Solvent, mix parts 1 and 2 in the following ratio:

- Part 1 – 1 gallon (U.S.) liquid – acid solution as received.
- Part 2 – 2 pounds of the crystalline and catalyst mix dissolved in 10 gallons (U.S.) of water.

g. Adhesion of the Coating Materials

The abrasion resistant coating materials shall retain adhesion to the substrate materials when subjected to Test Number 10, Article 5.10, of ANSI Standard Z26.1. The coating shall also pass a standard cross cut adhesion test. This test consists of scribing a grid of four horizontal and four vertical 1-inch long lines through the coating with a sharp steel blade. Pressure sensitive tape, Scotch brand Number 335-2, is then pressed firmly over the scribed area and is then pulled away at 90 degree angle to the sample sheet. The coating shall pass the test if no coating is removed from the substrate material.

h. Dimensional Tolerance

The overall dimensions of individual units as supplied shall be within 0.030 inch of the nominal dimension specified. The thickness of the plastic materials shall be within a tolerance of + 5% of the nominal thickness.

i. Flatness

When an individual light is placed on a truly flat surface, such as a surface plate, the material shall not indicate a bow of more than 0.031 inch per linear foot in any direction.

j. Edge Work

All edges shall be straight and perpendicular to the surface, and shall be sawed or routed and reasonably free of burrs in order to prevent cutting of the rubber glazing strips. Sharp corners shall be removed around the entire periphery.

k. Optical Quality

Optical quality of the plastic glazing materials shall be in accordance with Test Number 15, Article 5.15.2.2, of ANSI Z26.1. Under the specified procedure, no light and dark patches, existent over the entire area, shall appear in the shadow of the unmasked area of the specimen before the specimen shall have been moved a distance of at least 14 inches from the screen.

Protective covering shall be marked to permit orientation of extrusion grain for most favorable results at installation. This extrusion grain axis shall be horizontal in the installed light.

There shall be no detectable cracking of the coating as indicated by fine radial cracks at the point of impact when struck with 2 1/2 ft.-lb. energy from a dart having a 1/2-inch radius tip. The impacted specimen shall be examined for radial cracks visible by transmitted light.

No detectable cracks in the coating shall develop when the specimen is strained 2%. Stress may be applied by imposing 6,000 psi loading, using a tensile testing machine.

4. Material Quality

a. Foreign Material and Inclusion Defects

Less than 0.009 inch – allowed without population limit to the extent that they do not constitute a severe defect such as clustering.

0.010 inch to 0.019 inch – allowed up to 10 per square foot average over the area of the light.

0.020 inch to 0.050 inch – allowed up to three per square foot average over the area of the light.

0.051 inch to 0.065 inch – allowed one per square foot average over the area of the light.

0.066 inch to 0.150 inch – allowed one per edge, only in the outer 25% of the light.

0.151 inch and above – shall be cause for rejection.

Note: There shall be no black speck clusters of three or more above 0.019 inch in a 1-inch diameter circle.

Defects occurring in those areas of the lights which shall be covered by the glazing strips shall not be cause for rejection.

b. Fibers and Scratches

Fibers and scratches less than 0.060 inch in length are allowable without population limit to the extent that they do not constitute a severe defect such as clustering.

Fibers and scratches from 0.060 inch to 0.125 inch in length are allowed up to a maximum of two per square foot average over the area of the light.

Fibers and scratches from 0.126 inch to 0.250 inch in length are allowed up to a maximum of one per square foot average over the area of the light.

Fibers and scratches greater than 0.250 inch in length are not allowed.

Fine scratches which are detectable only when viewed in bright back lighting are acceptable.

c. Bubbles

There shall be no clusters of bubbles, no chain bubbles and no bubbles larger than 0.030 inches in diameter.

Bubbles 0.020 inch to 0.030 inch in diameter, if present, shall have a minimum separation of 3 inches.

In any 2-inch diameter area of glazing material, there shall be a maximum of 4 bubbles in the 0.011-inch to 0.020-inch diameter range.

d. Apparent Runs

0.125 inch and under – allowed without population limit to the extent that they do not constitute a severe defect such as clustering.

0.126 inch to 0.250 inch – allow four per square foot average over areas of light but not to the extent that they constitute a severe defect.

0.251 inch to 0.500 inch – allow one per square foot, providing it does not constitute a severe defect.

0.501 inch to 1.00 inch – allow one per edge only in the outer 25% of light area.

1.01 inch or above – shall be cause for rejection.

e. Orange Peel

"Orange Peel" in the surface shall be cause for rejection of the material if it exceeds the standards to be established between the Contractor and the Engineer prior to manufacture **[CDRL 20-013]**.

f. Quality Assurance

The Contractor shall be responsible for the performance of all inspection and test requirements. Except as otherwise specified, the Contractor may utilize the facilities of its supplier or any approved commercial laboratory.

g. Shipping

The material shall be carefully prepared for shipping and shall be properly protected to prevent damage. If a pressure sensitive masking is used, it shall be easily stripped from the material and not leave a gummy or sticky residue.

h. Identification

Each light shall be marked for identification in accordance with the requirements of ANSI Z26.1, Section 7 and 49 CFR 223 by the supplier in legible letters 0.1-inch high in the lower right hand corner as viewed from the inside of the vehicle. This identification shall be no closer than 3/4 inch to the edge. The identification shall give the product name, the manufacturer, the serial number, and FRA Type I or II designation. Marking shall be legible and permanent for this application and shall be applied in such a manner so as not to reduce the integrity of the coating. The light shall be installed so that the marking can be read from the inside.

i. Documentation

The Contractor shall certify that the shipped material complies with the requirements in TP20.08.B **[CDRL 20-014]**.

TP20.09 RUBBER FLOOR COVERING

A. General

Rubber floor covering shall contain a minimum of 38% (nominal, by weight of compound) Butadiene Styrene rubber, shall be non-staining, non-discoloring, and 100% non-oil extended. Only high quality, fine, hard clay shall be used as filler. No whitening (limestone) shall be used in the compound. The rubber shall be free from reground rubber, natural rubber, or coarse fillers.

At 68 degrees Fahrenheit, the rubber flooring shall bend 180 degrees around a 3/4-inch diameter mandrel without breaking, cracking, crazing, or showing any change in color. The rubber flooring material shall be fully homogeneous throughout and shall meet the requirements of ASTM F 1344.

B. Inspection Criteria

This Section addresses defects that shall be cause for rejection, their allowable limits, and repair methods where repairs are permitted.

1. Thin-Skinned Blister

A thin-skinned blister is a blister which, when finger pushed, will collapse upon itself. Any thin-skinned blisters which exceed the limits listed below shall be cause for rejection of the floor sheet.

Maximum Size – 0.030-inch high, 0.80-square inch area with longest dimension of 2 inches.

Maximum Population – three blisters in a 12-inch x 12-inch area with only one other blister within 3 feet of this area.

Repair Method – using a hypodermic needle, apply just enough Super Bond 420 or Bostik 1685 or equivalent, compress the blister and bring to a flush surface.

2. Thick-Skinned Blister

A thick-skinned blister is a blister which, when finger-pushed, will collapse and then returns to its original condition.

Maximum Size – 0.030-inch high, 0.80-square inch area with longest dimension of 2 inches.

Maximum Population – three blisters in a 12-inch x 12-inch area, and only one other blister within 3 feet this area.

Repair Method – no repair authorized, not an acceptable condition.

3. Lump

A lump is a blister without a void, consisting of solid material.

Maximum Size – 0.030-inch high, 0.80-square inch area with longest dimension of 2 inches.

Maximum Population – three 12-inch x 12-inch area, and only one other lump within 3 feet of this area.

Repair Method – no repair required.

4. Hole

A hole is a defect which is 50% or more through the material. Any holes found in the floor sheet shall be cause for rejection of the sheet.

5. Thin Area

A thin area is a defect where the sheet is of reduced thickness locally.

Maximum Size – 0.030-inch deep at the lowest point, 3 square inches with the longest dimension of 5 inches.

Maximum Population – one thin area in a 40-inch x 40-inch area, and no other thin area within 3 feet of this area.

Repair Method – rub with #00 steel wool to blend this area into the normal thickness material and then buff to a normal surface finish.

6. Color Distribution

Color distribution is an appearance judgment and shall be subject to the approval of the Engineer. If the base coloring is not within 5% between production runs or the coloring is not consistent over the entire surface, the roll shall be rejected.

TP20.10 WOOD AND PANELS

A. Lumber

Lumber shall be sugar or black maple, sweet or yellow birch only. Lumber shall be thoroughly air seasoned or kiln dried before using, so as not to have a moisture content of greater than 12%. Lumber shall be dressed on all surfaces to full dimensions. Lumber shall be straight-grained, free from dry rot, knots, checks, and other defects which may impair its strength and durability or mar its appearance.

B. Plymetal

The term "plymetal," as used in the Technical Provisions, means metal-faced plywood. All plymetal panels shall conform to the following requirements:

Mechanical Properties	**Minimum Metal to Wood Average Shear Value or 80% Wood Failure**
Dry shear	250 lbf/in²
Boil shear, 3 hr. boil, tested wet at room temperature	150 lbf/in²
Soak shear, 48 hr. soak wet at room temperature	150 lbf/in²
Creep or cold flow, under static load for 48 hrs., at room temperature	250 lbf/in²

The metal face of the plymetal panel that is faced with melamine shall be constructed in accordance with TP20.10 F. prior to the melamine-faced metal panel being laminated to the plywood core.

C. Plywood

All plywood shall be manufactured to conform with the requirements of Grade – Structural I of the National Bureau of Standards Voluntary Product Standard (American Plywood Association) PS 1-95, and then stored under cover. Scarf or finger jointed panels will not be allowed. All plywood shall be sealed with two coats of an epoxy paint on all edges and cutouts as soon as possible after fabrication. All exposed edges of panels, joints between panels, fastener heads, and openings of panels used in areas accessible to moisture shall be water-proofed and sealed with an approved coating prior to installation in the car.

D. Honeycomb Panels

The term "honeycomb panels" as used in the Technical Provisions refers to an assembly of honeycomb material bonded to melamine-faced metal panels or to metal panels.

Aluminum honeycomb material shall be commercial grade meeting the requirements of MIL-C-7438G. Bonding shall be sufficient to develop the full strength of the honeycomb material.

Stainless steel-faced, stainless steel honeycomb panels shall be constructed in accordance with the requirements of MIL-A-9067. The adhesive bond strength of the honeycomb core to the stainless steel face shall not be less than 15-lb/inch climbing drum strength when tested in accordance with MIL-STD-401. The adhesive bond strength of the integral stainless frame to stainless steel face shall not be less than 30-lb/inch climbing drum strength when tested in accordance with MIL-STD-401. Stainless steel honeycomb panels shall be tested in accordance with MIL-STD-401B to demonstrate the following requirements:

Mechanical Property	Minimum Requirements
Core shear yield at 200 degrees Fahrenheit	250 lbf/in^2
Flatwise tension at 200 degrees Fahrenheit	250 lbf/in^2
Beam flexure at 200 degrees Fahrenheit	75,000 lbf/in^2
Core shear fatigue at R.T.	150 lbf/in^2 @ 10^6 cycles
Flatwise tension at R.T.	250 lbf/in^2 @ 10^6 cycles
Beam flexure at R.T.	50,000 lbf/in^2 @ 10^6 cycles

No other honeycomb materials will be permitted.

E. Panel Contour Tolerance

Surfaces exposed to passengers shall not deviate from the specified contour by more than 3/32 inch in any 36-inch distance. The slope of any such deviation shall not exceed 3/32 inch in 12 inches.

F. Melamine-Faced Aluminum

Melamine-faced aluminum panels shall be constructed by laminating melamine to aluminum sheets. The melamine-impregnated papers shall be directly molded to the aluminum sheets at temperatures of no less than 270 degrees Fahrenheit and pressure no less than 1,000 psi. The surface characteristics, after manufacture, shall be no less than that required of type GP (General Purpose) in the NEMA Standards Publication Number LD-3, latest revision. The melamine and the required binder sheets shall be 0.015 +/- 0.005-inch thick. The aluminum sheets shall not be less than 0.025 inch in thickness when used as a facing on plywood. The aluminum sheets shall not be less than 0.081 inch in thickness when not laminated to a substrate such as plywood. Aluminum sheets shall be properly cleaned by etching, sanding, or other approved process to ensure full, permanent adhesion.

The use of contact adhesives to bond the melamine sheets to the aluminum backing is prohibited.

The bond between the melamine and aluminum sheets shall, as a minimum, meet the following requirements:

- Internal bond (ASTM D 952): 2,600 lbf/in^2

- Flexural strength - (S) (ASTM D 790)
  with grain: 26,500 lbf/in^2
  cross grain: 25,300 lbf/in^2

- Modulus of elasticity - (E) (ASTM D 790)
  with grain: 2.8 x 10^6 lbf/in^2
  cross grain: 3.1 x 10^6 lbf/in^2

- Tensile strength (ASTM D 638)
  with grain: 22,300 lbf/in^2
  cross grain: 20,300 lbf/in^2

G. Phenolic Composite Floor Panels

Phenolic composite floor panels shall be designed to withstand the following physical requirements with no visible or audible indications of delamination of the panel skin from the core, and permanent deformation of the top surface shall be less than 0.010 inch unless otherwise specified. There shall be no puncture or damage to fibers of the top surface. There shall be no separation of any internal core from the top or bottom skin. There shall be no fracture of the core. All test results are required to be submitted to the Engineer for approval **[CDRL 20-015]**.

- Indentation Resistance – The floor panel shall withstand a concentrated load of 300 pounds applied to a test dowel that has an overall 0.375-square inch surface area, with a 0.0625-inch radius on bottom edge of test dowel.

- Static Load Test – Average Loading – A representative sample section of the flooring (without rubber floor covering attached) shall be supported on beams spaced at the maximum spacing used on the car using production bonding and fastening techniques. A uniformly distributed load in accordance with the crush loading requirements of TP02 shall be applied to both sides of the joint (butt and/or shiplap). There shall be less than 0.088-inch deflection.

- Static Load Test – Maximum Loading – Using the identical floor panel-mounting configuration as described above, a uniformly distributed load of 200 lb. /ft² shall be applied to both sides of the joint (butt or shiplap).

- Small Area Static Load Test – Using the identical floor panel mounting configuration as described above, a 300 pound load shall be applied to a 1-inch x 3-inch contact area directly over the midspan, 6 inches from the outer carbody sidewall edge. The footprint shall be machined flat within 0.010 inch and the edges shall have a radius of not more than 0.125 inch. There shall be less than 0.200-inch deflection as a result of the load applied.

- Small Object Impact Test – Using the identical floor panel mounting configuration as described above, a 16-pound standard bowling ball shall be raised directly over the mid-

span, 24 inches from the edge of the panel and dropped from height of 60 inches. Permanent deformation of the top surface shall be less than 0.063 inch. Some core damage and some skin separation, radiating outward in a circular pattern with a maximum radius of 6.5 inches from the center of impact shall be allowed.

- Large Object Impact Test – Using the identical floor panel mounting configuration as described above, a 150-pound load shall be dropped from a height of 12 inches upon a 3-inch x 8-inch contact "footprint" pad located directly over the midspan, 24 inches from the edge of the panel. The "footprint" pad shall have a rubber pad on the downside surface with a Shore D 70 minimum. The pad shall have a 1-inch thickness machined flat within 0.060 inch with edges having a radius of not more than 0.030 inch. Permanent deformation of the top surface shall be less than 0.030 inch. Some core damage and some skin separation shall be allowed in a 6.5-inch diameter area.

- Rolling Load Test – Using the identical floor panel mounting configuration as described above, a four-wheeled cart with a load of 200 pounds per wheel shall be rolled on the panels laterally, longitudinally and in a circular path with a 2-foot radius. The wheels shall be 3 inches in diameter, 1-inch wide with a 0.125-inch radius on each edge with a Shore A durometer of 80.

TP20.11 SEAT CUSHION MATERIAL

A. General

Passenger and cab seat cushion fill material shall be low-smoke flexible foam constructed of inherently fire-retardant materials. The thickness shall be subject to approval during design review. The material shall have a polymerized or vulcanized homogeneous (free from foreign material) cellular structure with a porous surface and open cells. The cells shall be interconnecting and uniform in size. Cellular material may be molded in one piece or may be assembled by laminating to achieve the required thickness. Laminated cushions shall be bonded together. Cushion material shall be properly cured to prevent any objectionable odor.

B. Physical Properties

Non-silicone flexible low smoke foam shall meet the following physical property criteria when tested without upholstery material:

- Tensile Strength – 12.0 lbf/in² minimum when tested to ASTM D 3574 Test E

- Elongation – 150% minimum when tested according to ASTM D 3574 Test E

- Compression Set at 50% – A maximum of 10% when tested according to ASTM D 1055

- Flex Fatigue – Thickness loss 5% maximum when tested according to ASTM D 1055

- Tear Strength – 2.0 lbf/in² minimum when tested according to ASTM D 3574

Silicone foam shall meet the following physical property criteria when tested without upholstery material:

- Tensile Strength – 15 lbs. /in² minimum when tested to ASTM D 3574 Test E
- Elongation – 100% minimum when tested according to ASTM D 3574 Test E
- Compression Set at 50% – A maximum of 10% when tested according to ASTM D 1056
- Flex Fatigue – Thickness loss of 5% maximum when tested according to ASTM D 1055
- Tear Strength – 2.0 lbf/in² minimum when tested according to ASTM D 3574

TP20.12 SEAT UPHOLSTERY MATERIAL

A. Cloth Fabrics

1. General

Where specified or permitted, cloth fabrics used for seat upholstery shall be made of woven, transportation grade fabrics of wool, wool/nylon blend (90/10, 85/15), or an approved flame-resistant polyester. The maximum fabric shrinkage shall be 2% in either the warp or fill direction.

2. Physical Properties

Wool/nylon blend seat upholstery material shall be subjected to the physical tests of textile products required by the latest revision of the following ASTM methods, and the results shall not be less than the following values:

Test Number	Description	Criteria
D-3776	Fabric Weight	15.5 oz/sq yd without back coating
D-3775	Fabric Count	Warp – (ends) 88 epi Fill – (picks) 40 to 72 ppi
D-5034	Breaking Strength and Elongation	Warp – 200 lbs Fill – 200 lbs
D-2261	Tear Strength (Tongue)	Warp – 20 lbs Fill – 20 lbs
D-4034	Yarn Slippage	Warp – 30 lbs Fill – 40 lbs
D-3597	Color Fastness	Water – Class 4 minimum Solvent – Class 4 minimum Crocking – Class 4 minimum Light – Class 4 minimum
D-4966	Martindale Abrasion Test	20,000 cycles – no breaks

Flame-resistant polyester seat upholstery material shall be subjected to the physical tests of textile products required by the latest revision of the following ASTM methods, and the results shall not be less than the following values:

Test Number	Description	Criteria
D-3776	Fabric Weight	12 oz/sq yd without back coating
D-3775	Fabric Count	Warp – (ends) 88 epi Fill – (picks) 40 to 72 ppi
D-5034	Breaking Strength and Elongation	Warp – 270 lbs Fill – 200 lbs
D-2261	Tear Strength (Tongue)	Warp – 20 lbs Fill – 20 lbs
D-4034	Yarn Slippage	Warp – 75 lbs Fill – 65 lbs
D-3597	Color Fastness	Water – Class 4 minimum Solvent – Class 4 minimum Crocking – Class 4 minimum Light – Class 4 minimum
D-4966	Martindale Abrasion Test	20,000 cycles – no breaks

B. Fabric-Backed Vinyl

1. General

Where specified or permitted, vinyl used for seat upholstery shall be made of woven transportation grade fabric-backed vinyl with a minimum weight of 22.75 ounces per square yard.

2. Physical Properties

Fabric-backed vinyl used for seat upholstery shall be subjected to the physical tests of textile products required by ASTM, latest revision, and the results shall not be less than the following values:

Test Number	Description	Criteria
D751	Tensile Strength	Warp – 80 lbs Fill – 90 lbs
D751	Seam Strength	85 lbs
D751	Tear Strength (Tongue)	Warp – 6 lbs Fill – 7 lbs
D1117	Tear Strength (Trapezoid)	Warp – 18 lbs Fill – 20 lbs
AATCC-16A	Colorfastness – 200 hrs	No change
D751	Adhesion of Coating	10 lbs
Wyzenbeek 100k rubs #8 duck	Abrasion Resistance	No significant wear

TP20.13 FIBERGLASS-REINFORCED PLASTIC

A. General

Fiberglass-reinforced plastic (FRP) shall be a laminated material, consisting of a gel coated surface and a combination of reinforced fibers in a thermoset polymer resin matrix, where the reinforcement has an aspect ratio that enables the transfer of load between fibers, and the fibers are chemically bonded to the resin. FRP shall withstand, without any physical deformation or structural damage, the environmental conditions in TP02, and be resistant to acids, alkalis, and cleaning solutions recommended by the Contractor or used by RTD.

FRP shall be manufactured by an open molding or matched die molding process. Production techniques shall ensure that the glass fiber reinforcement is uniformly distributed throughout the final product in such a manner as to avoid resin-rich or resin-starved sections. An analysis shall be performed to confirm that the proposed construction method, glass content and laminate structure is adequate for its intended purpose and meets the strength requirements provided in the Technical Provisions. If Tedlar is applied to exposed interior surfaces, it shall be applied using the in-mold process.

FRP parts shall have a minimum thickness of 0.125 inch and shall have a greater thickness at attachment points and edges. If fasteners are used to attach and/or assemble FRP parts, the parts shall be reinforced in a manner approved by the Engineer to preclude the development of cracks. Exposed sharp edges shall not be allowed on any parts.

B. Construction

1. Resin

The resin shall be of good commercial grade, thermosetting, polyester, phenolic, vinyl-ester,, or, or acrylic material selected to meet the physical properties of the Technical Provisions and molding process requirements.

2. Reinforcement

The fiberglass reinforcement shall be mat, fabric, woven roving, continuous roving, spun roving, or swirl mat as required to meet the physical properties of the Technical Provisions and the molding process requirements. The proposed glass content shall be a minimum 20% by weight, and shall be confirmed through testing to ASTM D 2584.

3. Gel Coat

A gel coat shall be provided on all finished surfaces of FRP. The gel coat shall be resistant to scuffing, fire, weather, and cleaning agents. The gel coat shall have a minimum thickness of 0.016 inch and a maximum thickness of 0.030 inch. If the surface of the FRP panel is to be painted, a primer gel coat shall be used and the part shall be painted in accordance with TP20.24. If the FRP panel does not receive paint, then the gel coat shall be pigmented to match the car's color scheme.

Finished gel coated surfaces shall have a minimum gloss value of 85 when measured with a 60 degree glossometer and shall exhibit no print through of the reinforcements or have any appreciable orange peel.

4. Additives

Additives, fillers, monomers, catalysts, activators, pigments, fire retardants, and smoke inhibitors shall be added to the resin mixes to obtain finished products with the required physical, flammability, and smoke emissions characteristics of the Technical Provisions.

Antimony trioxide is prohibited as a component.

5. Strength Requirements

Independent laboratory test reports of production items shall be provided confirming that the production reinforced plastic material complies with the requirements of the following standards. The sample tested must be from a final production part and not simply the base laminate material. Test specimens shall be conditioned in accordance with ASTM D 618.

Mechanical Property	Test	Class I	Class II
Tensile Strength	D 638	10,000 lbf/in^2	18,000 lbf/in^2
Compressive Strength	D 695	18,000 lbf/in^2	24,000 lbf/in^2
Flexural Strength	D 790	15,000 lbf/in^2	30,000 lbf/in^2
Impact Strength	D 256	10 foot pounds per inch of notch	13 foot pounds per inch of notch
Hardness		45 Barcol	45 Barcol

Class I: Items which are non-structural or will not be exposed to any loads such as window masks, destination sign shrouds, ceiling cove panels, ceiling headers, etc.

Class II: Items which are structural or will be exposed to loads from passengers or impacts such as end bonnets, under floor equipment enclosures, door pocket panels, wainscot panels, toilet room modules, toilet shrouds, passenger seat back shrouds, windscreens, stair wells, etc.

TP20.14 THERMOPLASTIC SHEET

A. General

Thermoplastic sheet used in the construction of this vehicle shall withstand, without any physical deformation or structural damage, the operating environmental conditions and shall be resistant to cleaning solutions. Thermoplastic sheet shall be used as extruded or vacuum-formed and shall not contain plasticizers in polymer blend.

Thermoplastic sheet shall be homogeneous and extruded from virgin stock which does not include any regrind of vacuum formed parts. The exposed surface of this material shall conform to the color, texture, and gloss characteristics of the approved color scheme described in TP01.04. The applicable ASTM procedure, as approved by the Engineer, shall be used to measure each of the characteristics. Only UV stabilized pigments shall be used to create the specified color of the thermoplastic sheet. The color and surface finish of parts manufactured from this material shall be submitted and approved prior to the production run of any parts **[CDRL 20-016]**.

B. Quality

The finished parts shall be free of waves and quilting on both sides. Degraded polymer in the sheet shall not be allowed and, if present, shall be cause for rejection of the piece. Voids, lumps, and contamination shall also be cause for rejection of parts if the defects are larger than 0.010 inch and the population of these defects is greater than one defect in 4 square feet.

C. Strength Requirements

Independent laboratory test certificates shall be provided stating that the thermoplastic sheet complies with the requirements of the following standards **[CDRL 20-017]**. Extruded sheet in the surface finish specified shall be used for testing.

Mechanical Properties	ASTM Method	Value
Specific Gravity	D 792	1.20 to 1.45
Tensile Strength	D 638	5,500 lbf/in^2 minimum
Elongation	D 638	50%
Flexural Strength	D 790	8,000 lbf/in^2 minimum
Flexural Modulus	D 790	3.3 x 105 lbf/in^2
Hardness Rockwell "R" Scale	D 785	90 to 110
Heat Shrinkage - 15 minutes at 350 degrees Fahrenheit		10% maximum in machine direction 5% maximum in transverse direction
Heat Deflection (annealed) @ 264 lbf/in^2	D 648	165 degrees Fahrenheit minimum
Impact Strength Fabricated Parts Gardener Dart Drop 0.5-inch diameter ball: at 73 degrees Fahrenheit at -20 degrees Fahrenheit	D 3029	 320 in-lb minimum 80 in-lb minimum

TP20.15 PIPING AND TUBING

A. General

All piping valves, fittings, installation methods, and testing shall be in accordance with the Code for Pressure Piping, ANSI B31.1. All joints shall be easily accessible.

Following installation, all piping systems shall be cleaned to remove dirt, metal chips, oily contamination, and moisture. After cleaning, all piping systems shall be pressure tested in accordance with the latest edition of the Code for Pressure Piping, ANSI B31.1. All leaks shall be repaired and the system re-cleaned and retested until leak-free.

Pipes shall be supported throughout their length and at all connections to prevent vibration or noise and to limit stresses in the pipe to less than 50% of the pipe's fatigue endurance limit. Pipes and their connections shall not interfere with the removal of other components. Pipe routing and support shall be planned and accomplished in an efficient, organized manner to keep the total length and number of fittings and bends to an absolute minimum. All changes in direction shall be accomplished by bending the pipe to a radius of not less than that specified by AAR Specification Number 2518, Standard S-400. Direction-change fittings shall not be permitted in the trainline brake pipe or in the brake cylinder pipe. Support and clearances provided between adjacent pipes and between pipes and surrounding structure, equipment, or other appurtenances shall be sufficient to prevent chafing or contact due to any combination of car loading and deflection, car dynamics, and thermally-induced movement. The minimum clearance shall be ⅛ inch.

At all locations where pipe or tubing passes through holes in the floor, bulkheads, structure, or any fixed member, it shall be rigidly clamped to protect against possible damage or noise due to bearing, abrasion, or car dynamics-induced rattling. Clamps shall not be welded, brazed or otherwise permanently fastened to any pipe or tubing. Pipe and tubing interfaces with clamps shall

be insulated with an elastomeric or woven non-asbestos mineral fabric tape material to protect and sound-insulate the pipe or tubing.

Wherever carbody piping interfaces with vibration-isolated rotating equipment such as the air compressor and air conditioning compressor-condenser unit, approved flexible vibration eliminators shall be used. The pipe connection at either end of the flexible elements shall be rigidly clamped no farther than 2 inches from the flexible elements. All pipe clamps shall be inherently rigid and shall be firmly attached to car structure. Cantilevered clamps or clamp supports that are weaker than service-proven designs will not be accepted. All clamps shall be of a suitable material for the application.

Prior to constructing the cars, the Contractor, in cooperation with the manufacturers and suppliers, shall make a thorough analysis of the design of the pneumatic systems proposed for the cars to assure that every reasonable precaution has been taken in design and installation to prevent the cars from disablement in service as the result of adverse winter conditions. Before being granted permission to commence construction, the Contractor shall submit, to the Engineer, the details of its analysis and of any special preventative measures it proposes to incorporate in the design **[CDRL 20-018]**. The design layout, installation, and hanging methods, etc., shall be jointly inspected and reviewed with and approved by the Engineer.

B. Air Piping, Tubing, and Fittings

The main reservoir pipe and brake pipe shall conform to ASTM A53 or ASTM A106, Schedule 80 seamless pipe. Where ASTM A53 or ASTM A106 piping is provided, its application shall also comply in all respects to AAR Specification No. 2518, Standard 400, latest revision.

Type “K” annealed copper tube per ASTM B88, latest revision, may also be used, provided it is installed no lower than 2 inches below the floor sheet or structural member and is protected by means of equipment or approved steel guards from any potential impact damage from rail debris, especially in the truck and outboard of the bolster areas. Approved copper tube shall also comply with any relevant requirements of AAR Specification No. 2518. Where suitable protection in damage-prone areas is not possible or practical, approved steel or stainless steel piping sections shall be provided. All joints for copper tubing shall utilize fittings of wrought copper or non-porous cast brass in accordance with ANSI B16.22 and B16.18.

All air hoses shall conform to the requirements of AAR Specification M-618 with AAR approved reusable fittings meeting AAR Specification M-927.

All air piping must comply in all respects with the air brake supplier’s design and installation requirements. The diameter of the main reservoir pipe and brake pipes shall meet the brake supplier’s requirements; however, in no case shall these pipes be less than 0.875 inch Outside Diameter. All air pipes shall be sized in accordance with the function intended and may be either ASTM A53 or ASTM A106 schedule 80 pipe or seamless copper tubing as described previously.

Within 180 days of NTP and prior to manufacture of production cars, the Contractor shall provide the Engineer with a report containing written approval from the air brake supplier of the Contractor’s air brake piping fabrication, installation, and design concept **[CDRL 20-019].** The following information shall be contained in the report:

- All critical line sizes and materials including the main reservoir pipe, the brake pipe, and the brake cylinder piping.

- The installation details of all critical lines including routing, total length and volume, elevation and slopes, and major joint and direction change locations. A list of all proposed bend radii shall also be provided.

- Pipe processing details including welding, brazing, cleaning, and fabrication methods, as required by TP20.23.

- Locations of all major air brake valves and controls, relay, and emergency venting devices, and the proposed location and volume of all reservoirs.

- An air consumption analysis justifying the proposed air storage system design.

All air piping shall be installed in a manner to provide drainage away from devices, or branch pipes leading to devices, when the function of those devices could be impaired by the accumulation of water or ice.

All cutout cocks shall be of the vented type, except where function prohibits. All cutout cock handles and their arrangements shall be as described in Section 12.10.

Air piping on the trucks shall be 0.50 inch ASTM A53 or A106, Schedule 80, or approved equal. Low spots (traps) are strictly prohibited on the trucks. Truck piping shall not be run on the bottom of truck side frames, transom, or bolster, unless approved by the Engineer.

Where steel piping is used, all connections and joints where disassembly for service may be required shall utilize swivel type butt-welded flange fittings with an "O" ring type seal. The use of threaded fittings is expressly prohibited. Approval may be granted for the use of threaded fittings in extreme cases where adequate proof is provided that flanged fittings cannot be used.

C. Air Conditioning System Piping, Tubing, and Fittings

Air conditioning refrigerant lines shall be of seamless copper tubing, ASTM B88 type "K" or type "L" or ASTM B280, with wrought copper sweat type fittings. Refrigerant lines below the upper surface of the floor or subject to damage during operation shall be ASTM B88, type "K" only. Condensate drain lines shall be seamless copper tubing, type "K" or seamless stainless steel tubing.

Joints shall be kept to a minimum and all inaccessible runs of tubing shall be without joints. Finned tubing in evaporators and condensers shall be copper. Instead of elbows, tubing may be bent utilizing a bending tool designed specifically for bending of the tubing to be used.

Suction lines shall be designed and installed without traps. The suction line shall be sized for 3 psi (gauge) maximum system pressure drop and the liquid line shall be sized adequately to prevent flashing due to pressure drop.

Lines subject to condensation shall be insulated with an approved insulation, applied with an approved contact cement. The liquid line shall be insulated in all areas where required to provide additional mechanical or thermal protection. Insulation at all joints and fittings shall be mitered and sealed with an approved material. The insulation, adhesive, and sealant shall meet the Specification requirements for thermal, smoke emission, and flammability performance.

All piping and pipe subassemblies shall be deburred, cleaned, dried, and capped with tight fitting plastic caps, or approved equal on all openings after fabrication. Caps shall remain in place until immediately prior to incorporation into the final assembly.

Vibration eliminators shall be used in piping connections to the compressor unless deemed unnecessary by the Engineer. Tubing installations shall be designed so that ease of maintenance can be achieved, as approved by the Engineer.

Tubing installations shall be designed to allow any single length of tubing to be replaced without dismantling or removing surrounding equipment, piping, wiring, or other appurtenances. This requirement does not apply to piping within a self-contained modular HVAC unit.

D. Brazing and Soldering of Piping, Tubing, and Fittings

All brazing and soldering shall comply with the applicable parts of TP20.23 and the following requirements. All refrigerant piping and air system copper tubing shall be joined using silver solder conforming to Federal Specification QQ-B-654A, BAg-5 or BCuP-5. Refrigeration piping and tubing shall be internally swept with a continuous flow of a non-oxidizing gas such as dry nitrogen during brazing. Condensate drain tubing shall be joined using 95-5 solder or silver solder as above. Solder joints shall be wiped and have flux cleaned from tubing and fittings after soldering. After fabrication, the refrigeration and air systems shall each be cleared of all dirt and foreign matter, flushed with a degreasing agent and dried, all according to a procedure prepared for each by the Contractor and submitted to the Engineer for approval **[CDRL 20-020]**.

TP20.16 PRESSURE VESSELS

All pressure vessels shall conform to the latest revision of Section VIII of the ASME Boiler and Pressure Vessel Code for Unfired Pressure Vessels. Test reports shall be furnished for each pressure vessel, and each pressure vessel shall be stamped to record the test.

TP20.17 WIRE AND CABLE

A. General

The Contractor's design and construction shall ensure that the minimum number of wire types and sizes shall be used in the vehicle.

The wire manufacturer shall be submitted to the Engineer for approval **[CDRL 20-021]**.

Selection of wire sizes and insulations shall be based on the current-carrying capacity, voltage drop, mechanical strength, temperature, and flexibility requirements in accordance with applicable AAR, APTA, ICEA, ASTM, NEC, and MIL Specifications. However, in no case shall the properties of the wire and cable be less than those properties delineated in the Technical Provisions. Extra-fine wire stranding shall be utilized on applications subject to repetitive motion.

All applications of shielded cable shall be subject to approval.

The Contractor and each manufacturer of equipment through the Contractor shall submit to the Engineer for approval samples (at least 12 inches long), specifications, and certified qualification

test documentation of each size and type of wire and cable specified before utilizing said wire and cable **[CDRL 20-022]**.

B. Conductors

Conductors for irradiated, cross-linked polyolefin wire shall be soft, annealed tinned copper in accordance with ASTM B33. Minimum stranding shall conform to AAR Standard S501, S502 (Number 589), or ASTM B-172 Class K, or ICEA S-66-524/Nema WC7, Table L-7, Class K for AWG No. 10 or larger, as appropriate for the application.

Stranding and conductor construction for wire sizes AWG No. 12 to AWG No. 16 shall be in accordance with ASTM B-174, Class K, or ICEA S-66-524, Table L-8, Class M, as appropriate for the application.

Stranding and conductor construction for wire sizes AWG No. 18 and smaller shall be in accordance with ASTM B-174 Class L or ICEA S-19-81, Table L-8, Class M, or shall be 19-strand construction as appropriate for the wire size.

The use of solid wire shall not be permitted except for approved wire wrap applications.

Wiring shall be sized for the intended load, voltage drop, installation method, and applicable codes. Calculations of wire sizes shall be in accordance with APTA-RP-E-009-98, *"Recommended Practice for Wire Used on Passenger Equipment."* When the free air rating is used, the Contractor shall furnish data to show that the cables will not exceed their rated temperature at the rated current. Wire ampacities shall be derated to meet the temperature requirements of all devices to which the wire connects. When short-time ratings, short-time overload temperatures, and thermal time constants are used to determine cable size, the parameters used shall be submitted for approval.

In no case shall wire smaller than the following sizes be used:

- Wire which is pulled through conduits or wireways – AWG No. 14.
- Wire on electronic units, cards, and card racks – AWG No. 28.
- Wire within control compartments – AWG No. 18.
- Multi-conductor cables where current is not a factor in wire size selection – AWG No. 18.
- All other wire, including that which is not pulled through wireways and conduits – AWG No. 16.

The Engineer may approve smaller wire sizes for selected applications upon submission of appropriate applicable data for justification.

C. Insulation

1. General Wiring Insulation

For all general car body wiring, the insulation shall be a flame retardant, flexible, irradiated cross-linked polyolefin material having a continuous temperature rating of 230 degrees

Fahrenheit. The insulation shall be rated at 2,000 V, AC and dc, in the case of wires carrying a nominal voltage greater than 150 V AC or dc, and rated at 600 V, AC and dc, in the case of wires carrying a nominal voltage of 150 V or less, AC or dc. For wire sizes AWG No. 6 and larger, the insulation material shall be formulated for extra flexibility.

Insulation shall meet the following flammability requirements:

- Cross-linked polyolefin shall be tested in accordance with test method ICEA S-66-524/NEMA WC7, Paragraph 6.12.5. After five applications of 15 seconds each, with a 3-second rest period between applications, flame shall extinguish in 10 seconds or less.

- Other insulation materials shall be tested in accordance with IEEE-383.

The irradiated cross-linked polyolefin wire insulation shall be constructed and tested in accordance with the following requirements. Certificates of previously performed tests may be submitted provided the proposed and tested wires are identical.

a. Flexibility Tests for Cables

Flexibility tests for cable sizes up to AWG No. 2/0 shall be performed in accordance with AAR RP-585, paragraph 5.9.7.1, for the appropriate wire size.

Flexibility tests for cable sizes AWG No. 2/0 and larger shall be performed in accordance with AAR RP-585, paragraph 5.9.7.

b. Single Conductor Thermal Overload Test

A continuous current of 115 amperes shall be applied to an 18-foot length of AWG No. 10 test wire in 77 degrees Fahrenheit (25 degrees Celsius) still air. A 3000 V DC potential shall be maintained between the test wire and an AWG No. 18 bare copper wire wrapped snugly around the outer insulation surface of the test wire. Failure shall occur when a short circuit is established between the bare copper wire and the test wire. Minimum time to failure shall be 3 minutes.

c. Bundle Overload

A bundle overload test using a seven-wire bundle shall be performed in accordance with AAR Standard RP-585, paragraph 5.9.1.

d. Temperature Cycling Tests

This test shall be performed on an 8-foot length of AWG No. 10 wire with 2 kV insulation.

Thermocouples shall be attached to the outer jacket surface, and on the conductor, under a small incision in the insulation approximately 12 inches from one end of the sample. Both ends of the sample shall be securely clamped using hose clamps.

Prior to temperature cycling, the sample shall be conditioned for 2 hours at a temperature of 302 degrees Fahrenheit (150 degrees Celsius).

The sample shall then be temperature cycled between ambients of 257 degrees Fahrenheit (125 degrees Celsius) and -22 degrees Fahrenheit (-30 degrees Celsius) by

transferring the sample between an air-circulating oven, set at 257 degrees Fahrenheit (125 degrees Celsius) and an air-circulating cold box set at -22 degrees Fahrenheit (-30 degrees Celsius). The time during which the sample stays in each chamber shall be sufficient to allow both thermocouples on the sample to read the same temperature as the environment.

One cycle shall be defined as an approved dwell time at both 257 degrees Fahrenheit (125 degrees Celsius) and -22 degrees Fahrenheit (-30 degrees Celsius). The sample shall be subjected to a total of 250 cycles, with a visual observation at the end of each cycle for cracks and other damage. After 250 cycles, the sample shall be immersed in water for 6 hours with both ends out of the water, and then subjected to a dielectric test of 5 kV AC for 5 minutes, and also examined by microscope to verify that no cracks exist.

Flame-retardant, flexible, irradiated, cross-linked polyolefin insulation rated at 257 degrees Fahrenheit (125 degrees Celsius) may be used, provided that it meets the requirements of all the above tests and standards, modified to reflect the temperature rating-related characteristics. The revised values, and the use of such wire, shall be subject to approval by the Engineer. Cross-linked polyolefin insulation shall not be permitted for use on wires connected to heater elements or any other high-temperature device.

All insulation other than irradiated, cross-linked polyolefin shall meet the following test requirements, based on MIL-W-22759, and using the following parameters.

Dielectric – Test per MIL-W-22759/10B (for 1,000 V wire with tests at 9.5 kV impulse) or MIL-W-22759/6B (for 600 V wires with tests at 8 kV impulse).

Insulation Resistance – Test per ASTM D-470. Minimum accepted value shall be 1,000 mega-ohms per 1,000 feet, using a 1,000 V DC megohmmeter.

Spark test – 100% of all single conductor cables and all single conductor cables being used in a multiconductor cable shall be inspected by Impulse Dielectric Test or by chain electrode Spark Test. Spark Test Apparatus and Procedure shall be in accordance with MIL-W-22759. Spark Test voltages shall be equivalent to impulse test voltages by corresponding rms value at 3 kHz.

Impulse Test Voltage kV Peak	3 kHz Test Voltage kV rms
8.0	5.7
9.5	6.7
10.0	7.1

Air aging – Test per ASTM D-638. Age sample for seven days at 302 degrees Fahrenheit in air oven. Minimum tensile strength and elongation shall not be less than 85% of the unaged values. Also test per IEEE STD 383-1974 and ASTM D-573 for extended life.

Cold Bend – Test per NEMA WC3, except test temperature shall be -58 degrees Fahrenheit.

Weight Loss – Weight loss of the insulation material shall not exceed 1% when subjected to an oven temperature of 266 degrees Fahrenheit for 500 hours.

Chemical Resistance – An appropriate length sample shall be measured for insulation diameter and total weight to record initial values. The wire shall be immersed to within 3 inches of each end in the test fluid for 24 hours at 149 degrees Fahrenheit. During the immersion stage, the minimum bend radius of the wire shall be 10 times the diameter of the wire being tested. Upon removal from the test fluid, the specimen shall be cooled to room temperature for 1 hour and the diameter gauged and reweighed for comparison with the original values. The maximum diameter and weight increase shall not exceed 30%. Typical fluids for this test include:

- Humble Number 2214 Railroad Diesel Lubricating Oil and lubricants (100% solution);
- Humble Diesel 260 or Railroad T fuel oil (100% solution);
- Mineral oil (100% solution);
- Hydrochloric acid, nitric acid, sodium hydroxide, sulfuric acid (0.1% solution);
- Potassium hydroxide (0.1% solution);
- Petroleum distillates and other graffiti removers and cleaning compounds;
- Kerosene solvents (100% solution);
- Trisodium phosphate solution (50% solution);
- Skydrol 500 B hydraulic fluid (100% solution); and
- Water.

Temperature Cycling Testing – The test shall be performed on an 8-foot length sample of AWG No. 12 wire.

Thermocouples shall be attached to the outer jacket surface and on the conductor under a small incision in the insulation about 12 inches from one end of the sample. Both ends of the sample shall be securely clamped using hose clamps.

The sample shall be conditioned for 2 hours at a temperature of 302 degrees Fahrenheit (150 degrees Celsius). The sample shall then be temperature cycled between ambients of 257 degrees Fahrenheit (125 degrees Celsius) and -22 degrees Fahrenheit (-30 degrees Celsius) by transferring the sample between an air-circulating oven, set at 257 degrees Fahrenheit (125 degrees Celsius) and an air-circulating cold box set at -22 degrees Fahrenheit (-30 degrees Celsius). The time during which the sample stays in each chamber shall be sufficient to allow both thermocouples on the sample to read the same temperature as the environment.

One cycle shall be defined as an approved dwell time at both 257 degrees Fahrenheit (125 degrees Celsius) and -22 degrees Fahrenheit (-30 degrees Celsius). The sample shall be subjected to a total of 250 cycles, with visual observation at the end of each cycle for cracks and other damage. After 250 cycles, the sample shall be immersed in water for 6 hours with

both ends out of the water, and then subjected to a dielectric test of 5 kV AC for 5 minutes, and also examined by microscope to verify that no cracks exist.

Single Conductor Thermal Overload Test – A continuous current of 115 amperes shall be applied to an 18-inch length of AWG No. 12 test wire in 25 degrees Celsius still air. A 1,000-V DC potential shall be maintained between the test wire and an AWG No. 18 bare copper wire wrapped snugly around the outer insulating surface of the test wire. Failure shall to occur when a short circuit is established between the copper wire and the test wire. Minimum time to failure shall be 3 minutes.

Seven-Wire Bundle Thermal Overload Test – A seven-wire cable bundle shall be formed by twisting six insulated AWG No. 12 conductors around a center-insulated AWG No. 12 conductor.

A 120 ampere current shall be passed through the center conductor for seven minutes. After a test period, the cable bundle shall be examined for visible damage to the outer six conductors. Failure shall occur if any of the outer conductors split, rupture, or melt and adhere to the center conductor insulation.

Qualification and Production Tests – The tests required for the Technical Provisions concerning Qualification and Production shall be in accordance with tests in MIL-W-22759 for all lots produced.

All test reports covering Production and Qualification tests shall be submitted to the Engineer for approval with samples prior to any shipment of materials **[CDRL 20-023]**.

2. Wire Insulation for High Temperature Applications

Teflon, mineral-filled, abrasion-resistant insulation may be used on wire sizes AWG No. 12 to AWG No. 28. Otherwise, high temperature insulation shall be used where wiring is connected to heat-generating apparatus, where the ambient temperature can exceed 257 degrees Fahrenheit (125 degrees Celsius), or where Teflon is specified as a requirement. The insulation shall be rated at 1,000 V, AC and dc, in the case of wires carrying a nominal voltage greater than 150 V, AC or dc, and rated at 600 V, AC and dc, in the case of wires carrying a nominal voltage equal to or less than 150 V, AC or dc. The insulation shall have a continuous temperature rating of 302 degrees Fahrenheit (150 degrees Celsius) or greater and shall be in accordance with the following requirements:

For wire sizes AWG No. 16 and larger: abrasion resistant Teflon (Polytetrafluorethylene – PTFE) meeting MIL-W-22759/6B or 10B, as appropriate for the voltage level used, or silicone rubber meeting AAR RP-587C. Conductors for high temperature wire AWG No. 12 and smaller shall be soft, annealed nickel-plated copper constructed in accordance with MIL-W-22759/6B.

For wire sizes AWG No. 18 and smaller: abrasion resistant Teflon (PTFE) meeting MIL-W-22759/6B or 10B, as appropriate. When used for interconnecting of apparatus, this type wire shall be in bundles with a protective covering of high temperature-rated, low smoke-generating insulation.

The Contractor may propose other insulated wire, submitting specifications for approval in a specific high temperature application, including the design ambient temperature, routing,

rms ampere value, worst-case ampere value, worst-case temperature rise, stranding, and insulation material.

No high temperature insulated wire shall be used in conduit or raceways without specific approval. The Contractor shall submit all applications of high temperature wire insulation for review and approval.

3. Wire Insulation within Equipment

Insulation on wiring within replaceable modular units, electronic apparatus such as cards and card racks, and other equipment, as approved, shall be Tefzel (Ethylenetetrafluoroethylene – ETFE) per ASTM D 3159, and insulation construction per Military Specification MIL-W-22759/16 (AS), irradiated cross-linked polyolefin per TP20.17.C.1, or Teflon (Polytetra-fluorethylene – PTFE) type EE, per Military Specification MIL-W-16878/5.

4. Wire Insulation in Crowded Locations

Wire for connections to the control console, or in other locations where there are crowded concentrations of low voltage control wiring, may be insulated with Tefzel (ETFE) per ASTM D 3159 and insulation construction per Military Specification MIL-W-22759/16 (AS), except that the wall thickness shall be 0.025 inch. When used for this application, wires shall be bundled with a protective covering of irradiated, cross linked modified polyolefin or similar, approved, high temperature-rated, low smoke-generating insulation.

5. Thinwall Cable Exane 15 for Protected Areas

Dual layer thinwall wire, Rockbestos Surprenant EXANE 15, Spec. DAA1168A, or equal may be used in low voltage application in protected areas within the car. This wire may be used where space and weight are factors and the location is in protected areas. The use of this wire shall be limited to sizes AWG No.12 to AWG No. 22. This wire may also be used in multi-conductor cables where it is protected by an outside jacket. The thinwall wire shall be tested to RP-585 with the currents, weights, and mandrel sizes adjusted to reflect its construction.

D. Multi-Conductor Cables

1. General

Multi-conductor cables, where approved, shall be constructed using wiring as described in TP20.17.B and TP20.17.C. For high-temperature applications, the cable shall conform to MIL-C-27072, with Type V connectors, Style 4 sheaths, Class D jackets, if needed, and shields, if needed. All conductors in multi-conductor cables shall be color coded or otherwise permanently identified as approved. Materials used in the construction of multi-conductor cables shall meet the requirements below. In applications where current is not a factor in wire size selection, such as LED indicator lights or status displays, AWG No. 16 may be used between repeater devices and displays. For multi-conductor cables carrying low-voltage, high-speed, serial data, exceptions to the wiring requirements may be submitted for approval, based upon availability of wire to meet the application requirements.

2. Fillers

Where required to obtain a circular cross-section, fillers shall be made of non-hygroscopic materials compatible with the wire insulation and jacket, and shall be of the same or of a higher temperature rating than the wire insulation.

3. Tape

A binder tape shall be employed over the assembly of conductors in multi-conductor cables if needed to assist in cable manufacture, or as required to permit the cable to function as intended in its application. The binder tape material shall be non-hygroscopic and shall be of the same (or better) temperature class as the wire insulation, and shall be of a compatible material.

4. Shield

The shield, if required, shall consist of either tin-plated copper braid, concentrically-served copper, or aluminum/polyester tape with a drain wire, as appropriate for the application. Tape shields shall be permitted for fixed installations only. The shields shall have the following minimum properties:

- Copper shield shall be made of either tinned, coated copper strands which conform to ASTM B 33, or silver-coated copper strands which conform to ASTM B 298, as is appropriate for the wire insulation. Shield coverage shall not be less than 85%. Shield strand size and application shall be as recommended by the cable manufacturer for the particular application, but shall not be smaller than AWG No. 38.

- Aluminum/polyester tape shields shall consist of a helical wrap of aluminum/polyester tape with a nominal thickness of 0.0004-inch aluminum on a backing of 0.001-inch polyester. The tape shall have a minimum overlap of 10% of the tape width to ensure complete coverage. In contact with the aluminum side of the shielding tape shall be AWG No. 22 7/30 tinned copper drain wire conforming to ASTM B 33 and B 174.

5. Jackets

The overall jacket of multi-conductor cables shall be of flame-retardant, irradiated, cross-linked, modified polyolefin, Tefzel (ETFE), or Teflon (PTFE) to be fully compatible with the wire insulation and application as approved. The coupler cable shall have a jacket of low temperature arctic grade neoprene per MIL-C-13777, with a wall thickness suitable for 600 V. The jacket shall be extruded and vulcanized over the cabled conductors, and shall be centered, with a smooth appearance without objectionable roughness or irregularities, consistent with good industry practice. The nominal jacket thickness for polyolefin, Teflon, Tefzel and Neoprene shall be that shown below, with the minimum wall thickness not less than 80% of the nominal value.

Nominal Jacket Wall Thickness in Inches			
Cable Diameter Under Jacket	**Modified Polyolefin**	**Teflon or Tefzel**	**Neoprene**
0.000-0.250	0.045	0.010	0.072
0.251-0.500	0.045	0.015	0.087
0.501-0.750	0.060	0.021	0.100
0.751-1.000	0.080	0.021	0.100
1.000-1.500	0.080	0.025	0.115
1.501-2.000	0.11		0.135
2.001-2.500	0.13		0.152
2.501-3.000	0.14		0.195

E. Wire Wrap

Wire wrap connections may be used in selected electronic applications, where approved by the Engineer. Where used, the following standards shall be followed as a minimum:

- Only soft or annealed oxygen free solid copper conductor shall be used.
- Wire size shall be AWG No. 28.
- A silver conductor coating, with a minimum coating thickness of 40 micro-inches, shall be applied to the wire.
- Wire shall have "MIL-ENE" insulation, or approved equal, manufactured to MIL-W-81822/1A. The insulation shall have a minimum 300 V AC/V DC voltage rating and shall allow a 275 degrees Fahrenheit (135 degrees Celsius) maximum conductor temperature.
- Wrapping shall be "modified" wrap, nominal 7 1/2 turns, including 1 1/2 turns for strain relief.

F. Insulation Smoke Test

1. Scope

This test method describes the equipment and the procedure for preparing insulated wire samples from which the specific optical density (D_S) of smoke generated can be determined in the Aminco-NBS Smoke Chamber. This method shall be used for wire sizes up to and including AWG No. 12. For wire sizes above AWG No. 12, the standard procedure outlined in ASTM E 662 shall be used. Equipment calibration, standardization, and operation shall be in accordance with ASTM E 662, *Standard Test Method for Specific Optical Density of Smoke Generated by Solid Materials*. The performance criteria shall be $D_s(4.0) <= 200$ (flaming) and $D_s <= 75$ (non-flaming).

2. Apparatus

Aminco-NBS Smoke Chamber and Recorder.

Aminco 6 tube, 90 degree burner assembly for flaming mode testing. All burners shall be directed at the sample in one plane.

Notchless wire frame (Aminco AWG No. 20 wire frame with notches machined off).

Aminco troughless wire specimen holder assembly.

Air oven.

Humidification chamber.

Heavy duty aluminum foil 0.001 +/- 0.0005 inch.

Razor blade.

Tape measure.

3. Procedure

Determine the length of insulated wire required for testing. The individual sample length shall be calculated to produce a sample area of 35 square inches

Calculate the sample length as follows:

$$L = \frac{35}{3.1416 \, x \, d}$$

Where:
L = sample length
d = diameter of insulated wire (inches)

Cut and identify a minimum of three samples of the required length.

Condition samples prior to testing by pre-drying in an air oven for 24 hours at 140 degrees Fahrenheit +/- 5 degrees Fahrenheit, followed by humidification at 73 degrees Fahrenheit +/- 5 degrees Fahrenheit and a relative humidity of 50% +/- 5% for a minimum of 24 hours.

After conditioning, wind a sample uniformly around the wire frame so that the frame opening is uniformly covered.

Cover the wire-wrapped frame with aluminum foil across the back, along the edges, and over the front surface periphery with a single sheet of aluminum foil, with the dull side in contact with the wire.

Place the foil-wrapped wire in a trough-less sample holder such that the wire is vertically oriented. Insert millboard backing, spring, and retaining clip.

Carefully trim the aluminum foil from the front opening of the sample holder.

Adjust wire turns, if necessary, to assure that the sample holder opening is uniformly covered.

Perform smoke testing in accordance with ASTM E 662, noting any unusual behavior that occurs during the test; for example, self-ignition of the sample in the non-flaming test mode or any extinguishment of a burner triplet during the test.

Report the sample orientation, test conditions, results, and observations made during the test.

TP20.18 WIRING

A. General

All car wiring shall be in conformance with APTA RP-E-002 *"Recommended Practice for Wiring of Passenger Equipment"* and the AAR Manual of Standards, Section F S-538, *"Wiring Practice and Rolling Stock Standard,"* except where otherwise specified and except that all wire shall be as required in the Technical Provisions. Circuit protection shall be in conformance with Chapter 2 of NFPA 70, Article 240.

B. Wire Handling

All wiring shall be performed by qualified, experienced wiring personnel using appropriate tools for stripping insulation, cutting, tinning, soldering, harness making, attaching terminals, and other wire fabrication tasks. All wiring tools and equipment shall be used as recommended by the tool and equipment manufacturer.

Wire shall be protected from damage during all phases of equipment manufacture. Wire shall not be walked on, dragged across sharp or abrasive objects, kinked or twisted, or otherwise mishandled. The ends of wire shall not be permitted to lay on wet floors or other damp areas where moisture may be absorbed into the conductors.

When removing insulation, wire strands shall not be nicked or broken in excess of the limits of FAA Specification Number AC 43.13-1A, Section 449, "Stripping Insulation." Additionally, the following criteria shall apply:

Wire Size	Maximum Number of Nicked Strands*
Wires smaller than AWG No.10	None
AWG No.10 through AGW 1/0	7.4%
Above AWG 1/0 through 1600/24	4.4%
Above AWG 1600/24	Graduated Scale

*Definitions:

- A cutoff strand shall count as two nicked strands
- A nick is defined as 25% or more of the strand area damaged or cut more than 33% of its diameter.
- Longitudinal scratches in a copper strand are not considered cause for rejection.

C. Wiring Layout and Installation

1. Wire Harness

The layout of wiring, for both car and equipment, shall be designed in advance of its installation and in cooperation with the suppliers of the related equipment. Wiring shall be pre-fabricated into standard harnesses, wrapped or tied with spiral wrap or tie wraps. Harnesses shall be installed with identical arrangement and location in each car having similar equipment. Separate harnesses shall be provided for major circuit groups or types, or as required for specified circuit separation. All circuits and branches shall be separated by means of terminal boards to isolate portions from others for troubleshooting and searching for undesired grounds. All circuits subject to periodic high potential tests shall be so arranged that they can be conveniently set up for the tests. No exposed wiring within the interior of the car shall be permitted.

Alternative methods for fabricating and installing wiring which are standard Contractor practices will be considered for approval by the Engineer.

Harnessed wires shall not be installed in conduit. Wires from different conduits or other openings shall not be harnessed together with wires running within the box or entering the box through another entrance point. Each harness or group of wires between equipment enclosures shall contain a minimum of 10% spares, but no fewer than two spares for each wire size. All spares for all wiring shall be submitted for approval by the Engineer. Unless specifically approved by the Engineer, no spare wiring shall be used by the Contractor. All spares shall be installed in connectors on terminal boards or other means as approved by the Engineer.

2. Circuit Separation

Circuits shall be physically separated to reduce the possibility of unsafe conditions, interference, or equipment damage.

The following major circuit groups shall not be harnessed or bundled together, shall not run in the same conduit, and shall be physically separated and secured in enclosures, wire ducts, junction boxes, or other wire routing devices:

- High voltage circuits
- AC circuits
- Communication circuits
- Battery voltage level circuits
- Semiconductor gating voltage level circuits
- Conductors carrying in excess of 100 amps

Wires in circuits with potentials differing by 50 V or more shall be separated by a physical barrier. Whenever possible, the wires shall not be cabled together and shall not be placed in the same conduit, junction box, or enclosure. Where a raceway, duct, junction box, or

enclosure is divided into two or more distinct areas by metallic partitions, each area may be considered separately in this application.

Where it is impossible to avoid wires at different voltages in the same equipment enclosure, the wires shall be physically separated, bundled, and secured separately such that physical contact between wiring is not possible. All wiring within an enclosure shall be insulated for the highest voltage in the enclosure, unless approved otherwise. All wiring connected to apparatus shall be insulated for the highest voltage connected.

Wiring connected to transient-generating apparatus, such as unsuppressed contactor coils, shall not be run adjacent to wiring carrying signals to, from, or between semiconductor circuits, logic circuits, vital no-motion circuits, or communication circuits. In cases in which adequate physical separation is impossible, shielded wire shall be used for all conductors involved.

3. Wire and Cable Runs

Wire runs shall be continuous and unbroken between connection points, shall be supported at no greater than 24-inches spacing, and shall be protected at each support point against mechanical crushing and abrasion. A watertight bushing and drip loop shall be provided on all exposed cable entries. All cable bundles and wires shall be routed a minimum of 1 inch above the bottom of equipment enclosures.

All undercar wiring smaller than AWG No. 6 shall be run in closed wire ducts, conduits, or open wire mesh wireways in an approved manner. Wire and cable shall be secured within ducts or open wireways, including each entrance and exit point, to prevent chafing and movement. Wire ducts and conduits shall be of waterproof construction. Permanently retained watertight strain relief bushings, with insulated throat liners of an approved design, shall be used at locations where wires, cables, or harnesses enter or exit conduit, ducts, apparatus, and equipment enclosures. In addition, strain relief bushings on equipment enclosures shall include a permanently retained O-ring type seal.

Lead wires to resiliently-mounted electrical apparatus shall be carried in conduit to a point as close to the apparatus as possible. The length of the leads between the end of the conduit and each piece of apparatus shall be as approved. Short runs of cables or harnesses entering or leaving conduit and apparatus shall have an approved guard mounted to the carbody to protect the wires from mechanical damage. Lead wires to solidly-mounted electrical apparatus and equipment enclosures shall run in conduit connected to the apparatus or enclosure.

Any wiring run through the floor shall be run in ducts or conduit. Wiring, even if enclosed in loom, must not be run through partitions without suitable bushings at such points of passage.

Cables shall be laid in place with sufficient slack at the bends so that cables will clear the inside bend surface of the wireway/wire duct.

All wire and cable shall be free of kinks, insulation damage, insulation abrasions, and nicked strands. Wire installation shall not be subject to accumulations of water, oil, or other foreign matter.

Wires or cables shall not pass through or over the battery compartment and shall not pass over heat-generating equipment such as acceleration and braking resistors, even if the wires or cables are in conduit.

Exposed harnesses, short cable runs, and harnesses entering or leaving exposed raceways shall have approved, fire-resistant, flexible dielectric sleeving installed over the raceway edges and grommet-type insulation at penetration holes. Wiring shall be retained to the sleeving with tie-wraps.

a. Cable Cleating and Support

All cable and wiring, exiting wireways/wire ducts, and that which is not installed in conduit, shall be cleated using split-block cleats of molded neoprene rubber. Cables shall be cleated and bushed when passing through bulkheads and structural members. The cushioning neoprene shall be non-conductive, fire-retardant insulating material with a durometer of 50 to 60 meeting the requirements of TP20.07. Each cleat shall have a stiffener of at least 10-gauge material on the side away from the mounting bracket which shall act to spread the bolt clamping force over the entire length of the cleat. Bolts shall have lock nuts. The Contractor shall minimize the number of different configurations of cable cleats.

AWG No. 6 or larger insulated wire may be cleated in place without conduit, duct, or open wireway. However, in the areas over the truck in the wheel wash and not protected by underfloor-mounted equipment, the wire shall be mechanically protected by an open mesh, expanded metal, or other type of approved guard. The guard may be attached to the bottom of each cleat with the cleat clamping bolts or by other approved arrangement.

Cleats shall be designed to grip each cable individually and firmly, but without causing damage to cable insulation, including cold flow of the insulation. Each cable in the cleat shall have its own cutout sized to the correct wire diameter. Cleated cables shall be routed and supported such that they cannot, under any combination of forces and car movement, touch each other or any other part of the car, except the cleat cushioning material.

Wire and cable runs shall be continuous and unbroken between terminations and shall be supported at not greater than 24-inch intervals in ducts, open wireways, or cleats. Wires that run under the car shall be supported at not greater than 20-inch intervals. The wire shall be protected at each support point from mechanical crushing and abrasion.

Wire splices shall not be permitted, except with express written approval of the Engineer and in accordance with the wire splicing requirements of TP20.

Concealed wires, such as within conduits and wire ducts, shall be such that wires may be replaced or added to without the removal of other than access panels. It shall not be necessary to disconnect or disassemble conduit to accomplish this task.

Wiring run in loom shall not be carried over potential chafing hazards.

Wires entering any removable box shall be harnessed and secured to facilitate removal of the box.

All wires and cables shall be fully protected against contact with any surface other than that designed specifically to support or protect them. This applies to all current-carrying wires, cables, or buses on the vehicle.

b. Wire Securement and Termination

Wiring and cabling shall be readily accessible for inspection and maintenance. Extensive wiring and cabling in the vehicle interior is contrary to accessibility, even though access panels, false floors, and other portals may be provided. To control this requirement, the Contractor shall submit a complete wiring plan for approval **[CDRL 20-024]**.

All wiring shall be secured and protected against movement, chafing, and contact with conductive, sharp, or abrasive objects, including the inside surfaces of wire runs.

All wiring shall be located and secured such that normal equipment motion, maintenance access, heat sources, and the environment do not damage or reduce the life of the wiring.

Junction boxes with terminal boards shall be used, as required, for wire terminations. Harness connections to the boxes, as well as internal wiring to terminal boards, shall be as specified in TP20.19.B. Exterior junction boxes shall be weathertight.

In cases where it is necessary to anchor wires or cables to metallic parts of the car, cleats or approved stainless steel bottle clamps shall be used. Wires and cables shall not be allowed to chafe or rub against any part of the car or each other.

Wire and cable dress shall allow for sufficient slack at equipment terminals to provide for movement induced by shock and vibration, equipment shifting, alignment, cover removal, and component replacement. Sufficient lengths shall be provided at points of termination for additional reterminations without applying tension to the wire and without splicing as follows:

- AWG No. 10 and smaller – Three reterminations
- AWG No. 8 and larger – Two reterminations

A drip loop shall be provided on all exposed wires and cables to prevent fluid runoff into connected equipment.

Wire-tying devices shall be of such material and construction that they will adequately retain the wires for the life of the wiring and shall be resistant to ozone and ultraviolet light. Wire and cable ties shall be trimmed and located to eliminate any hazard to personnel from sharp edges. Wire-tying devices shall be snug, but shall not be so tight as to cause indentation and cold flow damage to insulation. Wire-tying devices shall be mechanically fastened to a permanent structure. Adhesive-installed mounting bases shall not be used for ties or for cable support.

All wire bundles and cables within an enclosure shall be supported by the use of tape rails, shall be spaced away from the equipment box structure, metal edges, bolt heads, and other interference points, and shall have electrical clearance from the covers, regardless of the insulation properties of covers. Wire bundles shall be located above or alongside the apparatus rather than at the bottom of the box wherever possible. In all cases, wire shall be a minimum of 1 inch above the bottom of the box. Wire entry into control or junction boxes shall not be permitted through the bottom of the box.

Truck wiring shall be designed to ensure sufficient slack and shall be provided with clamp supports and abrasion protection. T-splices shall not be permitted.

All jumpers, jumper heads, and jumper receptacles shall be sealed in an approved manner to prevent the entry of water at any operational speed of the car.

Any wiring needed to calibrate and test car functions shall be a part of the permanent car wiring to enable the maintenance staff to conveniently maintain the equipment. This wiring shall terminate in approved connectors in the respective control groups and cabinets.

Wire and cables that are subject to high currents in fault conditions or normal operation shall be secured against secondary damage from high magnetic forces. Propulsion inverter circuits are an example. This requirement shall include damage to bus bars or devices at which the cables terminate.

4. Circuit Shielding

Wire shields used in trainline circuits shall be continuous to the car's electrical jumper receptacle and coupler contacts. The wire shields shall be connected through all applicable connectors and junction boxes. Circuits shall be categorized. Shields in one circuit category shall not be interconnected with shields of another category. Shields used to protect against interference shall not carry signal currents.

Shields on low-level signal wires shall not be interconnected with shields on high-level signal wires in the same category. Each group of shields (other than at the electric jumper receptacles and couplers) shall be carried through on a connector pin or pins, or on terminal strips which shall be in the immediate proximity of the categorized group of circuits. Loops at the interconnections of shields shall not be permitted.

Coaxial cables used as constant impedance transmission lines shall be terminated as required by the circuit termination design and shall not be considered to be shielded conductors. Triaxial cables may be used as coaxial impedance transmission lines with the outer conductor employed as an RF shield.

The following three guidelines shall be applicable in so far as possible, but are not requirements:

- Shields used to suppress electromagnetic interference (EMI) at all frequencies shall be terminated only at the low potential side of the interference circuit, at the termination which exhibits maximum susceptibility.

- Shields used to protect against the effect of, or to exclude, EMI at frequencies below 150 kHz, shall be terminated either to the low potential side or at the balance point of the protected circuit at the termination which exhibits maximum susceptibility.

- Cables requiring both audio frequency (AF) and radio frequency (RF) shields shall be electrically isolated from each other. The resistance between these circuits shall be at least 500 mega-ohms when 500 V DC is applied. Double shielding shall be required on circuits that are both AF-susceptible and RF-susceptible.

D. Insulation Resistance

Refer to Section TP20.17.C.1 for insulation resistance requirements.

E. Marking and Designation

As part of the wiring plan described in TP20.18.C, the Contractor shall devise and submit for approval a wire and terminal marking and designation system that shall coordinate all electrical circuits in the car into a unified system **[CDRL 20-025]**. The system shall identify all wiring, including circuit return wiring, and terminals according to their respective circuit function(s) and shall accurately correlate these designations with the car schematic diagrams. Each circuit shall be individually designated from point to point. Common designations for return circuits will not be permitted.

Each wire and cable shall have printed on its outer surface the manufacturer's identification, conductor size, temperature rating, and voltage rating. For wire size 1/0 and larger, stranding shall be given in addition to the other parameters.

Except for spares and wires entirely within an equipment enclosure, each wire AWG No. 8 and smaller shall be permanently and legibly marked along its entire length. Wires larger than AWG No. 8 may have wire markers applied at each end of the wire. Blank spaces between markings shall measure approximately 1.5 inches. Spare wires and wiring entirely within an equipment enclosure may have a single wire marker at each end, subject to approval by the Engineer in lieu of continuous marking. No disassembly of the wire harness or the connection shall be required to read the wire marking. Wires shall be marked with an alpha-numeric circuit designation. A circuit designation shall change only when it passes through an active or passive component such as a relay coil or relay contact, fuse, circuit breaker, lamp, motor, or resistor. A circuit designation shall remain unchanged when it passes through a terminal strip or junction box stud regardless of the number of wires of that circuit are common to that point. There shall be no duplication of wire codes in unrelated circuits throughout the car. Where there is more than one identical assembly per car, each assembly shall be wired identically to the other(s) and wire marking shall be identical at each assembly.

For pre-manufactured multi-conductor cables and for cases where individual circuit identification markers may be approved, the following requirements shall apply:

- All wires and terminals shall be clearly identified. Wires attached to terminal studs shall have a marker indicating the terminal stud to which it is attached.

- Identification of the multi-conductor cable shall be by continuous wire marking printed on the cable jacket.

- The jacket shall be marked 6 inches from of the end of the cable, every 12 inches within 10 feet of the end of the cable, and every 24 inches over the remainder of the cable. The marking shall be stamped in two places, approximately 180 degrees apart, to facilitate reading the marking. Wires in multiple-conductor cables shall be color-coded.

- Identification of terminals shall be by white or yellow permanent markers with black printing.

- Wire markers shall meet the adherence and solvent resistance requirements specified by MIL-M-81531, latest revision, and shall withstand all combinations of ambient and equipment temperatures. Hand printing is prohibited.

- For cable identification, the Contractor shall use a basic identification system in conformance with ANSI/IEEE 200 and shall submit the system selected for review and approval by the Engineer.

F. Pulling Compound

Pulling compound shall be non-conductive, non-hygroscopic, non-odorous, and shall not support bacterial activity nor attract vermin.

G. Solder

Solder shall be in accordance with ASTM B 32, Grade Sn60. A non-corrosive flux shall be applied immediately before soldering.

H. Tape

Electrical tape shall be in accordance with AAR Standard S-540, or equivalent approved for railway practice. Electrical tape shall meet or exceed the voltage rating of wire where the tape is applied.

TP20.19 WIRE AND CABLE CONNECTIONS

A. General

All equipment enclosures and junction boxes, except primary power circuits, shall be fitted with terminal boards or connectors. Primary power circuits shall be fitted with compression terminals and knuckle joint connectors as described herein. Unused connector pin positions shall be sealed with either connector contacts or plastic sealing plugs designed for that purpose.

As part of the wiring plan described in TP20 and the connector requirements in TP05 and TP18, the Contractor shall submit the proposed design and product line samples of all connectors with sample wiring (pigtails) for approval **[CDRL 20-026]**. IPC/WHMA-A-620 Requirements and Acceptance for Cable and Wire Harness Assemblies shall be integrated into the wiring plan.

Terminal boards with M4 or No. 6 or smaller screws and quick-disconnect terminals, other than those stated herein, shall be permitted only with prior approval by the Engineer.

B. Terminal Boards and Terminal Points

Molded case, modular terminal blocks which utilize a spring clamp to hold the wire may be used for low voltage circuits. Each terminal block shall be properly identified with a permanent marking and each assembly shall be secured to the mounting (DIN) rail by end clamps which incorporate metallic hardware. All wires AWG No. 12 and smaller shall receive a ferrule. Plug-in style (split) terminal blocks will only be permitted if, as part of the design, these plugs will be used as a connector when performing maintenance, testing or replacement of a Line Replaceable Unit. All molded case, modular terminal blocks are subject to review and approval by the Engineer.

All other electrical terminal points and terminal boards shall be one of the following types:

- Stud type in accordance with MIL-T-55164A
- Binding head screw type in accordance with MIL-T-55164A, only where approved.

Stud-type terminal points shall have brass studs and connections, each of which shall be locked using a single brass nut with brass flat washer and a plated spring-type lock washer. Studs, nuts, and washers may also be made of corrosion-resistant, plated steel, where approved. Each board or connector shall have the necessary number of terminations plus a minimum of 10% spares, but no fewer than one spare shall be provided, unless otherwise approved. Binding-head, screw-type terminal boards will be permitted only where approved. All terminal boards shall be in accordance with Military Specification MIL-T-55164A.

Threaded studs shall have a minimum of 2 1/2 threads exposed beyond the final nuts. Adequate space shall be provided to permit connecting wire terminals with standard tools. All terminals shall be properly torqued to assure sound connections. Spacers shall not be used.

Jumpers between terminal board points shall be brass or plated steel. Wire jumpers between adjacent terminals of terminal boards will not be permitted.

An approved permanent marking strip on each terminal board shall be provided and attached adjacent to the wire junction point to identify the wires attached thereto, and the wires connected to terminal boards shall have the terminal point location printed on the wire.

A maximum of two terminals shall be connected to any one binding screw. A maximum of four terminals shall be connected to any one threaded stud, provided that there is no interference among terminal barrels. On terminal boards, the wiring shall be arranged so that no more than two terminals are connected to a stud from each side.

C. Wire Terminations

Terminals and connections used throughout the car shall be mechanical, solderless, crimp type as manufactured by AMP Incorporated or other approved manufacturer with a comprehensive line of terminals, connector pins, and application tools available. All terminals for the same wire size shall be crimped with the same model tool. The Contractor shall minimize the total number of crimping tool types needed for crimp connections. Terminals for AWG No. 10-22 wire sizes shall require no more than three tool models to provide certified crimp connections. As part of the wiring plan described in TP20.18.C, the Contractor shall submit the proposed product line for approval along with the proposed “pull-test” method for approval by the Engineer **[CDRL 20-027]**. For approval, terminals shall have been tested to Military Specification MIL-T-16366F for

temperature rise, voltage drop, vibration, current overload, and corrosion. Test results shall be submitted for approval on a part number basis.

Terminals and connections shall be attached to the wiring with proper crimping tools and dies as recommended by the manufacturer. The terminals used on AWG N0. 10 and smaller wires shall securely grip and hold the insulation of the conductor, unless otherwise approved. Terminals shall be ring lugs in accordance with Military Standard MS-25036; spade and hook-type terminals shall not be used. Corrosive protection shall be provided for all base materials.

Conductors subject to motion relative to the terminal shall be protected to prevent breakage of the conductor at or near the terminal. Sufficient slack shall be provided in all wires and cables to prevent breaking or pull out of bushings and terminals. Only one wire shall be crimped in any one terminal.

Wherever several wires are connected to terminals of a terminal strip on a device which is removable from the car for maintenance, the wires shall be terminated, with double-ring terminations which shall be screwed to an insulating fanning strip which shall serve to keep the terminations in the correct relative locations while disconnected from the device, unless otherwise approved by the Engineer.

D. Power Cable Terminations

Power cables shall be terminated with an approved compression terminal. Sufficient cable slack shall be provided to preclude breaking or pull-out from bushings or terminals and to allow two reterminations. Cable conductors shall be clean prior to installation of terminals. Compression terminals shall be applied using tools and procedures recommended by the terminal manufacturer for that purpose. Swaging tools shall be of a type that ensures complete swaging in every case.

E. Cable Connectors

All cable connector applications shall be subject to approval.

All cable connectors shall conform to MIL-C-5015 or an equivalent standard as approved. They shall employ removable crimp contacts of the correct size for the wire being terminated. Except as noted below, the connector contact area shall be plated with a minimum of 0.000030 inch of gold over a minimum of 0.000050 inch of low-stress nickel. For high current applications, the connector contact area shall be plated with a minimum of 0.00010 inch of silver. Adjacent connectors shall use either different inserts or different insert orientations to prevent erroneous connections. The receptacle portion of all cable connectors shall be rigidly mounted.

All cable connectors used in exterior locations shall be of the environmental watertight variety. Cable connectors shall be equipped with sealing gaskets on the front mating surface and on the back where the cable enters. The cable jacket shall be held by a clamp within the connector body. Unused connector pin positions shall be sealed with either connector contacts or plastic sealing plugs designed for that purpose.

Plastic-bodied connectors shall not be used in exterior locations. Quarter-turn, bayonet-lock, quick-disconnect type connectors shall not be used on trainline jumper cables.

Except as specified above, all cable connectors in exterior locations shall be quarter-turn, bayonet-lock, quick-disconnect type CIR connectors as manufactured by Litton-Veam SPA or approved

equal. Quarter-turn, bayonet-lock connectors shall conform to all provisions in MIL-C-5015, or other approved standard, except for the screw coupling requirement.

Connectors in high vibration or high motion areas, such as speed sensors and trainline jumpers, shall have the wire connections soldered and potted and shall have a watertight jacket molded over the cable and connector to form a unitized assembly. Trainline jumper connectors shall be as manufactured by Pyle-National or approved equal. The Contractor shall conduct an approved vibration test on the unitized assemblies.

Non-metallic body, non-environmentally sealed connectors are limited for use on non-vital, interior applications, such as lighting and speaker connections. These connectors must be threaded or include a positive locking mechanism. All such connectors must include a suitable means of strain relief for the wires. If a spring-clamp is used to terminate the wire within the connector body, a properly sized ferrule must be applied to the wire. The receptacle half of each connector assembly shall be rigidly mounted. Proposed connectors shall have a minimum of 2 years of successful service in similar applications and are subject to approval by the Engineer.

F. Quick-Disconnect Terminals

Approved, quick-disconnect terminals shall be utilized to facilitate maintenance and inspection. They shall provide positive terminal engagement and be shock- and vibration-proof. All terminals shall be provided with insulation equal to that of the wire. No “push-to-fit” (FASTON) type terminals shall be permitted unless specifically approved by the Engineer.

G. Grounding

1. Connections

Grounding connections to the carbody and equipment shall be made through silver-soldered or brazed copper pads of an area adequate for the anticipated maximum current that may be carried under any circumstances. Alternative ground pad material may be permitted in certain cases as approved by the Engineer. Transition (base) plates, if used, shall be of the same alloy group as the respective carbody part or apparatus. The base plate shall be welded to the carbody or apparatus. Grounding connections shall not be made to aluminum alloy members. All ground pads shall be visible and accessible for inspection and troubleshooting. The ground connections shall be attached by a bolt, washer, and nut designed for the purpose. Anti-corrosive grease shall be applied over the connection.

All equipment enclosures and shock-mounted equipment shall be grounded with flexible, grounding leads bolted between a car body grounding pad and the equipment’s grounding pad. Braided, strap-type leads shall be used where there is relative motion between the two items being connected. The ground strap termination method shall apply uniform pressure to the conductive surface and the current density shall not exceed the bonding requirements of TP20.

The Contractor shall submit, for approval, a complete grounding plan, which shall indicate the means by which it is proposed to prevent currents from passing through journal bearings and truck-center bearings **[CDRL 20-028]**. Refer to other sections of the Technical Provisions for ground brush and related requirements. Low voltage and high voltage circuits shall not be grounded at the same connection.

2. Bonding

All grounding and bonding jumpers and straps shall be sized to carry fault current and lightning discharge current for which the voltage drop shall not exceed 25 V. The bonding method employed shall not produce a DC resistance in excess of 0.0025 ohms, or more than 0.025 ohms at 150 kHz for any applied AC voltage. Grounding and bonding jumpers and brazed shunt straps shall be "extra-flexible."

H. Wire Splicing

Splicing of conductors shall be avoided and shall be permitted only with approval on a case-by-case basis. Splicing of conductors in conduit shall not be permitted. In the event a splice is approved, it shall be in a junction box, and the spliced joint shall be mechanically as strong and have the same conductivity as any other part of the conductor. The splice shall be an insulated permanent crimp splice in accordance with Military Specification MIL-T-7928G, Type II, Class I, and shall be installed with the crimping tool and die of the splice manufacturer. All splices shall be insulated with self-sealing, weathertight, seamless shrink tubing. The outside diameter of the spliced portion of the cable after the insulation is applied shall not exceed the outside diameter of the unspliced portion by more than 40%. Splices shall be identified in the integrated schematic.

TP20.20 CONDUIT

A. Types

All conduit and conduit couplings shall be of an ANSI-approved type. All exterior rigid conduit shall be standard weight, galvanized steel with threaded fittings. All conduit ends shall be deburred inside and out to remove sharp edges and all pieces shall be blown out with compressed air and cleaned before installation to remove filings and other foreign material.

Steel conduit shall be mild steel in standard lengths with threaded ends and hot-dipped zinc-coated exterior and interior surfaces. Conduit shall be free of burrs and projections, circular in cross-section, of uniform wall thickness, and shall conform to the requirements of ANSI Standard C-80.1. The threads per inch and length of threading shall conform to ANSI Standard B-2.1 for pipe threads.

Steel fittings shall be used to assemble steel conduit. Elbows, nipples, and couplings shall be made of the same grade of steel as that of the conduit. All fittings shall be treated, coated, and threaded according to the requirements for zinc-coated, rigid steel conduit and shall conform to UL 6.

Thin-walled Electrical Metal Tubing (EMT) or aluminum conduit, when permitted, shall be for interior use only. All applications of such conduits shall utilize compression-style connectors and couplings.

Flexible conduit, where permitted, shall be watertight and interlocking aluminum, or steel strip protected, with an approved rust-resistive coating. Flexible covering on conduit shall not contain polyurethane, nylon, or PVC vinyl. Cross-linked polyolefin may be used.

Liquid-tight flexible nonmetallic conduit, if required for special applications, may be used with the Engineer's approval. Liquid-tight flexible nonmetallic conduit shall not be used where subject to physical damage or in lengths longer than 6 feet.

Conduit shall be color coded – red for those carrying circuits greater than 100 V and yellow for those carrying circuits less than 100 V.

B. Size and Fill

Conduit shall be sized such that the sum of the cross-sectional areas of the conductors and their insulation does not exceed 40% of the cross-sectional area of the conduit for three or more conductors. For two conductors, a limit of 31% shall be used, while for a single conductor, a limit of 53% will be permitted. Where conduit having a length not exceeding 24 inches without bends of more than 15 degrees is used between enclosures, a maximum fill of 60% will be permitted.

C. Installation

A run of conduit between junction boxes and/or pulling outlets shall not contain more than the equivalent of four quarter-bends, 360 degrees total, including the outlet fittings. Bend radii at the inner surface of the bend shall be no less than eight times the nominal inside diameter of the conduit.

All conduit bends and offsets used shall be made by the use of special forms or tools and shall have the largest radius possible so that wires can be pulled without the use of tackle or power.

Conduit shall be securely clamped with all runs electrically grounded to make a continuous ground. Conduit installation shall not create problems associated with physical contact between dissimilar metals.

All conduits shall be arranged to prevent moisture traps and shall drain toward control boxes, except that all open-ended conduits shall be installed in such a manner as to ensure gravity drainage out the end. The conduit arrangement and installation shall be subject to approval by the Engineer **[CDRL 20-029]**.

TP20.21 CONDUIT FITTINGS AND JUNCTION BOXES

A. General

The conduit fittings and junction boxes for car wiring shall be as manufactured by the Contractor or by a supplier of a comprehensive line of parts. The Contractor shall submit the proposed product line for approval **[CDRL 20-030]**. All conduits and their connections to electrical equipment shall be installed to make a continuous ground. All conduit fittings and junction boxes shall be provided with gasketed covers as described in TP20.21.D. Gaskets for conduit fittings and covers shall prevent the ingress of dust, debris, and water encountered in the operating environment.

B. Boxes

All exterior junction boxes shall be fabricated of minimum 14-gauge steel or aluminum (where permitted). All exterior junction boxes shall be weatherproof and shall be connected in such a way that drainage from equipment groups will not pass through conduit into the junction boxes.

Interiors of all junction boxes shall be primed and then protected with a white, insulating coating as specified in TP20.24.

C. Conduit Interface

The open ends of conduit shall be provided with strain relief-type fittings with extended rubber bushings, bell-mouth fittings, or insulated throat box connections as approved by the Engineer. All conduit entries into removable equipment boxes shall be secured by means of a bolt-on watertight access panel.

D. Covers

All junction box covers shall be dust-proof and retained by compressive spring-type latches or captive screws as approved on a location-by-location basis. All fasteners used in junction boxes shall be stainless steel. All covers shall be designed to accept or mate with a bulb-type clamp-on seal.

TP20.22 WIREWAYS

All wireways shall be of rigid, stainless steel construction. The trays shall be adequately supported throughout their entire length in an approved manner. The trays shall be completely de-burred, leaving absolutely no sharp edges, before installation on the vehicles. Grommet clamps shall be provided at all locations where cables or wires enter or leave the wireways. Under no circumstances shall leads be draped over the edge of the wireways, with or without wireway edge protection. Heads of screws or bolts inside the raceways shall be flush with the metal surface. Metal wireways, elbows, couplings, and similar fittings shall be flush with the metal surface. Points of screws or fasteners shall not be directed toward the interior of wireways.

The wireways shall be routed such that they avoid:

- Sources of heat such as propulsion and braking resistors;
- Wheel splash; and
- Areas subject to damage by debris or foreign objects.

Wireways shall be located to provide access to the harnesses contained within for maintenance. They shall be provided with approved covers which may be interrupted for entry and exit of wires and cables. Edges of such interruptions shall be completely covered with protective bushings.

Wireways shall be designed to prohibit the collection of dirt and debris, and shall be perforated, without compromising their requisite strength, to permit ventilation and drainage. They shall preclude water entrapment.

Metal raceways and the elbows, couplings, and fittings shall be electrically and mechanically coupled while protecting wires from abrasion, and shall make a continuous ground with the car structure.

Bends in wireways shall be avoided; however, if required, approved protection shall be provided to avoid insulation chafing at the bends.

Wireways shall be permitted in approved exterior and ceiling locations only. They shall not be permitted in the carbody sidewall. Conduit only shall be permitted in the carbody.

Wireways shall not contain more than 30 current-carrying (i.e., power source as opposed to signaling) conductors at any cross-section. The sum of the cross-sectional areas of all conductors contained at any cross-section of a wireway shall not exceed 40% of the interior cross-sectional area of the wireway.

All wire and cable shall be securely fastened within wireways to eliminate movement and chafing.

TP20.23 WELDING AND BRAZING

A. General

The Contractor shall be responsible for the quality of its welding and brazing as well as that of its suppliers and subcontractors. Cleaning prior to welding shall be in accordance with applicable parts of Section 2, MIL-HDBK-132, *"Protective Finishes."*

B. Structural

All structural welding practices shall be according to requirements of the AWS D1.1, *"Structural Welding Code – Steel;"* AWS D1.2, *"Structural Welding Code – Aluminum;"* AWS D1.3, *"Structural Welding Code – Sheet Steel;"* AWS D1.6, *"Structural Welding Code – Stainless Steel;"* AWS D15.1, *"Railroad Welding Code;"* and the AWS Handbook. Requirements for dynamically loaded structures shall be applied. Cast steel welding shall be according to ASTM A 488/488M, *"Steel Castings, Welding, Qualification of Procedures and Personnel."* Resistance welding shall be in accordance with AMS-W-6858B. AWS D1.1 shall apply to steel of 1/8-inch and greater thickness. AWS D1.3 shall apply to steel less than 1/8-inch thickness.

Canadian Welding Bureau (CWB) Standard W59, *"Welded Steel Construction,"* may be used in lieu of AWS D1.1 and CWB Standard W59.2, *"Welded Aluminum Construction,"* may be used in lieu of AWS D1.2. In the case of the use of CWB standards, AWS standards shall be used for all materials and thicknesses where CWB standards do not apply.

Structural welding of ferritic and austenitic stainless steel shall be governed by AWS D1.6. ASME Section IX and ASME Section VIII, Part UHA shall apply when appropriate. AISI 201L (UNS 20103) and 301LN (UNS 30153) stainless steels shall be treated as P-Number 8, Group-Number 3, category for reference to ASME requirements. Ferrite number for welds shall be between WRC4 and WRC10, or as proposed by the Contractor and approved by the Engineer. Weld heat-affected zones (HAZ) and weld metal shall be limited to maximum allowable stress values in ASME Section VIII, Table UHA-23, for UNS S20100 stainless steel and Table UW-12 rating of welds. Fatigue allowable stresses shall not exceed the lesser of fatigue limits in AWS D1.1, Section 2.20.6, or 50% of the joint strength level calculated from ASME maximum allowable stress values. Higher values shall only be used if qualified by Contractor tests.

Regardless of the governing codes, all Welding Procedure Specifications (WPS) shall be fully qualified by test by the Contractor. Qualification shall be documented by Procedure Qualification Records (PQR). WPS and PQR shall be prepared by the Contractor and reviewed and accepted by the Engineer and a Certified Welding Inspector **[CDRL 20-031].** The use of AWS-B2.1 shall not be permitted and shall not be included or referenced in Welding Procedure Specifications (WPS) and Procedure Qualification Records (PQR). The use of any WPS purchased from AWS shall not be permitted. All WPS shall be fully qualified by the Contractor PQR-welding tests and subject to approval by the Engineer and a Certified Welding Inspector.

C. Welder Qualification

Welders shall make only those welds for which they have been qualified according to the requirements of the AWS, ASME Section IX, ASTM A 488/488M, or other approved qualifying procedures. (AWS B2.1 shall not be used, as noted above.) Records of welder qualification tests shall be made available for review.

D. Inspection

The Contractor shall visually inspect all structural welds in accordance with AWS D1.1 requirements.

In addition to the visual inspection specified for all welds, nondestructive surface inspection (dye penetrant or magnetic particle methods, as appropriate) shall also be used to inspect all first-production welds. The Contractor shall specify a sample nondestructive inspection rate for all subsequent welds. A record of all NDT inspections shall be included in the Car History Book.

On the first structure, all full penetration welds shall be nondestructively, volumetrically inspected (ultrasonic or radiographic methods) according to AWS D1.1 requirements. The Contractor shall specify a random sampling plan for volumetric inspection of subsequent, full-penetration welds for approval **[CDRL 20-032]**. The minimum acceptable inspection plan shall require inspection of one portion of a full penetration weld for every 200 production welds. The proposed test welds shall be selected from among welds that are most critically loaded as determined by calculation or load test. With approval, destructive sectioning and metallurgical examination may be substituted for some or all of the required volumetric inspection requirements for production welds.

If ring welds are used on the first structure, all ring welds shall be nondestructively inspected by magnetic particle or dye penetrant methods. Sample ring welds shall be sectioned and examined metallographically to determine HAZ hardness, which shall not exceed 400 VH (Vickers Hardness). The Contractor shall submit a random sampling plan for additional metallographic examinations of ring welds for approval **[CDRL 20-033]**. The minimum acceptable sampling plan shall require inspection of one ring weld sample for every 300 production ring welds.

E. Post-Weld Cleaning Requirements

All welds exposed to passengers or on sliding contact surfaces of truck frames and bolsters shall be completely cleaned of spatter.

F. Contractor Documentation

All welding procedures and documents, including Welding Procedure Specifications, Procedure Qualification Records, and Resistance Spot Welding Schedules, shall be submitted for approval before application, as required in TP20.23.B.

Specifications for purchase of welding electrodes, welding wires, and cover gases shall be submitted for approval before their application **[CDRL 20-034]**.

G. Special Welding

Procedures and qualification records for structural welding of stainless steel to LAHT, or other combinations of metals or conditions not covered by AWS specifications or codes, shall be submitted for approval. As part of the qualification of all dissimilar metal welds, sample welds

shall be sectioned and examined metallographically to determine HAZ hardness. The HAZ hardness shall not exceed 400 HV (Vickers Hardness).

Austenitic stainless steel electrodes or wire shall be used to join carbon or LAHT steels to stainless steels. Galvanized steel shall not be welded to stainless steel.

H. Resistance Welding

Resistance welding of stainless or carbon steels shall be according to AMS-W-6858, Class B for structural applications and Class C for non-structural applications. All resistance welding procedures shall be qualified per AMS-W-6858B; procedures and qualification records shall be submitted for review and approval **[CDRL 20-035].** Contractor-proposed deviations from AMS-W-6858, including, but not limited to, weld nugget diameter, tension shear strength, and minimum spacing, shall be submitted, reviewed and approved before application in production **[CDRL 20-036]**.

Design strengths higher than standard certification and production strength requirements shall be qualified according to AMS-W-6858, Figure 11b, for one thickness. This shall require a test lot size of 180 spot welds. Additional thickness combinations with the same increased strength ratio may be qualified by 25 spot weld shear tests plus three macro-sections. Twenty of the 25 shear test specimens may be recorded from production witness tests taken from 20 consecutive production days (not calendar days). The Contractor shall submit records of the settings, ultimate shear strength, weld diameter, and weld penetration for approval **[CDRL 20-037]**.

Surface indentation shall not exceed 20% of material thickness (t) or 0.01 inch, whichever is greater. However, for exterior resistance-welded areas exposed to passenger view, indentation shall not exceed 10% of t or 0.005 inch, whichever is greater. For exposed welds, the Contractor shall vary welding parameters and conditions within their acceptable ranges to minimize indentations. Surface burn and discoloration shall be removed by chemical cleaning, or an approved equal method, and sanding or polishing to match the surrounding surface.

Production witness welds shall be made and tested once each day and, whenever otherwise necessary, such as by change in any of the following:

- Operator;
- Material, material thickness, or combination of thicknesses;
- Electrodes; and
- Settings.

I. Resistance Spot Weld and Intermittent Weld Spacing

Spacing of structural resistance and spot welds shall be according to approved structural drawings. Spacing shall not exceed 2 inches plus twice the weld nugget diameter for any structural application, including carbody side sheets, roof sheets, and corrugation. Intermittent fusion-weld spacing pitch shall not exceed 5 inches for 2-inch (minimum) weld lengths, such that a minimum weld length of 40% of the overall joint length is achieved.

For any corrugation application, if the pitch of the corrugation nodes does not allow the above weld spacing, there shall be two spot welds between each node.

For non-structural applications, weld spacing shall be designed and qualified in accordance with the appropriate welding code requirements.

J. Toughness of Welded Assemblies

The Contractor shall prove all welded steel structures are above the ductile-brittle transition temperature for the specified environmental exposure. Specifically, the weld heat-affected zone (HAZ) and base metal shall resist service impact loads at the lowest specified operating temperature without brittle failure. In the absence of prior operating history, and if the Contractor's approved design does not require greater toughness, the minimum impact value for Charpy V-notch specimens shall be 15 ft-lbf of absorbed energy at the lowest specified operating temperature, or -20 degrees Fahrenheit, whichever is lower.

The Engineer shall have the right to require impact tests to verify the specified toughness. If tests are required, verification of HAZ toughness shall be done on a test sample welded according to PQR parameters. Base metal toughness shall be certified on a heat basis by the steel manufacturer or steel supplier; if these data are not available, the Contractor shall perform tests on each heat of as-received base metal.

K. Torch Brazing

All brazing, defined as heating above 840 degrees Fahrenheit, shall follow the recommendations of the latest AWS Welding Handbook, Volume 2 issue. Procedures and personnel who perform brazing work shall be qualified in accordance with AWS B2.2, *Standard for Brazing Procedure and Performance Qualification*.

L. Torch Soldering

All structural (not electrical) soldering, defined as heating below 840 degrees Fahrenheit, shall follow the recommendations of the latest AWS Welding Handbook, Volume 2 issue. Procedures and personnel who perform torch soldering shall be qualified through the preparation and testing of samples of production torch soldering. Test samples shall be prepared and submitted for approval before production torch soldering.

TP20.24 PAINTS AND COATINGS

A. General

The painted portion of the carbody, or any of its components, shall be painted as required by the Technical Provisions and in accordance with the specified color scheme. Austenitic stainless steel portions of the carbody shall not be painted, unless otherwise approved by the Engineer for cosmetic reasons. Where stainless steel is painted, procedures shall be as recommended by the paint manufacturer for the application, and surfaces shall be properly prepared to ensure adhesion.

All painting on the carbody or any component shall be performed in accordance with the paint manufacturer's recommendations. To the greatest extent possible, all exterior paint application for a given vehicle shall use the same paint batch to assure uniformity of appearance. The Contractor and its paint supplier shall submit a touch-up procedure and assure that a continuing supply of

touch-up paints in colors used on the car, suitable for spot application by spray, roller or brush, will continue to be available in the United States **[CDRL 20-038]**.

B. Materials and Preparation

Preparation of the substrate surface and application of painting materials by roller, brush, or spray shall be in accordance with the paint manufacturer's recommendations. Only primers recommended and approved by the paint manufacturer shall be used. All paint materials shall be used at the consistency recommended by the paint supplier. If thinners are necessary, they shall be approved by the paint manufacturer and shall be used only to the extent recommended. Painting shall be performed by experienced labor, using proper equipment under competent supervision following documented and approved procedures.

Painting materials for all surfaces shall provide a high quality finish resistant to corrosion, chipping, fading, and shall retain the gloss level. The coating shall be a two-part, high solids, low VOC, polyurethane paint system. All components of the paint system shall be provided by the same manufacturer. All paint and filler materials which are to be superimposed to form a finish system shall be mutually compatible and shall be warranted for use as a system by the manufacturer of the components.

Metal portions of the carbody not constructed of austenitic stainless steel shall, after fabricating, be prepared for painting by grit blasting and immediately painted with an approved epoxy primer, or washed with an alkaline solution, properly rinsed, phosphate coated, or painted with a coat of wash (etch) primer, and then coated with an approved epoxy primer. After erection of the framing structure and body sheets, all undercar metal, except stainless steel, shall receive a polyurethane finish as specified above. The exterior finish shall have a gloss level of 85 +/- 5 as measured with 60 degree Glossometer. The supplier shall submit color samples and corresponding gloss and tri-stimulus values for approval **[CDRL 20-039]**.

Alternate paint systems, such as base coat-clear coat systems, or direct-to-metal paint systems will be considered if the paint performance equals or exceeds that of a two-part polyurethane paint system.

C. Exterior Painting

All exterior surfaces that are to be painted shall be prepared as specified, and the paint shall be applied according to the paint manufacturer's recommendations. The paint shall be uniformly applied over all surfaces to be covered and shall be free from runs, sags, or other application defects. Painting shall be performed in a clean, dry atmosphere at an ambient temperature as recommended by the paint manufacturer.

Before painting any car surface that is exposed to view, all dents, gashes, nicks, roughness, or other surface imperfections or depressions shall be removed, in so far as possible, by straightening and shall then be properly prepared to receive a filler material. If the filler material is provided by a manufacturer other than the manufacturer of the paint system, a letter from the paint manufacturer shall be required stating that the filler is compatible with the paint system. These surfaces shall be properly cleaned and wash primed following straightening. Any remaining dents or other surface imperfections shall then be filled with an approved filler and sanded smooth. The maximum allowable filler thickness shall be as recommended by the filler manufacturer for the environment and service to which it is to be exposed, but in no case shall it exceed 0.125 inch.

The final painted surface shall be tested on the first car to the following criteria:

1. Hardness

Pencil Hardness tests shall be performed according to ASTM D 3363. The range of acceptance shall be between H and 2H and shall be the average of 10 readings taken from typical surface locations. To avoid a destructive test of a vehicle's surface, which would require the tested surfaces to be repaired, the use of a test structure for this test may be proposed for consideration.

2. Adhesion

Adhesion shall be tested per ASTM D 4541 and achieve a minimum rating as provided by the paint manufacturer. To avoid a destructive test of a vehicle's surface, which would require the tested surfaces to be repaired, the use of a test structure for this test may be proposed for consideration.

3. Thickness

The minimum and maximum dry film thicknesses shall be provided by the paint manufacturer. Dry film thicknesses beyond the manufacturer's recommendations will not be accepted. Non-destructive testing shall be performed to verify final dry film thickness.

4. Paint Cure

A solvent rub test shall be performed per ASTM D 5402. The test procedure requires no less than 50 double finger rubs with a cloth wetted in acetone or methyl isobutyl ketone to the painted surface. No paint color shall transfer to the cloth. After 72 hours, the painted surface must retain all original gloss and hardness characteristics.

D. Apparatus and Underfloor Equipment

All underfloor- and overhead-mounted apparatus (motors, control boxes, junction boxes, brake valves, and other equipment as specified) shall be primed and painted in accordance with the following requirements unless otherwise indicated. All other apparatus shall be painted in an approved color.

The exterior surfaces of undercar equipment enclosures and apparatus, other than propulsion control equipment, made from carbon steel shall be prepared, primed, and painted, as specified in TP20.24.B and TP20.24.C. The interior and exterior surface of all electrical equipment enclosures shall be coated with an approved insulating, thermosetting, resin-based, powder coating or polyurethane paint system. The interior of the boxes, including insides of covers, shall be white and the exteriors shall match the undercar paint scheme.

Parts of undercar equipment enclosures made from plastic or fiberglass shall be painted in accordance with the above requirements for metal portions, except that the paint system shall be compatible with the plastic used and an insulating coating need not be applied.

E. Painting Restrictions

Any equipment or parts of equipment which can be damaged or suffer impaired operation from painting shall not be painted and shall be corrosion resistant.

The following items shall not be painted:

- Copper tubing, piping, and fittings;
- Wearing surfaces;
- Couplers, including yoke and draft gear;
- Wire and cable;
- Power resistors;
- Heat transfer surfaces;
- Electrical insulators;
- Elastomeric parts;
- Grounding pads; and
- Conduit and fittings.

The following truck-related items shall not be painted:

- Wheels;
- Axles;
- Elastomeric parts;
- Grease fittings;
- Linkages;
- Threaded adjustment parts;
- Electrical equipment; and
- Wearing Surfaces.

F. Interior Painting

All exposed interior surfaces, including molding and trim, shall be as specified in TP04.03.B.2. Interior surfaces requiring painting shall be coated with an approved painted finish. Parts which are to be painted shall be cleaned and prepared in accordance with the recommendations of the paint manufacturer.

The Contractor and its paint manufacturer shall provide a touch-up procedure and assure that a continued supply of touch-up paint in the proper colors suitable for spot application will continue to be available in the United States.

G. Corrosion Protection

Concealed surfaces that can rust or oxidize shall be properly cleaned, receive a wash primer, shall be primed with an epoxy paint, and shall be painted with an approved finish. Where arc welding is performed on joints between stainless steel and other materials, the joint shall be de-scaled, cleaned, receive a wash primer coating, and shall be painted in accordance with TP20.24.

H. Acoustic Insulation

Acoustic insulating materials shall be applied to properly cleaned underframe, sides, ends, roof and floor sheets, in accordance with the supplier's recommendations as required in TP04.03. The materials shall be resistant to dilute acids, alcohols, grease, gasolines, aliphatic oils, and vermin. The material shall be unaffected by sunlight and ozone and shall not become brittle with age. It shall be Daubert Chemical Company's V-Damp 3680 sound deadening compound, 3M Corporation's 2552 Damping Foil,, or approved equal.

I. Paint Process Documentation

The Contractor shall prepare detailed paint coating and application documentation containing manufacturer's product data sheets, procedures for surface cleaning and preparation, priming, surfacing, and painting for the carbody and all equipment that is to be painted or powder coated, as well as touch-up and repair procedures. A detailed paint schedule showing the equipment painted, paint type and manufacturers' recommendations for thickness, and other pertinent parameters shall also be included. This document shall be submitted for approval prior to painting of any surfaces or components and shall be made part of the maintenance manuals **[CDRL 20-040]**.

J. Truck Painting

All truck components to be painted shall be given a full coat of primer prior to assembly. Following assembly, all exposed surfaces of each truck, including machined mounting surfaces not used, shall be cleaned with compressed air and solvent wiped to remove all dirt and grease. These surfaces shall then be sprayed with one coat of an approved black truck paint of a type which will not conceal cracks that may develop in service, and air-dried.

TP20.25 FLAMMABILITY, SMOKE EMISSION, AND TOXICITY REQUIREMENTS

A. General

All combustible materials used in the construction of the cars shall satisfy the flammability, toxicity and smoke emissions requirements of this Section and 49 CFR 238.103 and NFPA 130. In case of conflict, the more restrictive requirement shall prevail. The Contractor shall comply with all provisions of 49 CFR 238.103 (c), "*Fire Safety Analysis for Procuring New Passenger Equipment,"* and APTA RP-PS-005-00, *"Fire Safety Analysis of Existing Passenger Rail Equipment."* The Contractor shall prepare and submit a Fire Safety Analysis of the design and materials of construction of the vehicle as part of the design phase of the Contract to be submitted to the Engineer for review and approval prior to construction **[CDRL 20-041].**

Test reports from an independent laboratory indicating successful testing and demonstrating compliance with these requirements for all materials shall be submitted to the Engineer for review and approval. Testing shall be conducted after the Contractor's NTP and on a production batch of material intended to be used on the vehicle. Each laboratory shall have tested a standard test sample no greater than 30 days prior to performing the tests, the results of which shall be submitted to the Engineer. The Contractor shall be responsible for complete conformance with the requirements for itself and its subcontractors and suppliers. The Engineer may, at its discretion, require that the current batch of production material provided for this Contract be retested for conformance with these requirements.

B. Combustible Content

The design of the vehicle shall minimize the total combustible material content of the vehicle. A matrix showing the total weight of all combustible materials, where used, flammability, smoke emissions, test identity, test facility, test requirements, test results, nature, and quantity of the products of combustion, caloric content, and peak heat release rate per ASTM E1354 shall be submitted by the Contractor during detailed design review for the Engineer's approval **[CDRL 20-042]**. All materials with a surface area greater than 16 in^2 shall be tested at a flux radiance of $50KW/m^2$. As a minimum, the following materials, (if applicable), shall be tested: seating materials, wall coverings, window masks, fiberglass components, insulations, plymetal walls, floor covering, floor panels, non-metallic equipment boxes and covers, window glazing, lighting lenses and diffusers, window gaskets, door seals, HVAC in-air stream components, grills, door nosings, filters, electrical wire and cable insulation, and lavatory toilet seats.

C. Flammability and Smoke Emission

All materials used in passenger vehicles shall be tested to demonstrate compliance with the requirements set forth in 49 CFR 238.103 and NFPA 130. The Contractor shall submit test reports for all materials tested and a summary table to the Engineer for approval. The summary table shall be as described in 20.25.A.

D. Floor Assembly Fire Resistance Testing Criteria

The Contractor shall test the floor assembly in accordance with ASTM E 119 to demonstrate a 30 minute endurance rating. The comprehensive test report which shall include the test procedure, test facility, test results, and "before and after" photographs shall be submitted for review and approval by the Engineer prior to procurement of any flooring material for vehicle production **[CDRL 20-043]**.

The following test criteria shall be met:

- The test specimens shall be a full width vehicle section including side sills or that portion of the wall, which extends below the floor. Each specimen shall have a minimum exposed area of 100 square feet. If approved, the exposed area may be reduced to meet a length limitation imposed by the size of the test furnace, but the length shall not be less than 11 feet. No less than one typical penetration of each type spaced at a distance from each other no greater than that which will exist in actual construction shall be included in the test specimen. The specimens shall include typical floor splice configurations.

- The test specimens shall be loaded to simulate "crush" passenger loading conditions. Concentrated loads shall be applied to simulate underfloor equipment, as applicable.

- The test specimens shall include at least three typical transverse supports.

- The test specimens shall represent the actual construction utilized in production. This shall include the floor covering, floor boards, floor structure, all types of penetrations, thermal and acoustic insulation, and floor pans.

- The conditions of acceptance for this test shall be those required for unrestrained assembly.

- Drawings of the floor fire test sample, with their respective production drawings, shall be submitted to the Engineer for approval prior to construction and testing of the sample. Any deviations from the approved configuration shall require approval and re-testing **[CDRL 20-044]**.

E. Toxicity

Materials and products identified by state agencies, Federal agencies, and the American Conference of Governmental Industrial Hygienists (ACGIH) as containing toxic properties or to emit toxic products of combustion in excess of the limits defined in the Technical Provisions shall not be used. Materials and products generally recognized to have highly toxic products of combustion shall not be used.

All materials used in the car construction, except for materials used in small parts such as knobs, rollers, fasteners, clips, grommets, and small electrical parts that would not contribute significantly to fire propagation or to smoke or toxic gas generation and are distributed throughout the car, shall be tested for toxicity using Boeing Specification Support Standard BSS-7239. Alternative test protocols may be proposed for the Engineer's consideration providing that the results are reported as noted below. Materials shall meet the following maximum toxic gas release limits (ppm) as determined per BSS-7239:

Carbon Monoxide (CO)	3,500 ppm
Hydrogen Fluoride (HF)	200 ppm
Nitrogen Dioxide (NO_2)	100 ppm
Hydrogen Chloride (HCL)	500 ppm
Hydrogen Cyanide (HCN)	150 ppm
Sulfur Dioxide (SO_2)	100 ppm

The tests shall be conducted in the flaming mode after 240 seconds using the NBS Smoke Density Chamber for sample combustion. The gas sampling may be conducted during the smoke density test. The test report shall indicate the maximum concentration (ppm) for each of the above gases at the specified sampling time and shall be submitted to the Engineer for review and approval **[CDRL 20-045]**.

F. Electrical Fire Safety

Electrical equipment, wiring and apparatus shall conform to NFPA 130, Section 8, except where more restrictive requirements are imposed by the Technical Provisions.

TP20.26 AIR FILTERS

A. General

Where possible, the air filters provided shall be the same as those used by RTD for similar applications. The Contractor shall make every attempt to minimize the number of different filters required for the vehicles.

B. HVAC and Equipment Ventilation Filters

Filters shall be selected in accordance with the manufacturer's recommendations for the specific equipment involved. Filters shall be the throw-away type with integral frames, except that reusable filters may be approved for specific applications where throw-away filters are not available. Filters shall be designed to meet the performance requirements of each installation and shall be subject to approval by the Engineer. All filters shall be freely accessible for replacement and maintenance. Filters shall meet the requirements of UL 900, Class 2.

C. High Pressure Air Filters

Air filter assemblies with replaceable filter elements shall be provided in the line that connects each subsystem to the main air supply system. The filtering capability, flow rate, and overall size shall be appropriate for the application so that the filter replacement interval is greater than one year. It shall be possible to gain access to filter elements for replacement without disconnecting or loosening pipe fittings. Filters shall be provided for each of the following systems and any others if operated from the air supply system:

- Each air brake control assembly;
- Each leveling valve;
- Door controls; and
- Horn and bell.

D. Low Pressure Air Filters

Replaceable media type filters shall use resin-bound, spun-glass fiber materials having an uncompressed thickness not less than 3 1/2 inches. They shall be non-absorptive of fluids and gases, shall be processed in such a manner that material density increases progressively from inlet to exit side, and shall be coated with not less than 24 grams per square foot of a dust-retaining, viscous adhesive film. This film shall be stable at temperatures up to 150 degrees Fahrenheit. The filter medium shall be cut not less than 1/2-inch oversize to ensure adequate sealing between the edge of pad and its integral frame.

TP20.27 ELECTRICAL AND ELECTRONIC DESIGNS

A. General

Except as otherwise noted herein, electronic equipment shall conform to IEC 60571, *Electronic Equipment Used on Rail Vehicles*, Class TX, unless otherwise approved by the Engineer. All standard type tests shall be performed.

B. Reliability Standards

A standardized MIL-HDBK-217F reliability part stress prediction shall be performed on all electrical and electronic control systems. This reliability prediction shall be based on the "Ground Mobile" environment. Use of alternative reliability database information may be permitted for parts not contained in MIL-HDBK-217F, subject to approval. Submittal of the reliability prediction shall be identified in the Reliability Program Plan described in TP24.01. The prediction shall be used during design and development to compare competing designs, perform design tradeoffs, detect overstressed parts, and identify high failure rate items.

A documented closed-looped Failure Reporting and Corrective Action System (FRACAS) (per *Reliability Toolkit: Commercial Practices Edition*) shall be established and maintained to provide for the identification, tracking, and repair of all product/process failures. Early elimination of failure causes or trends will contribute significantly to reliability growth and continuous process improvement.

All semiconductor devices shall be derated to operate within the acceptable region for electrical and temperature stress as specified in *"Reliability Toolkit: Commercial Practices Edition."* If there is a conflict between guidelines given elsewhere in this Specification and the Reliability Toolkit, the more restrictive condition shall govern.

All electronic assemblies shall undergo Environmental Stress Screening (ESS). The temperature cycling regimen shall be in accordance with table 7.5-2, unit column, of the Reliability Design Toolkit: Commercial Practices Edition, from the Reliability Analysis center, except as indicated below. The temperature extremes may be limited to –13˚F to + 158˚F, at the discretion of the supplier. A minimum of 20 complete temperature cycles shall be conducted. The ESS shall be performed with the equipment operational, powered, and oriented as per the ultimate application. Input signals and output loads to simulate the maximum power dissipating condition in the equipment shall be applied during the rising temperature and maximum temperature portions of the temperature cycle. The equipment shall be given a full functional test before and after the ESS, and monitored for failure throughout the ESS. In the event of equipment failure, the repaired equipment shall be given another complete ESS test. Alternatives to this baseline ESS may be acceptable at the discretion of the Engineer. Assemblies consisting exclusively of components rated at 50 amperes or greater are exempt from this requirement.

C. Ability to Repair

Where practical, all electrical assemblies, including such items as PC boards, shall be designed for repair by the maintenance staff in their electronics laboratory (Refer to MFBTE in TP25.01.B).

Assemblies shall not be sealed, potted, or constructed to prohibit repair. Assemblies that must be potted or sealed or can not be repaired by design shall have a minimum 10 year warranty. Such non-repairable components shall be designed to allow unit replacement.

All USB connections and all other removable storage devices shall have a sacrificial pass through connector that can be easily replaced without requiring special tools or unsoldering.

D. Hardware

Refer to TP20.02.B for general hardware requirements. All hardware associated with electronic and electrical control systems shall be protected against moisture, oxidation, and common airborne contaminants.

E. Enclosures/Racks

All circuit boards that are rack-mounted shall plug into racks containing the mating half of the circuit board connector. The circuit board rack shall mount in an enclosure conforming to requirements in the Technical Provisions. The rack, circuit board, and circuit board hardware shall be designed as an integrated system. All circuit boards and connectors in a rack shall be keyed to ensure the correct board is installed in the proper location.

The rack and enclosure shall provide environmental and EMI shielding, as required, to meet the requirements of the Technical Provisions.

Printed circuit boards shall be positively retained by means of keeper bars or other approved method. The enclosure or rack cover shall not be used to retain the circuit boards, unless specifically designed to do so. Hinges and latches shall be of stainless steel.

Each circuit board shall be fitted with an ejector or hand grip to assist in board removal. The rack and the edge of each board, or the card ejector, shall be labeled with corresponding numbers to identify board location within the enclosure. A brief functional designation shall also be included on each label.

The enclosure/rack shall not be connected to the power supply return or signal common.

Where it is necessary to use printed circuit boards that are not plug-in and not mounted in an enclosure, the following additional requirements shall apply:

- The PC board shall be protected from mechanical damage and hostile environments such as arc discharge and contact with high voltage.

- If the PC board is part of a high voltage circuit, special care shall be used in its design with regard to strike distance and creepage in the rail vehicle environment. This shall include the distance between PC board components and with respect to any grounded mounting surfaces. APTA RP-E-004-98 shall apply where applicable.

- Any test points required in routine testing or fault isolation to the user replaceable level, shall be easily accessible without disassembly or tools.

If replacement of the PC board is required (as part of secondary maintenance), no special tools or soldering shall be required. Each PC board use and application of this type shall be subject to Engineer approval.

F. Optical Fibers

Any application of optical fibers shall be approved by the Engineer prior to implementation. This approval is not intended to discourage the use of optical fibers; rather, it is meant to verify reliability and maintainability of the proposed application. In no case shall the on-car repair of an optical fiber require sophisticated or complex polishing and alignment. The connections between optical fibers and car-replaceable units shall be via approved "quick-disconnects."

TP20.28 SEMICONDUCTOR STANDARDS

A. General

Semiconductors shall be selected to withstand all continuous and transient voltage and power demands present in the circuit application without damage or shortened life. All circuit designs shall provide for the presence of high current switching equipment on the vehicle and the resultant induced voltages and currents in electrical equipment.

B. Ratings

Semiconductors, except diodes (see below), operated from the battery supply or those connected to trainlines, shall have minimum breakdown ratings of four times the maximum achievable circuit voltage. Suppression devices shall be provided as necessary to protect the devices and limit the circuit voltage.

Diodes operated from the battery supply, used as suppression devices, or connected to trainlines shall have a minimum breakdown rating (PIV) of 1,000 V. Diodes with less than 1,000 V PIV rating may be used if adequate circuit transient protection is also provided.

All discrete semiconductors operated from inverters or other isolating devices shall have a minimum breakdown rating of two times the maximum circuit voltage, except where specifically detailed otherwise. Suppression devices shall be provided as necessary to protect the devices and limit the circuit voltage.

All semiconductors shall be operated at less than 50% of the maximum continuous-current rating or 50% of the maximum continuous-power rating, whichever is more restrictive. High power/current devices may be exempt from this requirement with prior approval on a case-by-case basis. The Contractor shall submit complete device information, including all manufacturers' application recommendations, and calculated current and power demands with all waiver requests for approval by the Engineer. If approved, such waivers shall not reduce other requirements, including reliability.

Integrated circuits operated from the battery supply through inverters or other isolating devices shall be operated within the voltage and current ratings specified by the manufacturer, derated to less than 50% of the maximum stress level at the maximum operating temperature of the device as specified by the manufacturer.

Where the supplies to integrated circuits are regulated and surge protected, the voltage rating shall be 15% below the manufacturer's recommended maximum. In addition, the maximum power shall be limited to 50% of the manufacturer's specified maximum at the maximum operating temperature.

All semiconductor junction temperatures shall be limited to 302 degrees Fahrenheit (150 degrees Celsius) (or to the maximum rated temperature for the device, whichever is less) or less at maximum ambient temperature and at maximum rated output power.

Silicon semiconductors shall be rated for operation over the temperature range of -40 degrees Fahrenheit to 185 degrees Fahrenheit and shall be hermetically sealed to the extent required by the application.

Gallium arsenide and similar optical semi-conductors shall be rated for operation over the temperature range of -40 degrees Fahrenheit to 185 degrees Fahrenheit.

C. Availability and Identification

All semiconductors shall be available from at least two manufacturers and available from U.S. distributors. Single source devices, such as high voltage power devices, microprocessors, ASIC's and related support chips shall be used only if approved by the Engineer. Such devices shall be essential to the proposed equipment, shall meet the proven service requirements, and shall be supplied by veteran manufacturers committed to support the devices.

Each device shall be labeled to identify both the manufacturer and the complete part number. Operational characteristics of the device shall be published and available to the Engineer.

D. Burn-in

Refer to Reliability Standards section in TP20.27.B.

E. Other Prohibitions

Electronic equipment shall utilize stock components and shall function properly with the component manufacturer's full range of tolerances, such that after-purchase screening or testing of components shall not be required.

Matching of components shall be permitted only if the components are normally available from the manufacturer in matched sets.

Germanium semiconductors shall not be used.

TP20.29 PRINTED CIRCUIT BOARD STANDARDS

A. General

Printed circuit boards shall be designed, constructed, and inspected to ANSI/IPC-D-275, latest revision, except where more stringent requirements are noted herein. Within ANSI/IPC-D-275, printed circuit board classes are designated. Printed circuit boards supplied under this Specification shall be Class 2, minimum, with the exception of wayside computers that are not utilized in vehicle operation. Class 3 requirements shall apply to all vital equipment.

Circuit board material shall be per NEMA Standard LI 1, Type FR-4 (MIL-P-13949, Type GF), for boards which have no components, whose power dissipation is greater than 2 watts, and when said board is not mounted adjacent to components dissipating greater than 2 watts. Otherwise, circuit board material shall be per NEMA Standard LI 1, Type FR-5 (MIL-P-13949, Type GH).

Printed circuit boards shall have a minimum thickness of 1/16-inch base material. All conductor material shall be copper and shall be firmly attached to the board and shall be resistant to blistering and peeling when heated with a soldering iron.

All printed circuit boards shall be designed for ease of testing per ANSI/IPS-D-275, *"Testability Design Check List."*

Traces shall be made as wide as practical, with the minimum width being based on a 50 degrees Fahrenheit (10 degrees Celsius) temperature rise.

Components with pins shall be mounted only on one side. Connections shall be made to the other side or internal layers via plated through holes. SMT devices may be mounted on both sides if part of an approved existing design.

All circuit boards shall be inherently stiff or shall be reinforced to prevent damage due to vibration or handling. Circuit boards larger than 100 square inches shall be centrally stiffened unless otherwise approved.

All printed circuit boards with the same function shall be interchangeable between equipment groups without adjustment.

All printed circuit boards shall be of the "plug-in" type, with positive support against vibration. Single board applications, where approved by the Engineer, may be of a "non-plug-in" type.

Printed circuit boards shall be designed for insertion and removal with power applied, except where power is removed by a switch adjacent to the card rack and except where the mechanical construction would generally prohibit removal and insertion with power applied. Where a switch is used, it shall be labeled with a warning regarding its proper use.

B. Marking

All circuit boards shall be labeled with a part number, serial number, and descriptive nomenclature.

All components shall be labeled on the board with component drawing references and such other information as may be required to repair and troubleshoot the board. The component and wiring sides of the board shall each be marked to indicate capacitor and diode polarity, and at least two leads or one lead and a graphic symbol indicating orientation of all transistors and thyristors.

Integrated circuits and other multi-terminal devices shall have an index mark on the component side of the board, visible with the component inserted, to indicate proper keying and insertion; additionally, the first pin on all IC packages shall be identified on the wiring side of the board. The labels used to identify components on the printed circuit board shall match those used in the schematic drawings for that particular component.

C. Component Mounting

Components shall be fastened to the board in such a manner as to withstand repeated exposure to shock and vibration. Large components shall be mechanically supported in addition to the solder connections. Power resistors shall be mounted on standoffs so that the resistor bodies do not

contact the board, spaced far enough away from the board so that resistor-produced heat will not discolor or damage the board.

D. IC and Device Sockets

IC and device sockets are prohibited except for components that must be removed for reprogramming or initial calibration procedures or devices that are available only for mounting in sockets. All socket applications shall be subject to approval by the Engineer. All other components shall be soldered in place.

Where approved, IC sockets shall comply with approved standards such as MIL-S-83502 and MIL-S-83734, as applicable for the device, and shall be made of the following materials:

- The bodies shall be molded from diallyl phthalate, PTFE Teflon, or approved equal.
- The contacts shall be fabricated from beryllium copper and shall be plated with a minimum of 0.000030 inch of gold over a minimum of 0.000050 inch of low-stress nickel in the area of contact with IC pins.

E. Conformal Coating

Both sides of the assembled printed circuit boards shall be coated with a clear insulating and protective coating material conforming to MIL-I-46058, latest revision, or approved equal.

The coating shall be easily removed with a brush-applied solvent or penetrated by a hot soldering iron when a component must be unsoldered. The coating solvent shall not adversely affect board-mounted components.

All IC sockets, connectors, and test points shall be masked when the coating is applied.

F. Keying

All printed-circuit boards shall be mechanically “keyed” to prevent insertion into the wrong slot or in the wrong orientation. Further, circuit boards in safety-related control systems, such as friction brakes, cab signal, ATC, ATS, and systems which can cause damage or unsafe train operation if the vehicle is operated with a card removed, shall be connected through a safety circuit to disable the vehicle if a circuit board is removed.

G. Circuit Board Connectors

Printed circuit board connectors shall be heavy-duty, high-reliability, two-part type with a history of successful service in rail applications and shall be approved by the Engineer prior to commencing design.

Connectors which comply with MIL-C-55302 or DIN 41612 Level 1 or 2 and which have plated contacts as described below are considered to comply with the requirements of this Section.

The connector contact area shall be plated with a minimum of 0.000030 inch of gold over a minimum of 0.000050 inch of low-stress nickel.

Card-edge connectors are prohibited.

H. Testing

Sufficient clearance shall be provided between components to allow testing, removal, and replacement without difficulty.

Test points shall be provided in appropriate locations on modules and printed circuit boards. A negative-return test point shall also be provided. The test points for manual testing shall either accept and hold a standard 0.080-inch diameter tip plug or shall be a turret lug similar to Cambion Number 160-1026-01-05 or approved equal, with sufficient clearance to permit it to accept a standard oscilloscope probe clip and shall be identified by appropriate markings.

TP20.30 ELECTRICAL DEVICES AND HARDWARE

A. General

All electrical devices shall be rail industry proven.

B. Contactors and Relays

All contactors and relays shall meet or exceed the requirements of MIL-R-6106 and MIL-R-5757 respectively, with the following qualifications.

- Devices shall be tested for proper functioning in orientations up to 30 degrees from the orientation in which they are mounted in the vehicle in each of the three possible rotations: pitch, yaw, and roll.

- If adequate documentation exists to demonstrate that, during functional and operational testing of the vehicle, the contactors underwent normal duty cycle tests, it shall be considered as an acceptable alternative to burn in.

- In selected applications, contactors and relays shall comply with the requirements of MIL-R-6106 (for ratings of 10 amperes or greater) and MIL-R-5757 (for ratings of less than 10 amperes) but need not be qualified to these documents if all of the following requirements are met:

    - The device is service-proven in the same application.

    - The device is service-proven in rail service.

    - All other requirements of the Technical Provisions are met.

    - The Engineer approves of the application.

All devices shall be constructed and utilized in a fail-safe manner; that is, all failures shall be such that passengers, crew, and equipment are not placed in jeopardy.

All devices shall be installed so that they are fully accessible for inspection, repair-in-place, or removal and replacement. All contactor terminals shall be fully accessible for troubleshooting. Contactors and relays shall incorporate means of visually determining whether they are picked up

or dropped out. Relays on printed circuit boards or within electronic assemblies may be exempted from the requirement for a visual indication, as approved by the Engineer.

There shall be a maximum of two wire terminations on any one contact of any device.

The coils of all devices shall be suppressed to protect the low-voltage network from transients.

Under no circumstances shall either the main or auxiliary contact tips of the devices be placed in parallel for the purpose of carrying a current load at or above the manufacturer's contact tip rating.

Contact tip ratings shall be stated for the worst condition of reduced surface contact which may result from tip misalignment and wear during normal operation.

Contactor installation shall be such that the arc spray is directed by an arc chute away from ground and any other electrical devices proximate to the contactor.

Devices shall be constructed in a heavy-duty fashion suitable for railroad service. The Engineer reserves the right to review and approve the design and selection of all contactors and relays.

Contactor tip replacement shall not exceed 10% of the total number of tips at 90-day intervals.

All contactors shall be constructed so that the main contact tips make and break with a wiping motion that prevents deposits and pitting.

All DC contactors shall have series-fed blowout coils. The Contractor shall demonstrate the ability of each contactor type to reliably interrupt current over the full design operating range.

All devices shall be readily identifiable by means of a permanent, durable marking strip giving the device circuit designation. No identification shall be obscured, or partially obscured, by wire routing. The identification strip shall be mounted adjacent to the mounting of said device.

Bifurcated contacts shall be used in low voltage applications, whenever necessary, due to dry contacts or low current switching requirements.

All time delay relays shall be of the R-C delay or solid state type. No mechanical or pneumatic time-delay devices will be permitted.

Where plug-in relays are approved, the relay shall be positively retained by means of a retaining clip or bar. This device shall be captive, of rugged construction, and shall be easily positioned for relay installation and removal without the need for special tools. When the relay is removed, the retainer shall itself be retained so that it cannot come in contact with devices which may have exposed energized electrical circuits, and it shall not interfere with the operation of any other device when in this position.

Adequate gap and creepage distances shall be maintained from high voltage contactor tips and low voltage coil and auxiliary contacts, to prevent entry of high voltage arcs or transients into low voltage circuits. The same shall also apply to grounded mounting surfaces. APTA RP-E-004-98 shall apply where applicable.

Relays shall not be affected by the accumulation of airborne dust.

C. Switches

Under no circumstances shall poles of switches be placed in parallel in order to carry currents in excess of the contact pole rating given by the manufacturer.

Switches shall be provided with a "keying" feature so that after installation, the body of the switch is constrained from mechanical rotation.

All switches provided shall be of the highest quality procurable and shall be fully suitable for the RTD service environment. All control switches subject to water splash (switches mounted near windows or doors, or mounted on the Operator's console) shall be environmentally sealed. Toggle and push button switches shall be per MIL-S-3950, MIL-S-8805, MIL-S-83731, or equal. All safety-critical switches, such as those for door operation, shall be designed to withstand a high potential test of 1500 V for 1 second, in a clean, dry condition, without false conduction. The design and selection of all switches shall be subject to review and approval.

There shall be a maximum of two wires connected to each terminal of the device.

Switches shall be individually replaceable by disconnecting only the mounting fasteners and electrical connections of the switch to be replaced.

In addition to the above requirements, all switches and pushbuttons shall meet the following requirements:

- Contact resistance shall be less than 0.1 ohm at 3 V DC and 10 milliamps.
- Open circuit resistance shall be 50 mega-ohms minimum.
- Resistance to case shall be 1000 mega-ohms minimum at 500 V DC.

D. Circuit Breakers

1. General

All circuit breakers shall be rugged and fully suitable for the service intended and of the highest quality procurable. Design and selection of all circuit breakers shall be subject to review and approval. All circuit breakers of the same rating shall be of the same manufacture and model throughout the vehicle. Circuit breakers shall be Din Rail mounted when ever possible.

The "ON," "OFF," and "TRIPPED" positions of all circuit breakers shall be permanently marked on the handle or the case of the circuit breaker. The circuit breaker, when tripped, shall assume a distinct position between the "ON" and "OFF" positions to permit determination that it has been tripped by either its overcurrent or shunt trip elements. All circuit breakers shall be mounted in the vertical direction with the "ON" position up.

Circuit breakers shall be individually replaceable by disconnecting only the mounting fasteners and electrical connections of the breaker to be replaced.

Electrical connections to circuit breakers shall either be threaded to accept machine screws or use a threaded stud. Wires to circuit breakers shall use ring terminals.

Circuit breaker terminals shall not be used as junction points.

Each input-power circuit shall be protected by an individual circuit breaker. Separate circuit breakers shall be provided for major assemblies or functions. No circuit breaker shall protect more than one circuit, nor shall any one circuit be protected by more than one circuit breaker.

All circuit breakers shall be sized by current rating and trip time to protect both the associated equipment and the minimum size wire used for power distribution within the protected circuit without causing nuisance trips.

High voltage circuit breaker poles may be connected in series, if necessary, to achieve the stated voltage interruption requirements.

Each circuit breaker pole shall be equipped with a means of arc extinguishment to prevent flashover.

The continuous current rating of thermal-magnetic trip circuit breakers shall be selected in accordance with ANSI C37.16 for the load and type of service specified.

All thermal-magnetic trip circuit breakers shall conform to the requirements of ANSI C37.13 and ANSI C37.14.

Circuit breaker current rating shall be clearly and permanently marked and shall be completely visible after installation.

Electrically operated circuit breakers shall be arranged for operation from the low voltage DC supply.

2. High Voltage Circuit Breakers

All high voltage circuit breakers shall have not less than three poles connected in series.

All distribution-type, high voltage circuit breakers shall be Westinghouse Series C, FDB frame, Heinemann type GH, or approved equal.

The trip elements shall be thermal-magnetic, or magnetic, connected in series.

The circuit breaker handle shall protrude from the circuit breaker panel cover sufficiently to be manipulated in all positions.

3. Low-voltage Circuit Breakers

Low voltage circuit breakers shall be either one-pole or two-pole devices, depending on the intended function. Trip elements shall be thermal-magnetic, or magnetic, as appropriate for the application.

All low voltage circuit breakers shall be:

- General Use – Westinghouse Series C, Quicklag C frame, Heinemann Series AM, or approved equal, with front connection or other approved access arrangement, and approved labeling.

- Fast Operation – Airpax type IMLK, dust sealed, magnetic breaker, or Airpax type UP, hermetically sealed, magnetic breaker, or approved equal.

E. Fuses

Fuses shall be used only where specifically called for in the Technical Provisions or where the use of circuit breakers is not technically feasible, and only with specific approval. Fuses may be considered in applications as follows:

- To protect solid state equipment from catastrophic damage.

- Where current or voltage levels prohibit circuit breakers.

Fuses shall be permanently identified in a location adjacent to the fuse. The rating of each fuse shall be permanently and clearly marked directly on each fuse.

Fuses shall be readily accessible. Fuses mounted in exterior equipment boxes shall be accessible from the side of the car.

Fuse holders shall contain fuse retention devices at both ends.

Air gap and creepage distances shall be subject to approval by the Engineer. APTA RP-E-004-98 shall apply where applicable.

Fuses used in nominal 600 V DC circuits shall be rated for no less than 1,000 V DC. Voltage ratings for fuses in other high voltage circuits shall be submitted for review and approval by the Engineer.

High voltage fuses shall be mounted in totally enclosed, dead front fuse holders, with no exposed high voltage connections. The fuse shall be extracted from the circuit when the fuse holder is opened and the exposed fuse shall be safely isolated.

Where circuits use multiple fuses or fuses and circuit breakers, the coordination between the protective devices shall be subject to approval by the Engineer.

F. Bus Bars

Bus bars shall be fabricated from OFE (Oxygen Free Electronic CDA C10100) or ETP (Electrolytic Tough Pitch CDA C11000) copper. Bus bar conductivity shall be 100% IACS. All bus bar joints shall be silver or tin plated.

Current densities, other than at joints, shall not exceed 1,000 amperes per square inch, and in any case shall not exceed a value which would cause a bus bar temperature rise greater than 86 degrees Fahrenheit (30 degrees Celsius). Current densities in brazed joints shall not exceed 150 amperes per square inch.

Bus bars shall be properly brazed together at joints unless bolted connections are found to be absolutely necessary for maintenance purposes and have been approved. The overlap at bus bar joints shall be no less than 10 times the thickness of the bus material. Bus bar connection bolts shall be torqued to obtain a uniform bus bar connection pressure of 200 psi. Bolting hardware shall be plated steel with Belleville washers to maintain connection pressure. Current densities in bolted joints shall not exceed 300 amperes per square inch for a minimum bolt size of 5/16 inch. The bus temperature rise including joints shall not be greater than 86 degrees Fahrenheit (30 degrees Celsius) above 104 degrees Fahrenheit (40 degrees Celsius) ambient.

Except for connection areas, bus bars shall be safety-insulated, using a high-dielectric powder coating, heat shrink tubing or other approved means. Tape will not be acceptable. Bus bars that are behind insulating panels are exempt from this requirement.

G. Capacitors and Resistors

Hermetically-sealed dry tantalum capacitors in metal cases shall be used in place of aluminum electrolytics, except for very high values which are not commercially practical or available, in which case long-life grade aluminum electrolytics shall be used.

Commutating capacitors shall be of paper or plastic film type, shall incorporate a non-toxic impregnant, and shall be chosen to give a service life of at least 20 years. Filter capacitors shall have high ripple current rating for long life.

Capacitors shall be derated 20% for voltage based on the nominal supply voltage and maximum case temperature. If filter capacitors are exposed to low ripple voltages, lesser values of derating may be accepted if it can be shown that reduced operating temperatures can be achieved due to reduced dissipation. However, the sum of the DC and AC ripple voltages shall always be less than the capacitor's voltage rating at a maximum case temperature of 185 degrees Fahrenheit (85 C).

Except for braking power resistors, all resistors shall be derated 50% for power dissipation. Applications for approval of less derating may be submitted on a case-by-case basis.

H. Transformers and Inductors

Transformers and inductors shall be derated 10% for current. Transformers shall:

- Have vacuum-impregnated windings.
- Be rated to withstand at least twice the maximum peak-to-peak voltage that they shall be subjected to in operation.
- Not emit audible noise in excess of 60 dB referenced to 20 micropascals at a distance of 2 feet while operating at rated voltage and load.
- Be designed to minimize radiated and induced EMI.
- Be located, orientated, mounted with cable connections and routing in accordance with the overall EMI/EMC control plan for the vehicle.

I. Switch, Circuit Breaker, and Fuse Panels

All switch, circuit breaker, and fuse panels shall be of the dead front type, mounted in the specified equipment enclosures.

Each switch and circuit breaker panel shall carry the necessary apparatus, arranged to be easily accessible to connections and designed to prevent operating or maintenance personnel from coming in contact with energized parts when operating the switches or circuit breakers. Furthermore, all energized portions of the protected circuitry shall be completely concealed so that no danger of shock exists from touching the panel or any appurtenances or devices mounted thereto.

Each switch, breaker, fuse, and indicating light shall be provided with a nameplate of raised or recessed lettering on the dead front, clearly identifying the circuit to which each applies, its circuit designation, operating voltage and instructions such as "Do Not Operate Under Load" as appropriate. The dead front panel shall conform to NFPA 70, Article 384. The dead fronts shall be made of moisture-proof, electrically insulating, laminated phenolic or fiberglass of approved quality suitable for switchboards. Asbestos shall not be used.

A wiring gutter shall be provided along the top, sides, and bottom, for the routing of high voltage leads to their designated circuit breakers.

The panel shall be secured by approved, captive fasteners and shall be configured for easy removal so that maintenance is not impeded.

Power distribution to circuit breakers and switches shall be from a bus bar or bus circuit. Distributing power by successive or "daisy-chained" connections between device terminals shall not be permitted.

J. Battery Backup Circuits

Backup batteries shall not be permitted, unless specifically approved by the Engineer.

TP20.31 SOFTWARE AND SYSTEMS

A. Introduction

This section includes requirements at the systems level as well as the software and related hardware levels. The systems level includes the overall car level and the individual systems level.

All software and related electronic hardware for this project will be subject to the requirements of this Section. This includes, but is not limited to, software systems for control, monitoring, diagnostics, portable or bench test equipment, and wayside data analysis. No exception is made for items that may be considered firmware. Systems that include data files or configurable or customizable components such as Field Programmable Gate Arrays (FPGAs) are also covered by sections below.

The specific application requirements for the individual systems are given in the corresponding sections of this Specification while the common requirements, pertaining to all such software systems, are given in this section. The additional requirements given here include both design requirements and documentation requirements.

The major goals for this section are to assure that these software systems are complete, reliable, require few if any changes late in the development cycle, are provided on schedule, and are changeable in the future without compromising design integrity. To achieve these goals, plans are needed early in the project; the requirements for software must be analyzed; and designs developed that meet all the requirements.

Documents will need to be read by numerous people during development and in the future for problem analysis and changes. It is important therefore, that all documentation be easily and unambiguously readable.

The Sections below require first, that the overall functions and design structure of the car systems be described along with the internal breakdown of the systems into software and hardware components.

Later sections below give requirements for the software and hardware components, as well as for how they shall be documented.

The software must be treated as an integral part of the total system design and will be reviewed as part of each design review for the corresponding systems.

Refer also to TP20.27 for electrical and electronic designs for related requirements.

B. Systems Requirements

1. General

The Contractor shall be responsible for the overall design and for the integration of the systems into the complete system. Each subsystem is then designed and documented by the supplier with the Contractor reviewing the designs and testing the completed systems.

2. System Documentation

a. Contractor's Overall SFD

The Contractor shall provide an Overall System Functional Description (Overall SFD) that shall define hardware and software components, the partitioning of the system functions, and the allocation of requirements (including derived requirements) to the individual Systems **[CDRL 20-046]**.

The Overall SFD shall include a diagram of the complete system to be provided. It shall show the systems with their interfaces including network interfaces. The terminology of the overall SFD document must be consistent with that of this Technical Specification. The locations of systems within car types shall also be shown in diagrams and described.

The allocation of requirements to multiple systems shall be clearly described with attention to derived requirements such as the allocation of timing requirements that together provide an overall timing requirement.

The SFD shall define any commonalities between the systems such as networking standards or other shared interfaces. Interface Control Documents (ICDs) shall be

used to document all significant interfaces between systems. The ICDs shall conform to IEEE Standard 1558-2004.

b. Individual Systems SFD Required Content

For each individual System, the Supplier shall provide a System Functional Description (SFD) that completely describes the system and how it meets the project requirements **[CDRL 20-047]**. These SFD documents shall be consistent with the overall SFD and with this Technical Specification.

The SFDs shall provide the following:

- A Context Diagram showing the system and the external interfaces. There shall be an accompanying description of the functions of the system and the relationships of the system with other systems.

- A Decomposition Diagram and Description shall describe all System Components and internal interfaces.

- Each hardware component shall be described to the level of its major components, LRUs, LLRUs, the functions to be performed, and the interfaces.

- The components shall be further broken down to the Hardware Components, Software Configuration Items, the internal interfaces within the system, and the external interfaces to other car level systems.

- Each Software Configuration Item (SCI) shall be described in its own numbered section. The major functions of the item shall be described and the relationship to the hardware including memory and input/output hardware. The power fail shutdown strategy shall also be described.

- A Software Summary Table with a row for each software item in the system, including proposed Commercially Available (CA) items. Columns will give the Software Configuration Item name and ID, Software Requirements Specification (SRS) name and ID, Software Design Description (SDD) name and ID, and section within the SFD where the SCI is described. The use of the proposed CA items and their classification as CA for the purposes of this project shall require approval.

- The descriptions of hardware items shall also include explanations of how the hardware requirements of TP20.31.D will be met.

- For Programmable Components such as FPGAs, Complex Programmable Logic Devices (CPLDs) and other similar devices, all specific hardware and programmed functions and the interfaces shall be defined in a dedicated section. This description must include the requirements to be met by the item. The detail design for those items shall then be documented separately in the schematics or in an appropriate application file and output file of the software tool used to program the device. If data files are to be separate configuration items, their content must be described also.

- A Failure Detection and Remedial Action description section shall describe monitoring the correct operation of system components, the failures to be detected and the remedial actions for minimizing the impact of the failure. Failure detection should facilitate root cause analysis, and repair. The description shall include the detection of sensor input failures and compensation schemes such as the use of alternate sensors or performance restrictions.

3. Testability

All features and functions of software systems shall be testable on a systems level. Specific approval is required for any feature that is not testable on a systems level. For features that are only testable with special equipment, all such equipment shall be supplied by the Contractor as special test equipment and subject to the conditions defined in TP25. This equipment shall provide the logic, sequencing, and emulation necessary to verify that the software functions as intended. In lieu of separate equipment, appropriate test functions may be provided within the Portable Test Equipment (PTE).

Type tests of all processor systems shall verify the proper operation of all software features, including component failure detection and diagnostics.

All Test Plans and Procedures shall be submitted for approval prior to conducting the tests. Each requirement of the SRS must be traceable to the test procedures using the Software Requirements Traceability Matrix.

C. Software

1. General

The requirements of this section pertain to all subcontractors providing software (including Programmable Devices). If the Contractor will supply software for this project then they shall also be considered a Supplier and shall also be subject to the requirements of this section pertaining to suppliers.

2. Contractor and Supplier Qualifications

Please see TP21.

3. Software General Features

Software shall perform the following basic functions:

- Implement the desired control scheme or other functions such that the specified performance is achieved;

- Monitor all inputs for unsafe, erroneous, or unknown conditions or combinations of conditions and take appropriate actions to preserve proper functioning and capture appropriate information to facilitate root cause analysis and repair.

- Sample all input conditions and hardware status at rates sufficient to detect and remedy all unsafe or damaging conditions in the shortest possible time. Sampling

rates and program execution times shall be such that the control system is not the limiting factor in response to unsafe or damaging conditions. All software shall be designed to ensure that the timing requirements for safety-related tasks are always met;

- Limit all output commands to safe levels regardless of any combination of input conditions;

- Perform self-diagnostic routines and respond promptly, safely, and predictably to detect faults. The self-diagnostics shall include tests for program corruption and for integrity in read/write memories;

- Respond safely and predictably when powering up or recovering from power interruptions. All power interruptions likely to have corrupted temporary storage shall be detected and cause the system to re-initialize all affected routines and temporary data. Detection of power interruptions may be by hardware; and

- Permit thorough interrogation of all input, output and internal conditions by external diagnostic equipment.

Software shall be designed to minimize the interaction between functions and reduce the coupling such that changes in one area are unlikely to affect other functions.

Programs shall be modular with separation of Operating/Executive Software from Application Software. Software version numbers shall be included within the code. Programs stored in sets of memory devices must self-test to assure that the correct complement of devices is installed. Software shall have verifiable version control based on a Cyclic Redundancy Check (CRC-32) polynomial published in the software documentation and verifiable using an approved utility. This utility shall be included in the Portable Test Units (PTUs), Software Workstations, and the Monitoring and Diagnostic System (MDS). When requested by these systems, the software shall calculate the CRC-32 value and report it along with the version ID. The CRC-32 value shall be provided with each release of a Software Item.

Processor system parameters shall be adjustable via PTE. Appropriate parameters shall be suggested by suppliers during design review for approval. User defined/changeable parameters shall be structured as tabled data. Allowable ranges of parameters shall be stated in the SRSs as requirements and the software shall restrict setting of parameters to these ranges.

In the event that vital functions are performed in processor based systems, functionality shall be verified in accordance with IEEE STD 1483-2000, "IEEE Standard for the Verification of Vital Functions in Processor-Based Systems Used in Rail Transit Control."

4. Software Activities

The Contractor and Suppliers must plan the activities they intend to perform in order to provide the required software and related documentation. These plans must be documented and submitted to the Engineer so agreement can be reached regarding them. Plans must include schedule and resource allocations sufficient to meet the submittal requirements of

TP20.31E. Progress will then be monitored according to the plan to assure on-time delivery and assure the software meets all requirements.

Software Requirements Analysis must be performed for each Software Configuration Item to determine all requirements from all sources. These requirements must be documented in a SRS and submitted for review **[CDRL 20-048]**. The requirements analysis must be performed early in the process to allow for general review and to allow the design to be developed considering all requirements. A Software Test Plan shall also be developed at the time of requirements documentation.

Software Design shall include software design activities and documenting the design.

The reuse of software from previous applications is encouraged, as long as the full set of requirements in this Contract are addressed. This includes both functional requirements and requirements of this Section such as those for language, structure, maintainability and error detection. Requirements analysis, design, and the documentation of both are needed regardless of the intended re-use of the software items.

Suppliers must conduct formal reviews for each document with review minutes submitted along with the documents.

5. Software Documentation

a. General

Software documentation shall be in accordance with IEEE Standard P1558-2004, "Standard for Software Documentation for Rail Equipment and Systems" and with the additional requirements set forth within this document. The IEEE 1558 requirements shall be for a type 5 system as defined within that standard.

Some software items may be classified by the Engineer as Commercially Available (CA) for this project. Requirements for these items are given in TP20.31.C.7.

The Engineer shall be provided with sufficient documentation to fully comprehend and analyze the operation of the equipment in which the software is to be installed, and to enable RTD to maintain and modify the software to correct problems, adapt it to changing requirements, add new features, and port it to a new hardware platform. The documentation shall describe how all requirements will be met. Software documentation training shall be included within the formal Training Program. Submissions shall also conform to the requirements of the procurement contract documents.

b. Application Software

All documents shall be in a common format within the set of Contractor documents and within each of the sets of documents from each Supplier. These formats shall use a consistent set of graphical and textual techniques to fully describe the software functionality and implementation.

(1) Contractor Requirements

The Contractor shall develop and submit for approval a Software Project Management Plan (SPMP) **[CDRL 20-049]**, a Software Quality Assurance Plan

(SQAP) **[CDRL 20-050]**, a Software Configuration Management Plan (SCMP) **[CDRL 20-051]**, and a Software Verification and Validation Plan (SVVP) **[CDRL 20-052]**, all in accordance with IEEE Standard 1558-2004. The scope of these documents shall cover the entire software development for the project. These documents shall be submitted for approval soon after the NTP, before the submittal of other documents. The Contractor Software Verification and Validation Plan (SVVP) shall describe the integration of the subsystems to cover the requirements of the entire system.

A SVVR is also required with each Software release presenting the Contractor V&V activities.

(2) Supplier Requirements

Each Supplier shall create and submit the full set of documents conforming to IEEE Standard 1558-2004 Standard for Software Documentation for Rail Equipment and Systems for type 5 as defined in that standard.

All documentation shall use names and acronyms consistent with those in the Technical Specification and as defined in the Contractor Overall SFD and the SFD for the specific system.

The problem reporting and corrective action process shall provide reports allowing the easy tracing of problems found at each development stage back to changes in the requirements, design or implementation in source code.

Each software release shall be provided in a package including a consistent set of up to date and approved software documents, test reports, the updated SVVR, and a Software Version Description (SVD).

All references to documents shall specify the location to a sufficiently specific section of text so the reader can easily and unambiguously understand the intention.

All source code, including comments, shall be submitted to the Engineer for review, in English **[CDRL 20-053]**. Source code must be well structured, modular, and clearly documented to allow easy comprehension and straightforward traceability to the SDD documents. Software comments shall also include explanation of all significant memory addresses such as interrupt vectors, I/O addresses, and memory locations for RAM, ROM, and other memory devices. Files and modules must have headers describing functions, interfaces and change histories. Proposals for escrow agreements will be considered according to the Terms and Conditions of the Contract.

All documents shall be submitted as separate documents, not as internal to other submittals such as the PDR, In-Progress Design Review (IPDR), or CDR documents. All documents shall use names and acronyms consistent with those in the Technical Specification and as defined in the Contractor overall SFD and in the individual system SFDs.

The Contractor shall assure that the documentation produced provides for the straightforward traceability of requirements of the Technical Specification throughout the design documentation and final test. This shall include derived requirements as well.

c. Additional Requirements for Software Documents

(1) Software Project Management Plan (SPMP)

Details shall be provided describing monitoring of Supplier activities to assure development tasks are completed and delivery requirements are met for each development phase from concept development to warranty support. It shall describe the procedures used to conduct software quality audits for suppliers and subcontractors.

The Software Project Management Plan (SPMP) for both the Contractor and the suppliers shall include a schedule showing the key tasks defined for the software development. A Gantt chart shall be used showing the timing and interrelationships between tasks. This schedule shall include the allocation of time and resources for realistically addressing issues raised by internal reviews, Contractor reviews, or by the Engineer. The SPMP shall include detailed descriptions of the metrics to be used to monitor and control the project. This shall include monitoring project progress on intermediate tasks and triggering management corrective actions with sufficient time to assure required deliveries are met.

(2) Software Verification and Validation Plan (SVVP)

The supplier's Software Verification and Validation Plan (SVVP) shall require internal peer reviews of each version of the documents and source code. Summary reports of these reviews shall be included in the Software Verification and Validation Report (SVVR) **[CDRL 20-054]**. For each review it shall also state the version of the items reviewed as well as the versions of all related documents used in the review. These include, for example, documents that the subject item was reviewed against such as an SRS or related interface specifications used in the review of an SDD. The review reports shall list for each V&V task, the type of review, the participants, the date, the versions of all items used in the review, and all issues from the review. The SVVP shall also require reviews of each design or source code change prior to test and release. Summaries of these also are required in the SVVR.

Similarly, test reports summaries shall be included in the SVVR and they shall identify the version of the SCI tested, the test plans, and the versions of all additional items used in the test.

(3) Software Requirements Traceability Matrix (SRTM)

The Software Requirements Traceability Matrix (SRTM) **[CDRL 20-055]** shall provide cross-referencing between the requirements of the SRSs and the corresponding sections of the SFD, SDD/ICD/DBDD, and the Software Verification Test Procedures. It shall include one table for each SCI and within each table there shall be a row for each SRS requirement. The columns shall include the unique identifier for the individual requirements defined in the SRS, a short description of the requirement, the reference to the corresponding SFD

section, the reference to the SRS section, the reference to the SDD section or sections, and the reference or references to the Software Validation tests. Since the references are dependent on the version of the documents referenced, the specific versions of all referenced documents must be stated for each table.

All references to documents shall specify the location of the pertinent text to a sufficiently specific section so the reader easily and unambiguously understands the intention.

(4) Software Version Description

SVDs shall be provided with each version of the software items. Each SVD shall include a description of the problems addressed or features added, changes to software requirements, changes to the software design, source code changes, a record of review results for each pertinent document, test plan changes, and test results. They shall also include the CRC-32 value which can be used to confirm the integrity of the software loaded in the target. The SVD shall be submitted to the Engineer before use of the software in official tests or on the cars.

6. Operating Systems and Languages

Software may be written in a high or low level language although high level languages are preferred. The language, and its implementation for the selected microprocessor system, shall be commercially available in English. No proprietary languages or code generating systems will be allowed. All languages and operating systems must have an acceptable installed base and be approved. Languages shall be described in the Methods, Tools, and Techniques sub-clause of the Suppliers SPMP.

7. General Market Software

Some software supplied under this procurement may be purchased by the suppliers from external sources and commercially available to a wide variety of users. Examples include operating systems and data base software for fault analysis. The Contractor shall submit for approval a list of software that is commercially available to the general public and which the Contractor and supplier would like to be considered Commercially Available (CA) for this project. This category does not include supplier software.

For Commercially Available software, software documentation requirements are limited to: the original data storage/transfer media (DVD, CD-ROM, or other), functional and usage details, all provider manuals, and licenses required for RTD site use for all hardware provided under this Contract. The requirements and interfaces shall be documented in a specific section of the SFD for that component of the system. References to specific sections of the provider's documentation shall be included for all requirements. The Contractor shall incorporate training on how the software is to be used in the specific situation for which it was provided as part of the Training Program. For all other software, the documentation requirements of TP20.31.C.5.b. apply, as deemed by RTD.

8. Time and Date Processing

All systems provided under this procurement, whether acting separately or in combination, must properly process all times and dates within the required span of years from 2000 to

2060 inclusive. This includes all carborne software as well as all PTUs, Development Systems, Operating Systems, and workstations.

The Contractor and suppliers will be required to provide robust designs and to establish, through both analysis and test, that the product shall process times and dates correctly. An example of a design that may not correctly process dates throughout the required range is storing time as seconds past 1-1-1970. This technique must either not be used or, if used, must be extended to correctly process all required times and dates. Requirements for proper time and date processing must be included in the SRSs for all systems for subsystems that use times and dates.

Since setting the time and date is required as part of the normal operating of the cars and is required for system validation testing, the design must allow for setting of the time for the whole system or for any part of the system, forward or back, to any time in the required span of years, without incorrect operation or loss of data.

D. Hardware

1. Hardware Platform

Microprocessor-based systems shall be based on an established family of microprocessors in wide use in the control system industry. They shall be supported by a full range of software development languages and diagnostic programs similar to that available for the Intel 80XXX, Motorola 68XXX and Echelon 31XX family of devices. Any use of commercially-available computer boards must be specifically approved. Such approval will be based upon a technical review of the product, product documentation, and a commercial assessment of product availability.

2. Electrical Isolation and Pre-processing

The control system shall be powered by dedicated transformer-isolated power supplies driven from the low voltage distribution network.

All control system input and output signals shall be through isolation buffers. High voltage inputs and outputs shall be isolated external to the microcomputer card rack. Low voltage (battery and logic voltage level) inputs and outputs shall be isolated via buffer cards in or external to the microcomputer card rack. The isolation buffers shall:

- Protect and isolate the control system from damage due to over-voltage, under-voltage, transient, shorts and opens;
- Perform necessary voltage translations;
- Remove noise and undesired signals;
- Limit, pre-process, discriminate and format those signals that would otherwise require excessive processor time; and
- Consist of optical isolators, transformers isolators, and other circuits appropriate to the application. Voltage divider circuits will not be allowed.

3. Program and Data Storage

Program code and fixed data shall be stored in a form of PROM. All EPROM windows shall be covered with labels that are opaque at the Ultraviolet (UV) erasing wavelengths. Flash PROM circuitry shall be configured so that reprogramming can be disabled via a semi-permanent hardware mechanism. After the software design has stabilized in revenue service, flash PROM reprogramming shall be disabled such that an approved hardware "key" is required for future reprogramming. The approved hardware key shall be a small electronic device that attaches to a parallel port or other similar interface on the laptop Personal Computer (PC) used for the PTE. The laptop PC will interact with the hardware key so as to allow loading of software to the car subsystem only if the hardware device is attached. If a system does not disable flash PROM reprogramming in the equipment itself, then it is acceptable to use a PTE where both a password and an approved hardware key are necessary for the PTE to download software. The same hard key shall be used for all systems. One hard key shall be provided for each PTE. The device shall be designed such that duplicate units may not be unofficially obtained. The device shall be programmable by RTD to limit its use (e.g., contains a time limit, only works with a specific PTE, etc.)

4. Batteries

Other than the main car battery, batteries are restricted as follows: One battery per car for the purpose of maintaining a clock and calendar, one battery per Event Recorder System, and any others only with approval. Rechargeable batteries shall be sized to retain data for at least six months without charging and shall be located such that leakage cannot damage any control system components. Battery life shall be no less than five years, regardless of type. The system shall annunciate the need for battery replacement such that the battery continues to perform its function until it can be replaced at the next periodic maintenance. Batteries shall not be connected by soldering. Non-rechargeable batteries shall be replaced within six months prior to car acceptance. Upon detection of the loss of input power to non-battery backed systems, the processor system shall store all necessary RAM control data to non-volatile memory.

5. Expansion Capacity

The hardware shall be designed to allow program expansion without hardware modification. The memory needs of the installed software shall not utilize more than 70% of the installed memory. This requirement applies individually to each type of memory installed, whether it is EPROM, EEPROM, Flash PROM, RAM, or other type. Peak processing time demands shall not be greater than 50% of the available processor time, except as indicated below. The Contractor may petition for relief from the 50% requirement, based on product maturity and the lack of potential for expansion or modification. In no case may more than 75% of the peak processor time be utilized, and generally, more margin will be required. The hardware shall include spare input and output channels of each type used within the system, except for major output drivers, the quantity of which is fixed by the overall system design (e.g., traction motor semi-conductors and sign character drivers). In addition, the architecture and assembly construction shall allow for the installation of additional I/O hardware. Simple singular functions performed by dedicated embedded processors may utilize up to 75% of the available processor time and do not have to provide spare I/O capability.

E. Deliverables and Schedule

All documents submitted shall utilize revision bars in the margin and/or underline/strike through to highlight changes from one revision to the next. A change history shall be included in each document describing each change and giving the reason for the change.

The correspondence between design reviews and software submittals shall be as shown below. Issues with documents must be resolved and documents approved to close each design phase.

Review Type	Software Related Documentation
Time of Conceptual Design	Software Quality Assurance Plan (SQAP) for Contractor
	Software Project Management Plan (SPMP) for Contractor
	Contractor Preliminary Overall System Functional Description (SFD)
	CMMI Reports, Improvement Plans, Interim Plans, and Monthly Progress Reports as required by TP21.
Preliminary Design Review	System Functional Description (SFD) for each System
	Updated Overall System Functional Description (SFD) and Interface Control Documents (ICDs) for Contractor
	Software Quality Assurance Plans (SQAP) for Suppliers
	Software Project Management Plan (SPMP) for Suppliers
	Preliminary Software Configuration Management Plan (SCMP) for Contractor and Suppliers
	Preliminary Software Requirements Specification (SRSs) and ICDs
	Preliminary Software Test Plans (STPs)
	Preliminary Software Design Descriptions (SDDs) and Preliminary Database Design Descriptions (DBDDs) if needed at the Architecture Design level
	Preliminary Software Verification and Validation Plan (SVVP) for Contractor and Suppliers
	Preliminary Software Requirements Traceability Matrix (SRTM) for each system

Review Type	Software Related Documentation
In-Process Design Review	Updated individual System Functional Descriptions (SFD)
	Overall System Functional Description (SFD) and Interface Control Documents (ICDs) for Contactor
	Software Requirements Specifications (SRSs) and ICDs
	Software Test Plans (STP) and Software Test Procedures (STPr)
	Software Design Descriptions (SDDs) – Architecture Design and Detailed Design and Database Design Descriptions (DBDDs) if needed
	Software Verification and Validation Plan (SVVP) for Contractor and Suppliers
	Software Requirements Traceability Matrix (SRTM) for each system
	Revisions to other documents for changes or updates
Critical Design Review	Updated Systems Functional Descriptions (SFD)
	Updated Contractor Overall System Functional Description (SFD) and Interface Control Documents (ICDs)
	Updated Software Requirements Specifications (SRSs)
	Updated Software Test Plans (STPs) and Software Test Procedures (STPr)
	Software Design Descriptions (SDDs) – Architecture Design and Detailed Design and Database Design Descriptions (DBDDs) if needed
	Source Code Files
	Updated Software Configuration Management Plan (SCMP) for Contractor and Suppliers
	Updated Software Verification and Validation Plan (SVVP) for Contractor and Suppliers
	Software Requirements Traceability Matrix (SRTM) for each system
	Revisions to other documents for changes or updates

After original approval, changes to the software shall be formally submitted for approval prior to implementation of the changes in the source code. New versions of software must be accompanied by revised, reviewed and released software documentation, as well as SVVRs.

F. Configuration Control

The Contractor shall develop and maintain a Software Configuration Control Plan (SCCP) for tracking software changes relative to cars, PTEs, and Bench Test Equipment (BTE) on RTD property. The Contractor shall include in the plan a data base management system capable of maintaining the history of all software and status changes making it possible to determine which versions currently resides in which equipment, on which cars, and also which versions were used in the past. The data base management system shall be capable of generating various reports showing the configuration of a typical vehicle system in terms of software history and status, and also a fleet configuration report capable of determining the exact software configuration of a particular operating train on a car-by-car basis. The Contractor shall make these reports available to RTD.

The SCCP shall be submitted for approval by CDR.

The Contractor shall submit a final software configuration for each car at the time of acceptance or conditional acceptance in an electronic format as approved

All software shall be identified by a name and a version number. The name shall identify the equipment into which the software is installed. Every change to software shall be reflected in an update to the version number. Refer to TP20.31D.3 for related requirements.

G. Software Maintenance and Related Tools

RTD retains the right to modify the performance of its equipment, at its own risk. Since this involves software modifications, all software for this vehicle shall be delivered to RTD on DVD, CD-ROM, or other approved media. Software more likely for RTD to alter (e.g., for messages, performance, platform length) shall be provided directly to RTD. Software that is less likely to be used by RTD may be held in escrow. RTD will audit any software in escrow according to the Terms and Conditions of the Contract.

The Contractor shall define and describe all items needed for software workstations including all of the software source files and software development tools used by the suppliers. The complement of equipment shall include all compilers, assemblers, linkers, in-circuit emulators, version control software, development kits, and other such tools that are used for software development. All such software and hardware tools shall then be provided and included in the escrowed material. Development tools and software which are identified must be the version as used by the suppliers. Specific instructions for compiling and linking the specific software shall be provided. The instructions shall include software modifications and tests of all non-Commercially Available software (non-CA software) on this procurement.

The instructions and software documentation is to be demonstrated, and proven to create program files that then must match the programs stored within the system's equipment.

All associated manuals and software licenses shall be provided.

TP20.32 CONTRACT DELIVERABLE REQUIREMENTS LIST

CDRL #	Title	Car Type	Reference Paragraph
20-001	Material Safety Data Sheets	All	TP20.01.A
20-002	Recommended Cleaning Agents	All	TP20.01.E
20-003	Plan for Joining Dissimilar Materials	All	TP20.02.A.1
20-004	Fastener Plan	All	TP20.02.B.4
20-005	Stainless Steel Inspection Plan	All	TP20.03.D
20-006	Structural Steel Inspection Plan	All	TP20.04.B
20-007	Structural Casting Qualification Report	All	TP20.05.B
20-008	Casting Inspection Report	All	TP20.05.C
20-009	Aluminum Sheet Extrusion and Forging Test Reports	All	TP20.06.A
20-010	Elastomer Test Results	All	TP20.07.B
20-011	Safety Glass Material Conformance Report	All	TP20.08.A
20-012	Plastic Glazing Weather Testing Report	All	TP20.08.B.3.e
20-013	Plastic Glazing Orange Peel Standard	All	TP20.08.B.4.e
20-014	Plastic Glazing Material Conformance Report	All	TP20.08.B.4.i
20-015	Phenolic Composite Floor Panels Test Results	All	TP20.10.G
20-016	Thermoplastic Color and Surface Finish Approval	All	TP20.14.A
20-017	Thermoplastic Strength Requirements Certificates	All	TP20.14.C
20-018	Pneumatic System Piping Design Standards	All	TP20.15.A
20-019	Air Brake Supplier Approval Certification of Car Piping	All	TP20.15.B
20-020	Refrigeration and Pneumatic Pipe Cleaning Procedure	All	TP20.15.D
20-021	Wire Manufacturer Approval	All	TP20.17.A
20-022	Wire Samples and Certified Qualification Test Reports	All	TP20.17.A
20-023	Wire and Cable Production and Qualification Test Reports	All	TP20.17.C.1
20-024	Wiring Plan	All	TP20.18.C.3.b
20-025	Wire Marking and Designation	All	TP20.18.E
20-026	Connector Product Line	All	TP20.19.A
20-027	Crimp Product Line and “Pull Test” Method	All	TP20.19.C
20-028	Grounding Plan	All	TP20.19.G.1
20-029	Conduit Arrangement Plan	All	TP20.20.C
20-030	Conduit and Fitting Product Line	All	TP20.21.A
20-031	Welding Procedure Specifications and Procedure Qualification Records	All	TP20.23.B
20-032	Full Penetration Weld Sampling Plan	All	TP20.23.D
20-033	Ring Weld Sampling Plan	All	TP20.23.D
20-034	Welding Electrode, Welding Wire, and Cover Gas Specifications	All	TP20.23.F

CDRL #	Title	Car Type	Reference Paragraph
20-035	Resistance Spot Weld Procedures and Qualifications	All	TP20.23.H
20-036	Resistance Weld Deviations	All	TP20.23.H
20-037	Spot Weld Parameters And Strength	All	TP20.23.H
20-038	Paint Touch Up Procedure	All	TP20.24.A
20-039	Paint Color Samples	All	TP20.24.B
20-040	Paint Coating and Application Documentation	All	TP20.24.I
20-041	Fire Safety Analysis	All	TP20.25.A
20-042	Combustible Content Matrix	All	TP20.25.B
20-043	Floor Fire Test Approval	All	TP20.25.D
20-044	Floor Fire Test Drawing	All	TP20.25.D
20-045	Toxicity Test Report	All	TP20.25.E
20-046	Overall System Functional Description (SFD)	All	TP20.31.B.2.a
20-047	System Functional Description (SFD)	All	TP20.31.B.2.b
20-048	Software Requirements Specification	All	TP20.31.C.4
20-049	Software Project Management Plan (SPMP)	All	TP20.31.C.5.b
20-050	Software Quality Assurance Plan (SQAP)	All	TP20.31.C.5.b
20-051	Software Configuration Management Plan (SCMP)	All	TP20.31.C.5.b.(1)
20-052	Software Verification And Validation Plan (SVVP)	All	TP20.31.C.5.b.(1)
20-053	Software Source Code and Related Necessary Files	All	TP20.31.C.5.b.(2)
20-054	Software Verification and Validation Report	All	TP20.31.C.5.c.(2)
20-055	Software Requirements Traceability Matrix	All	TP20.31.C.5.c.(3)

End of Section